Questions 33 to 37

(I) T stays as PG; S stays in hostel.
U stays in hostel and Q stays at home.
Now, R does not stay as PG. So, P stays as PG.
Clearly, R stays at home.

(II) S studies Physics; R studies Philosophy and T studies Mathematics. Now, P who stays as PG does not study Statistics or History. So, P studies English.

	Place of stay	Subject
P	PG	English
Q	Home	Statistics or History
R	Home	Philosophy
S	Hostel	Physics
T	PG	Mathematics
U	Hostel	Statistics or History

33. (*e*) : P studies English.
34. (*a*) : Clearly the incorrect combination is English — Hostel.
35. (*b*) : S stays in hostel and R stays at home.
36. (*c*) : Q studies History or Statistics.
37. (*b*) : Q and R stay at home.

Questions 38 to 42

(I) Kunal stays close to school. So, the other three — Rohit, Ashish and John stay far from the school.

(II) Rohit, Kunal and Ashish do not study in class VI. So, John studies in class VI.
Rohit and Kunal are not good at all subjects and John is in class VI. So, Ashish is good at all subjects and studies in class V.
Clearly, Rohit and Kunal study in class IV.

(III) Rohit is good at Mathematics, Kunal is weak in Mathematics. Ashish is good at all the subjects. Clearly, John is weak in all the subjects.

	Stay	Class	Good at	Weak in
Rohit	Far	IV	Mathematics	Hindi, Science, Social Science
Kunal	Close	IV	Hindi, Science, Social Science	Mathematics
Ashish	Far	V	All subjects	
John	Far	VI		All subjects

38. (*c*) : Ashish is good at all the subjects.
39. (*d*) : John is weak in all the subjects.
40. (*b*) : Kunal and Ashish are good at Hindi.
41. (*a*) : Rohit and Ashish are good at Mathematics.
42. (*d*) : Other than Rohit and Ashish, John stays far from school.

Questions 43 to 45

(I) D plays Polo; E plays Badminton; A plays Cricket. C does not play Tennis or Hockey. So, C plays Volleyball.
B does not play Tennis. So, he plays Hockey. Clearly, F plays Tennis.

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Questions 24 to 28

The given information can be analysed as follows :

Let 'O' denote optional and 'C' denote compulsory.

	A	B	C	D	E	F
History	C	C	C	O	–	–
Physics	–	–	–	–	C	O
Mathematics	–	–	O	–	O	C
English	O	–	–	C	–	–
Chemistry	–	O	–	–	–	–

One compulsory subject of F is Mathématics. F has Physics as one of the subjects. So, Physics is optional of F. F has Mathematics as optional and Physics as one subject.

So, Physics is the compulsory subject of E. A and D have the same subjects — History and English. D has History as optional subject and so English is the compulsory subject of D. Subjects of A and D are reverse in regard of optional and compulsory.

So, A has History as compulsory subject and English as optional.

Chemistry is the optional subject of only one teacher. So, it is the optional of B, which only remains.

We know that History is the compulsory subject of three teachers. So, it is compulsory for A, B and C.

D is the teacher having English as her compulsory subject.

So, D is the only female teacher.

24. (*a*) : C's compulsory subject is History.

25. (*d*) : D is the only female teacher.

26. (*e*) : E has same subjects as those of F but the compulsory and optional subjects of E are reverse of those of F. So, the answer is 'none of these'.

27. (*c*) : Clearly, E has the same subject combination as that of F.

28. (*d*) : A, B and C have History as the compulsory subject.

Questions 29 to 32

(I) **Choice :** Jayant likes Chocolates; Tanmay likes Bengali sweets. Namita does not like Dry fruits or Ice cream. So, Namita likes Pastries. Asha does not like Ice cream. So, she is fond of Dry fruits. Finally, Kamal likes Ice cream.

(II) **Date of Birth :** The one who likes Pastries *i.e.*, Namita is born in the middle of months given *i.e.*, in March. Tanmay is born in next month after Namita *i.e.*, in April. Jayant's birthday is in February.

	Choice	Date of birth
Jayant	Chocolates	February
Kamal	Ice cream	January or May
Namita	Pastries	March
Asha	Dry fruits	January or May
Tanmay	Bengali sweets	April

29. (*d*) : The choice of Asha is Dry fruits.

30. (*e*) : The correct combination for Jayant is February — Chocolates.

31. (*a*) : The choice of Kamal is Ice cream.

32. (*c*) : Kamal was born in January or May.

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(II) D has yellow car; E has white car; B has a red car; A has a black car.
Now, C does not have a blue car. So, colour of C's car is green.
Clearly, F owns a blue car.

	Game	Colour of Car
A	Cricket	Black
B	Hockey	Red
C	Volleyball	Green
D	Polo	Yellow
E	Badminton	White
F	Tennis	Blue

43. (*b*) : C plays Volleyball.

44. (*b*) : F owns a blue car.

45. (*b*) : Clearly, the incorrect combination is Green — Tennis.

46. (*c*) : The four vocal musicians and one actress in the group are given. Two dancers are Shailja and Tanuja. Two violinists are Girija and Vanaja. Since Jalaja and Shailja cannot be violinists, so, remaining two violinists are Tanuja and Pooja. Clearly, Tanuja is both a violinist and a dancer.

TYPE 2 : SEATING/PLACING ARRANGEMENTS

In this type of questions, some clues regarding seating or placing sequence (linear or circular) of some persons or items is given. The candidate is required to form the proper sequence using these clues and answer the questions accordingly.

Ex. 1. ***Read the following information carefully and answer the questions given below :***

Six persons A, B, C, D, E and F are sitting in two rows, three in each.
E is not at the end of any row.
D is second to the left of F.
C, the neighbour of E, is sitting diagonally opposite to D.
B is the neighbour of F.

1. Which of the following are sitting diagonally opposite to each other ?
(*a*) F and C (*b*) D and A (*c*) A and C (*d*) A and F (*e*) A and B

2. Who is facing B ?
(*a*) A (*b*) C (*c*) D (*d*) E (*e*) F

3. Which of the following are in the same row ?
(*a*) A and E (*b*) E and D (*c*) C and B (*d*) A and B (*e*) C and E

4. Which of the following are in one of the two rows ?
(*a*) FBC (*b*) CEB (*c*) DBF (*d*) AEF (*e*) ABF

5. After interchanging seat with E, who will be the neighbours of D in the new position ?
(*a*) C and A (*b*) F and B (*c*) Only B (*d*) Only A (*e*) Only C

Solution : The given information can be analysed as follows :
E is not at end. So, E must be in the middle of one of the rows.
D is second to the left of F. So, order of the row must be D — F.

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C is neighbour of E and is sitting diagonally opposite to D means C is under F in the other row *i.e.*, D — F

— E C

B is the neighbour of F.

So, the arrangement must be D B F

A E C

1. Other than D and C (given), A and F are sitting diagonally opposite to each other, as seen in the arrangement. So, the answer is (*d*).
2. Clearly, E is opposite to B in the other row. So, E is facing B and the answer is (*d*).
3. Clearly, from amongst the given alternatives, A and E are in the same row. So, the answer is (*a*).
4. Clearly, from amongst the given alternatives, D, B and F are in the same row. So, the answer is (*c*).
5. Clearly, neighbours of E are A and C. So, on interchanging the seat with E, the new neighbours of D will be A and C. So, the answer is (*a*).

Ex. 2. Eight books are kept one over the other. Counting from the top, the second, fifth and sixth books are on Plays. Two books on Plays are between two books on Composition. One book of Plays is between two books on Poetry while the book at the top of the book of Literature is a book of Composition. Which book is fourth from the top ?

(*a*) Plays (*b*) Poetry (*c*) Composition (*d*) Literature

Solution : We analyse the given information as follows :

Let C denote 'Composition', P denote 'Plays', Po denote 'Poetry' and L denote 'Literature'.

1	2	3	4	5	6	7	8
–	P	–	–	P	P	–	–
–	–	–	C	P	P	C	–
Po	P	Po	–	–	–	–	–
–	–	–	–	–	–	C	L

So, the arrangement becomes :

1	2	3	4	5	6	7	8
Po	P	Po	C	P	P	C	L

Clearly, the fourth book from the top is on Composition. So, the answer is (*c*).

Ex. 3. ***Read the following information and answer the questions that follow :***

(*i*) Six friends A, B, C, D, E and F are sitting in a closed circle facing the centre.

(*ii*) E is to the left of D.

(*iii*) C is between A and B.

(*iv*) F is between E and A.

1. Who is to the left of B ?

(*a*) A (*b*) C (*c*) D (*d*) E (*e*) None of these

2. Who is to the right of C ?

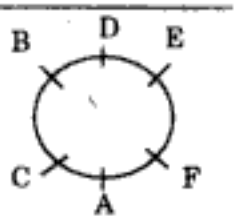
(*a*) A (*b*) B (*c*) D (*d*) E (*e*) F

3. Which of the above given statements is superfluous ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) (*iv*) (*e*) None of these

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Solution : Clearly, in the circle the arrangement is as shown :



1. (*c*) : Clearly, D is to the left of B.

2. (*a*) : Clearly, A is to the right of C.

3. (*e*) : Since all the statements are necessary to determine the arrangement, none of them is superfluous.

EXERCISE 6B

1. Four girls are sitting on a bench to be photographed. Shikha is to the left of Reena. Manju is to the right of Reena. Rita is between Reena and Manju. Who would be second from the left in the photograph ? **(Bank P.O. 1994)**

(*a*) Reena (*b*) Shikha (*c*) Manju (*d*) Rita

2. There are five different houses, A to E, in a row. A is to the right of B and E is to the left of C and right of A. B is to the right of D. Which of the houses is in the middle ? **(C.B.I. 1995)**

(*a*) A (*b*) B (*c*) D (*d*) E

3. In a March Past, seven persons are standing in a row. Q is standing left to R but right to P. O is standing right to N and left to P. Similarly, S is standing right to R and left to T. Find out who is standing in the middle.

(*a*) P (*b*) Q (*c*) R (*d*) O

(Assistant Grade, 1996)

4. Five children are sitting in a row. S is sitting next to P but not T. K is sitting next to R who is sitting on the extreme left and T is not sitting next to K. Who are sitting adjacent to S ?

(*a*) K and P (*b*) R and P (*c*) Only P

(*d*) P and T (*e*) Insufficient information

5. Five girls are sitting in a row. Rashi is not adjacent to Sulekha or Abha. Anuradha is not adjacent to Sulekha. Rashi is adjacent to Monika. Monika is at the middle in the row. Then, Anuradha is adjacent to whom out of the following ?

(*a*) Rashi (*b*) Sulekha (*c*) Abha

(*d*) Monika (*e*) Cannot be determined

Directions (*Questions 6 to 8*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1994)**

(A) There are five friends.

(B) They are standing in a row facing South.

(C) Jayesh is to the immediate right of Alok.

(D) Pramod is between Bhagat and Subodh.

(E) Subodh is between Jayesh and Pramod.

6. Who is at the extreme left end ?

(*a*) Alok (*b*) Bhagat (*c*) Subodh

(*d*) Data inadequate (*e*) None of these

7. Who is in the middle ?

(*a*) Bhagat (*b*) Jayesh (*c*) Pramod

(*d*) Subodh (*e*) Alok

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8. To find answers to the above two questions, which of the given statements can be dispensed with ?

(*a*) None (*b*) A only (*c*) B only (*d*) C only **(*e*) D only**

9. Five persons A, B, C, D and E are sitting in a row facing you such that D is on the left of C and B is on the right of E. A is on the right of C and B is on the left of D. If E occupies a corner position, then who is sitting in the centre ?

(*a*) A (*b*) B (*c*) C (*d*) D (C.B.I. 1995)

Directions (*Questions 10 to 14*) : *Study the given information carefully and answer the questions that follow :* (L.I.C. 1994)

(*i*) A, B, C, D, E, F and G are sitting on a wall and all of them are facing east.

(*ii*) C is on the immediate right of D.

(*iii*) B is at an extreme end and has E as his neighbour.

(*iv*) G is between E and F.

(*v*) D is sitting third from the south end.

10. Who is sitting to the right of E ?

(*a*) A (*b*) C (*c*) D (*d*) F (*e*) None of these

11. Which of the following pairs of people are sitting at the extreme ends ?

(*a*) AB (*b*) AE (*c*) CB (*d*) FB (*e*) Cannot be determined

12. Name the person who should change places with C such that he gets the third place from the north end.

(*a*) E (*b*) F (*c*) G (*d*) D

13. Immediately between which of the following pairs of people is D sitting ?

(*a*) AC (*b*) AF (*c*) CE (*d*) CF (*e*) None of these

14. Which of the conditions (*i*) to (*v*) given above is not required to find out the place in which A is sitting ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) All are required (*e*) None of these

15. In the Olympic Games, the flags of six nations were flown on the masts in the following way : (Assistant Grade, 1996)

The flag of America was to the left of Indian tricolour and to the right of the flag of France. The flag of Australia was on the right of the Indian flag but was to the left of the flag of Japan, which was to the left of the flag of China. Find the two flags which are in the centre.

(*a*) India and Australia (*b*) America and India

(*c*) Japan and Australia (*d*) America and Australia

16. Mr. A, Miss B, Mr. C and Miss D are sitting around a table and discussing their trades.

(1) Mr. A sits opposite to cook.

(2) Miss B sits right to the barber.

(3) The washerman is on the left of the tailor.

(4) Miss D sits opposite Mr. C.

What are the trades of A and B ?

(*a*) Tailor and Barber (*b*) Tailor and Cook

(*c*) Barber and Cook (*d*) Washerman and Cook

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17. Sitting in a row in front of a camera, Mr. X is on the left of the person sitting in the centre but is on the right of Mr. Y. Mr. P is on the right of Mr. Z and Mr. R is on the right of Mr. P. Mr. R is the second person from the person sitting in the centre. Who is the person sitting in the centre ? **(C.B.I. 1993)**

(*a*) Mr. X (*b*) Mr. Y (*c*) Mr. Z (*d*) Mr. R

Directions (*Questions 18 to 22*) : *Study the given information carefully and answer the questions that follow :* **(U.T.I. 1993)**

(*i*) Eleven students, A, B, C, D, E, F, G, H, I, J and K are sitting in the first row of the class facing the teacher.

(*ii*) D who is to the immediate left of F is second to the right of C.

(*iii*) A is second to the right of E, who is at one of the ends.

(*iv*) J is the immediate neighbour of A and B and third to the left of G.

(*v*) H is to the immediate left of D and third to the right of I.

18. Who is sitting in the middle of the row ?

(*a*) B (*b*) C (*c*) G (*d*) I (*e*) None of these

19. Which of the following groups of friends is sitting to the right of G ?

(*a*) CHDE (*b*) CHDF (*c*) IBJA (*d*) ICHDF (*e*) None of these

20. Which of the following statements is true in the context of the above sitting arrangements ?

(*a*) There are three students sitting between D and G.

(*b*) K is between A and J.

(*c*) B is sitting between J and I.

(*d*) G and C are neighbours sitting to the immediate right of H.

21. In the above sitting arrangement, which of the following statements is superfluous ?

(*a*) (*i*) (*b*) (*ii*) (*c*) (*iii*) (*d*) (*iv*) (*e*) None is superfluous

22. If E and D, C and B, A and H and K and F interchange their positions, which of the following pairs of students is sitting at the ends ?

(*a*) D and E (*b*) E and F (*c*) D and K

(*d*) K and F (*e*) None of these

Directions (*Questions 23 to 27*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1995)**

(*i*) Eight persons E, F, G, H, I, J, K and L are seated around a square table — two on each side.

(*ii*) There are three lady members and they are not seated next to each other.

(*iii*) J is between L and F.

(*iv*) G is between I and F.

(*v*) H, a lady member, is second to the left of J.

(*vi*) F, a male member is seated opposite E, a lady member.

(*vii*) There is a lady member between F and I.

23. Who among the following is seated between E and H ?

(*a*) F (*b*) I (*c*) J

(*d*) Cannot be determined (*e*) None of these

24. How many persons are seated between K and F ?

(*a*) One (*b*) Two (*c*) Three

(*d*) Cannot be determined (*e*) None of these

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25. Who among the following are the three lady members ?
(*a*) E, G and J (*b*) E, H and G (*c*) G, H and J
(*d*) Cannot be determined (*e*) None of these

26. Who among the following is to the immediate left of F ?
(*a*) G (*b*) I (*c*) J
(*d*) Cannot be determined (*e*) None of these

27. Which of the following is true about J ?
(*a*) J is a male member. (*b*) J is a female member.
(*c*) Sex of J cannot be determined. (*d*) Position of J cannot be determined.
(*e*) None of these

Directions (*Questions 28-29*) : *On the basis of the information given below, answer questions 28-29.*

(A) P, Q, R, S and T are sitting in a circle facing the centre.
(B) R is immediate left of T.
(C) P is between S and T.

28. Who is to the immediate left of R ?
(*a*) P (*b*) Q (*c*) S (*d*) T (*e*) Cannot be determined

29. To find the answer to the above question, which of the following statements can be dispensed with ?
(*a*) None (*b*) B only (*c*) C only (*d*) B or C only (*e*) None of these

30. Six friends A, B, C, D, E and F are sitting in a closed circle facing the centre. A is facing D. C is between A and B. F is between E and A. Who is to the immediate left of B ? **(I. Tax & Central Excise, 1996)**
(*a*) A (*b*) C (*c*) D (*d*) E

Directions (*Questions 31-32*) : *Read the following information carefully and answer the questions that follow :* **(Bank P.O. 1995)**

A, B, C, D, E and F are seated in a circle facing the centre. D is between F and B. A is second to the left of D and second to the right of E.

31. Who is facing A ?
(*a*) B (*b*) D (*c*) F
(*d*) Either F or B (*e*) None of these

32. Who among the following is facing D ?
(*a*) A (*b*) C (*c*) E
(*d*) Cannot be determined (*e*) None of these

Directions : *On the basis of the information given below, answer questions 33-34.* **(S.B.I.P.O. 1995)**

Eight friends A, B, C, D, E, F, G and H are sitting in a circle facing the centre. B is sitting between G and D. H is third to the left of B and second to the right of A. C is sitting between A and G and B and E are not sitting opposite to each other.

33. Who is third to the left of D ?
(*a*) A (*b*) E (*c*) F
(*d*) Cannot be determined (*e*) None of these

34. Which of the following statements is not correct ?
(*a*) C is third to the right of D. (*b*) A is sitting between C and F.

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(*c*) D and A are sitting opposite of each other.
(*d*) E is sitting between F and D.
(*e*) E and C are sitting opposite of each other.

35. A group of eight members sit in a circle. D is between A and F and is opposite to G. E is to the right of A but on the left of C, whose right hand neighbour is G. B enjoys having H to his left and F to his right. Find the member who is diagonally opposite to A. **(Central Excise, 1995)**

(*a*) B (*b*) F (*c*) G (*d*) H

Directions (*Questions 36 to 38*) : *Study the given information carefully and answer the questions that follow :* **(U.T.I. 1993)**

(*i*) There are seven books one each on Psychology, Hindi, English, Sociology, Economics, Education and Accountancy, lying on a table one above the other.
(*ii*) Sociology is on the top of all the books.
(*iii*) Accountancy is immediately below Education which is immediately below Sociology.
(*iv*) Economics is immediately above Psychology but not in the middle.
(*v*) Hindi is immediately below Psychology.

36. Economics is between which of the following books ?

(*a*) Accountancy and Education (*b*) Psychology and Hindi
(*c*) English and Psychology (*d*) Psychology and Sociology
(*e*) None of these

37. Which three books are between Accountancy and Hindi ?

(*a*) English, Economics and Psychology
(*b*) Economics, Psychology and Education
(*c*) Economics, Psychology and Hindi
(*d*) Cannot be determined
(*e*) None of these

38. If Sociology and English, Accountancy and Hindi and Education and Psychology interchange their positions, which book will be between Psychology and Sociology ?

(*a*) Accountancy (*b*) Psychology (*c*) Hindi
(*d*) Economics (*e*) None of these

39. In a shop, the items were arranged in a shelf consisting of six rows. Biscuits are arranged above the tins of chocolates but below the rows of packets of chips, cakes are at the bottom and the bottles of peppermints are below the chocolates. The topmost row had the display of jam bottles. Where exactly are the bottles of peppermints ? Mention the place from the top. **(Central Excise, 1996)**

(*a*) 2nd (*b*) 3rd (*c*) 4th (*d*) 5th

40. In a pile of reading material, there are novels, story-books, dramas and comics. Every novel has a drama next to it, every story-book has a comic next to it and there is no story-book next to a novel. If there be a novel at the top and the number of books be 40, the order of the books in the pile is :

(*a*) nscd (*b*) ndsc (*c*) csdn (*d*) dncs

(Hotel Management, 1996)

Directions (*Questions 41-42*) : *The following questions are based on the information given below :* **(S.B.I.P.O. 1995)**

(*i*) Seven books are placed one above the other in a particular way.

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(*ii*) History book is placed exactly above Civics book.

(*iii*) Geography book is fourth from the bottom and English book is fifth from the top.

(*iv*) There are two books in between Civics and Economics books.

41. How many books are there between Civics and Science books ? To answer this question, which other extra information is required, if any, from the following ?

(*a*) There are two books between Geography and Science books.

(*b*) There are two books between Mathematics and Geography books.

(*c*) There is one book between English and Science books.

(*d*) The Civics book is before two books above Economics book.

(*e*) No other information is required.

42. Out of the following, which three books are kept above English book ? To answer this question, which of the other informations, if any, is required ?

(*a*) The Economics book is between English and Science books.

(*b*) There are two books between English and History books.

(*c*) The Geography book is above English book.

(*d*) The Science book is placed at the top.

(*e*) No other information is required.

43. In a pile of 10 books, there are 3 of History, 3 of Hindi, 2 of Mathematics and 2 of English. Taking from above, there is an English book between a History and Mathematics book, a History book between a Mathematics and an English book, a Hindi book between an English and a Mathematics book, a Mathematics book between two Hindi books and two Hindi books between a Mathematics and a History book. Book of which subject is at the sixth position from the top ?

(*a*) English (*b*) Hindi (*c*) Mathematics (*d*) History

Directions (*Questions 44 to 48*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1991)**

In a car exhibition, seven cars of seven different companies *viz.* Cardilac, Ambassador, Fiat, Maruti, Mercedes, Bedford and Fargo were displayed in a row, facing east such that :

(1) Cardilac car was to the immediate right of Fargo.

(2) Fargo was fourth to the right of Fiat.

(3) Maruti car was between Ambassador and Bedford.

(4) Fiat, which was third to the left of Ambassador car, was at one of the ends.

44. Which of the following was the correct position of the Mercedes ?

(*a*) Immediate right of Cardilac (*b*) Immediate left of Bedford

(*c*) Between Bedford and Fargo (*d*) Fourth to the right of Maruti

(*e*) None of these

45. Which of the following is definitely true ?

(*a*) Fargo car is between Ambassador and Fiat.

(*b*) Cardilac car is to the immediate left of Mercedes.

(*c*) Fargo is to the immediate right of Cardilac.

(*d*) Maruti is fourth to the right of Mercedes.

(*e*) None of these

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46. Which cars are on the immediate either sides of the Cardilac car ?
(*a*) Ambassador and Marufi (*b*) Maruti and Fiat (*c*) Fiat and Mercedes
(*d*) Ambassador and Fargo (*e*) None of these

47. Which of the following is definitely true ?
(*a*) Maruti is to the immediate left of Ambassador.
(*b*) Bedford is to the immediate left of Fiat.
(*c*) Bedford is at one of the ends.
(*d*) Fiat is second to the right of Maruti.
(*e*) None of these

48. Which of the following groups of cars is to the right of the Ambassador car ?
(*a*) Cardilac, Fargo and Maruti (*b*) Maruti, Bedford and Fiat
(*c*) Mercedes, Cardilac and Fargo (*d*) Bedford, Cardilac and Fargo

Directions (*Questions 49 to 53*) : *Study the following information carefully and answer the questions given below it :*

All the roads of a city are either perpendicular or parallel to one another. The roads are all straight. Roads A, B, C, D and E are parallel to one another. Roads G, H, I, J, K, L and M are parallel to one another.

(*i*) Road A is 1 km east of road B.
(*ii*) Road B is $\frac{1}{2}$ km west of road C.
(*iii*) Road D is 1 km west of road E.
(*iv*) Road G is $\frac{1}{2}$ km south of road H.
(*v*) Road I is 1 km north of road J.
(*vi*) Road K is $\frac{1}{2}$ km north of road L.
(*vii*) Road K is 1 km south of road M.

49. Which is necessarily true ?
(*a*) E and B intersect. (*b*) D is 2 km west of B.
(*c*) D is at least 2 km west of A. (*d*) M is 1.5 km north of L.
(*e*) I is 1 km north of L.

50. If E is between B and C, which of the following is false ?
(*a*) D is 2 km west of A.
(*b*) C is less than 1.5 km from D.
(*c*) Distance from E to B added to distance of E to C is $\frac{1}{2}$ km.
(*d*) E is less than 1 km from A.
(*e*) D is less than 1 km from B.

51. If road E is between B and C, then distance between A and D is :
(*a*) $\frac{1}{2}$ km (*b*) 1 km (*c*) 1.5 km (*d*) 1.5-2 km (*e*) 2-2.5 km

52. Which of the following possibilities would make two roads coincide ?
(*a*) L is $\frac{1}{2}$ km north of I. (*b*) C is 1 km west of D.
(*c*) I is $\frac{1}{2}$ km north of K. (*d*) D is $\frac{1}{2}$ km east of A.
(*e*) E and B are $\frac{1}{2}$ km apart.

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53. If K is parallel to I and K is $\frac{1}{2}$ km south of J and 1 km north of G, which two roads would be $\frac{1}{2}$ km apart ?

(*a*) I and K (*b*) J and G (*c*) I and G (*d*) J and H (*e*) K and J

Directions (*Questions 54 to 58*) : *Read the following information carefully and answer the questions given below it :* **(M.A.T. 1997)**

Seven friends Kamla, Manish, Rohit, Amit, Gaurav, Pritam and Priya are sitting in a circle. Kamla, Manish, Rohit, Amit, Pritam and Priya are sitting at equal distances from each other.

Rohit is sitting two places right of Pritam, who is sitting one place right of Amit. Kamla forms an angle of 90 degrees from Gaurav and an angle of 120 degrees from Manish. Manish is just opposite Priya and is sitting on the left of Gaurav.

54. Who is the only person sitting between Rohit and Manish ?

(*a*) Pritam (*b*) Amit (*c*) Gaurav (*d*) Kamla

55. Gaurav is not sitting at equal distances from

(*a*) Rohit and Pritam (*b*) Amit and Kamla

(*c*) Manish and Pritam (*d*) All of the above

56. Gaurav is sitting ..... of Priya.

(*a*) to the left (*b*) to the right (*c*) two places right (*d*) None of these

57. The angle between Gaurav and Manish in the clockwise direction is

(*a*) 150° (*b*) 180° (*c*) 210° (*d*) None of these

58. Which of the following statements is not correct ?

(*a*) Pritam is between Manish and Kamla.

(*b*) Manish is two places away from Priya.

(*c*) Gaurav is sitting opposite Pritam.

(*d*) All of the above

ANSWERS

1. (*d*) : Shikha is to the left of Reena and Manju is to her right. Rita is between Reena and Manju. So, the order is : Shikha, Reena, Rita, Manju. In the photograph, Rita will be second from left.

2. (*a*) : B is to the right of D. A is to the right of B. E is to the right of A and left of C. So, the order is : D, B, A, E, C.

Clearly, A is in the middle.

3. (*b*) : Q is left to R and to the right of P *i.e.* P, Q, R.

O is to the right of N and left of P *i.e.* N, O, P.

S is to the right of R and left of T *i.e.* R, S, T.

So, the order is : N, O, P, Q, R, S, T.

Clearly, Q is in the middle.

4. (*d*) : S is sitting next to P. So, the order S, P or P, S is followed. K is sitting next to R. So, the order R, K is followed because R is on the extreme left. T is not next to P or K.

So, the arrangement will be R, K, P, S, T.

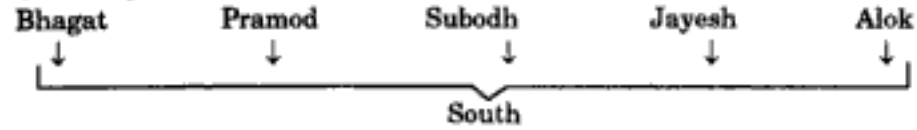
Clearly, P and T are sitting adjacent to S.

5. (*a*) : Clearly, the order is : Anuradha, Rashi, Monika, Sulekha, Abha. Anuradha is adjacent to Rashi.

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Questions 6 to 8

Note : The boys are standing facing south. So, consider 'left' and 'right' accordingly.
Jayesh is to the right of Alok *i.e.* J, A.
Pramod is between Bhagat and Subodh *i.e.* B, P, S.
Subodh is between Jayesh and Pramod.
So, the sequence is :



6. (*a*) : Alok is at the extreme left end.
7. (*d*) : Subodh is in the middle.
8. (*b*) : Statement (A) is superfluous.
9. (*d*) : D is on the left of C *i.e.* C, D.
B is on the right of E *i.e.* B, E.
A is on the right of C *i.e.* A, C.
B is on the left of D *i.e.* D, B.
From the above statements, the correct order is : A, C, D, B, E.
Clearly, D is sitting in the centre.
Note : It is given that A, B, C, D, E are sitting facing you. So, your right and left will be considered as left and right respectively.

Questions 10 to 14

C is to the right of D.
D is third from south. So, B will be at the extreme end from north because it should have E as its neighbour. G is between E and F. So, the sequence is :

B →
E →
G →
F → East
D →
C →
A →

10. (*e*) : G is sitting to the right of E.
11. (*a*) : A and B are sitting at the extreme ends.
12. (*c*) : G should change place with C to make it third from north.
13. (*d*) : D is sitting between C and F.
14. (*d*) : All the statements are required to determine the correct sequence.
15. (*a*) : Clearly, the correct sequence is :
France, America, India, Australia, Japan, China.
The two flags in the centre are of India and Australia.
16. (*b*) : Clearly, C and D sit opposite to each other. So, if A sits opposite to cook, B shall be the cook.
Now, B is to the right of barber. So, one of the rest, say C will be barber. Clearly, then D on the opposite side shall be washerman or tailor. But, washerman is left of tailor and D is to the left of A. So, D is washerman and A is tailor. Thus, A and B are Tailor and Cook.
17. (*c*) : Mr. X is on the right of Mr. Y and on the left of person in the centre *i.e.* Y, X, —
Mr. P is on the right of Mr. Z and Mr. R is on the right of Mr. P *i.e.* Z, P, R.
R is second from the person sitting in the centre.
So, the order is Y, X, Z, P, R. Clearly, Z is sitting in the centre.

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Questions 18 to 22

D is to the left of F and second to the right of C *i.e.* C — D F.

A is second to the right of E *i.e.* E — A.

J is the immediate neighbour of A and B and third to the left of G *i.e.* A J B — G.

H is to the left of D and third to the right of I *i.e.* I — H D.

The above four orders may be combined to obtain the correct order as under :

E, K, A, J, B, I, G, C, H, D, F

18. (*d*) : I is in the middle.

19. (*b*) : C, H, D, F are to the right of G.

20. (*c*) : B is sitting between J and I.

21. (*e*) : All the statements are needed to determine the correct sequence.

22. (*c*) : The new sequence formed on interchanging seats is :

D, F, H, J, C, I, G, B, A, E, K

D and K are sitting at the ends in the new arrangement.

Questions 23 to 27

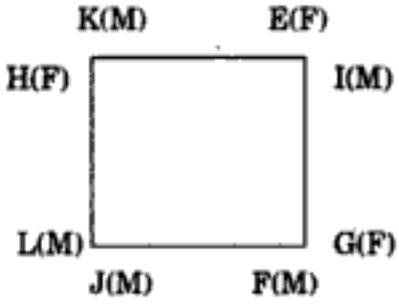
J is between L and F *i.e.* the order is L J F.

G is between I and F *i.e.* the order is F G I.

Thus, the sequence becomes L J F G I.

H is second to the left of J *i.e.* H L J F G I.

Writing the above sequence in form of a square table, we have :



M = Male,
F = Female

E is seated opposite F. Since G is between F and I, so G is a female member.

23. (*e*) : K is seated between E and H.

24. (*c*) : Three persons — H, L and J are seated between K and F.

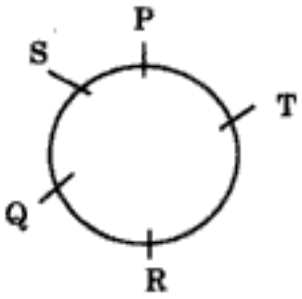
25. (*b*) : The three lady members are E, H and G.

26. (*c*) : J is to the immediate left of F.

27. (*a*) : Clearly, J is a male member.

Questions 28-29

Clearly, in the circle the arrangement is as shown :

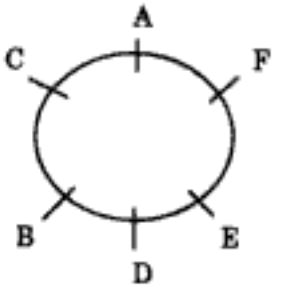


28. (*b*) : Q is to the immediate left of R.

29. (*a*) : All the statements are necessary.

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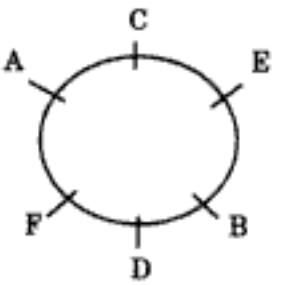
30. (*b*) : Clearly, in a circle the arrangement is as shown.



Thus, C is to the immediate left of B.

Questions 31-32

Clearly, the circular arrangement is as shown :



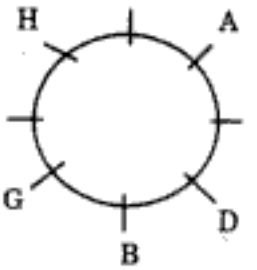
31. (*a*) : Clearly, B is facing A.

32. (*b*) : C is facing D.

Questions 33-34

B is between G and D *i.e.* the order is G B D. H is third to the left of B and second to the right of A.

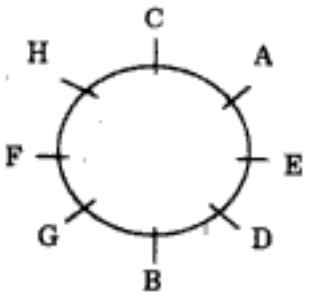
So, forming a circle we have :



C is between A and G. But E is not opposite B.

So, C is between A and H.

Thus, the final arrangement becomes :

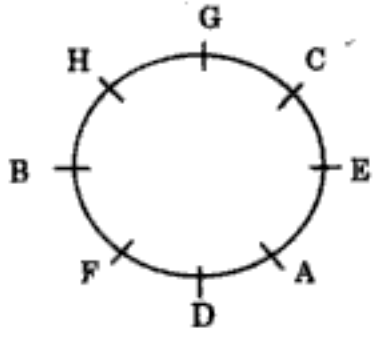


33. (*c*) : F is third to the left of D.

34. (*d*) : Clearly, E is not sitting between F and D.

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35. (*d*) : Clearly, in a circle the arrangement is as shown :



H is diagonally opposite to A.

Questions 36 to 38

Sociology is on top of all books. Education is below Sociology. Accountancy is below Education.

Economics is above Psychology and Psychology is above Hindi.

Economics is not in the middle. So, in the middle lies the seventh book *i.e.* English.

Thus, the sequence from top to bottom is :

Sociology, Education, Accountancy, English, Economics, Psychology, Hindi.

36. (*c*) : Economics is between English and Psychology.

37. (*a*) : There are three books between Accountancy and Hindi — English, Economics, Psychology.

38. (*c*) : The new sequence formed on interchanging places is English, Psychology, Hindi, Sociology, Economics, Education, Accountancy.

Clearly, Hindi is between Psychology and Sociology.

39. (*d*) : Jam bottles are at the top.

Biscuits are below chips, chocolates are below biscuits, peppermints are below chocolates and cakes are at the bottom.

So, the sequence from top to bottom is :

Jam bottles, Chips, Biscuits, Chocolates, Peppermint, Cakes.

40. (*b*) : Clearly, the sequence is : novel, drama, story-book, comic *i.e.* ndsc.

Questions 41-42

Clearly, the sequence of the five books mentioned, from top to bottom is :

——, History, Civics, Geography, English, Economics, ——

41. (*c*) : Clearly, (*c*) gives us the clue that Science book is placed at the bottom. Thus, we know that there are three books between Civics and Science.

42. (*e*) : Clearly, History, Civics and Geography are three books kept above English. To find this, no information other than the given ones is required.

43. (*b*) : Starting from above, English is between History and Mathematics *i.e.*, H, E, M.

History is between Mathematics and English *i.e.*, H, E, M, H, E.

Hindi is between English and Mathematics *i.e.*, H, E, M, H, E, Hi, M.

Mathematics is between two Hindi books *i.e.*, H, E, M, H, E, Hi, M, Hi.

Two Hindi books are between Mathematics and History *i.e.*, H, E, M, H, E, Hi, M, Hi, Hi, H. Clearly, Hindi book is at sixth position from top.

Questions 44 to 48

We analyse the given information as follows :

Arranging the cars from left to right as per the information, we have :

Fargo and Cardilac

Fiat, Car, Car, Car, Fargo

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Maruti between Ambassador and Bedford.

Fiat, Car, Car, Ambassador.

Knowing that Fiat lies at one of the ends, we have from left to right :

1	2	3	4	5	6	7
Fiat	Bedford	Maruti	Ambassador	Fargo	Cardilac	Mercedes

44. (*d*) : Clearly, Maruti is in the third place and Mercedes in the seventh *i.e.* Mercedes is fourth to the right of Maruti.

45. (*b*) : Clearly, the Cardilac on the sixth place, is to the immediate left of the Mercedes, on the seventh place.

46. (*e*) : On either side of the Cardilac are the Fargo and the Mercedes.

47. (*a*) : Clearly, Maruti in the third place, is to the immediate left of Ambassador, in the fourth place.

48. (*c*) : To the right of the Ambassador are the Fargo, Cardilac and Mercedes.

Questions 49 to 53

Clearly from statements (1) and (2), figure 1 follows; from statement (3), figure 2 follows; from statement (4), figure 3 follows; and from statement (5), figure 4 follows; and from statements (6) and (7), figure 5 follows.

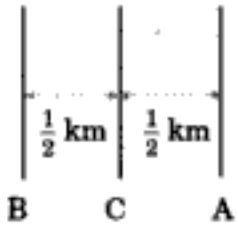


Fig. 1

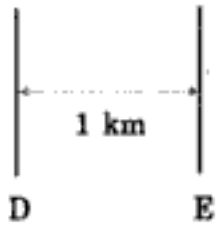


Fig. 2

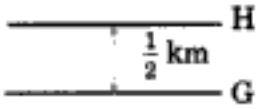


Fig. 3

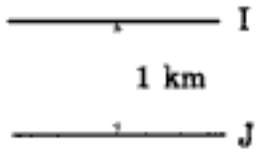


Fig. 4

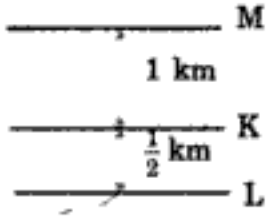
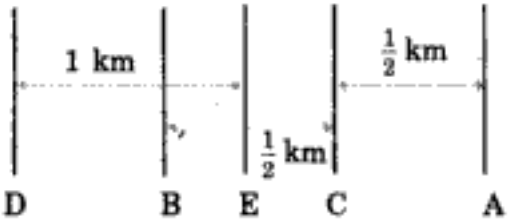


Fig. 5

49. (*d*) : It follows from figure 5 that distance of M from L = LK + KM = $\frac{1}{2}$ + 1 = 1.5 km and M is to the north of L.

50. (*b*) : If E is between B and C, we will have the following figure :



Thus, the statement that D is 2 km west of B is false.

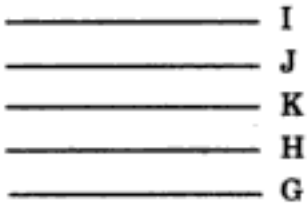
51. (*d*) : From the figure in Q. 50, the distance between A and D

$= AB + ED - BE = \left(1 + 1 - \frac{1}{4}\right) = 2 - .25$ *i.e.* between 1.5 to 2 km.

52. (*e*) : Clearly seeing the figure in Q. 50, we find that if E and B are 1/2 km apart, there is a possibility that E coincides with C.

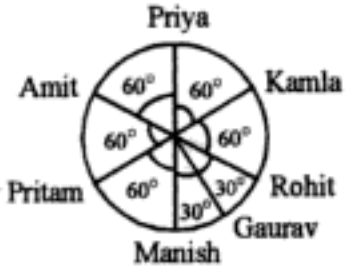
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53. (*e*) : Clearly as per the information combined with figures 3 and 4, the figure is as shown. The two roads J and K, K and H and H and G are 1/2 km apart. So, the answer is K and J.



Questions 54 to 58

Clearly in a circle the arrangement is as shown :



54. (*c*) : Gaurav is sitting between Rohit and Manish.

55. (*d*) : Gaurav is not at equal distances from Rohit and Pritam or Amit and Kamla or Manish and Pritam.

56. (*d*) : Gaurav is three places left or four places right of Priya.

57. (*d*) : The angle between Gaurav and Manish in clockwise direction is 30°.

58. (*d*) : Clearly, all the statements follow from the diagram above.

TYPE 3 : COMPARISON TYPE QUESTIONS

In such type of questions, clues are given regarding comparisons among a set of persons or things with respect to one or more qualities. The candidate is required to analyse the whole information, form a proper ascending/descending sequence and then answer the given questions accordingly.

Ex. 1. ***Read the following information and answer the questions given below it :***

There are five friends — Sachin, Kunal, Mohit, Anuj and Rohan.
Sachin is shorter than Kunal but taller than Rohan.
Mohit is the tallest.
Anuj is a little shorter than Kunal and little taller than Sachin.

1. Who is the shortest ?
(*a*) Rohan (*b*) Sachin (*c*) Anuj (*d*) Kunal (*e*) None of these

2. If they stand in the order of their heights, who will be in the middle ?
(*a*) Kunal (*b*) Rohan (*c*) Sachin (*d*) Anuj (*e*) None of these

3. If they stand in the order of increasing heights, who will be the second ?
(*a*) Anuj (*b*) Sachin (*c*) Rohan (*d*) Kunal (*e*) None of these

4. Who is the second tallest ?
(*a*) Sachin (*b*) Kunal (*c*) Anuj (*d*) Rohan (*e*) None of these

5. Who is taller than Anuj but shorter than Mohit ?
(*a*) Kunal (*b*) Rohan (*c*) Sachin
(*d*) Data inadequate (*e*) None of these

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Solution : Let us denote the five boys by the first letter of their names, namely S, K, M, A and R.

Then, $R < S < K < M$ and $S < A < K$.

$\therefore$ $R < S < A < K < M$

1. (*a*) : Rohan is shortest.

2. (*d*) : Anuj is in the middle.

3. (*b*) : In the order of increasing heights *i.e.* shortest to tallest, Sachin is second.

4. (*b*) : Kunal is second tallest.

5. (*a*) : Kunal is taller than Anuj but shorter than Mohit.

Ex. 2. Read the information given below and answer the questions that follow :

(*i*) There is a group of five girls.

(*ii*) Kamini is second in height but younger than Rupa.

(*iii*) Pooja is taller than Monika but younger in age.

(*iv*) Rupa and Monika are of the same age but Rupa is tallest between them.

(*v*) Neelam is taller than Pooja and elder to Rupa. **(Bank P.O. 1996)**

1. If they are arranged in the ascending order of height, who will be in third position ?

(*a*) Monika (*b*) Rupa (*c*) Monika or Rupa

(*d*) Data inadequate (*e*) None of these

2. If they are arranged in the descending order of their ages, who will be in fourth position ?

(*a*) Monika or Rupa (*b*) Kamini (*c*) Monika

(*d*) Data inadequate (*e*) None of these

3. To answer the question "who is the youngest person in the group", which of the given statements is superfluous ?

(*a*) Only (*i*) (*b*) Only (*ii*) (*c*) Only (*v*)

(*d*) Either (*i*) or (*iv*) (*e*) None of these

Solution : We first find the sequence of heights :

By (*iii*), we have : $M < P$.

By (*v*), we have : $P < N$.

Now, Rupa is tallest and Kamini is second in height.

So, the sequence of heights is : $M < P < N < K < R$.

Now, we determine the age sequence :

By (*ii*), we have : $K < R$.

By (*iii*), we have : $P < M$.

By (*iv*), we have : $R = M$.

By (*v*), we have : $R < N$.

So, the sequence of ages is : $N < R = M < K < P$ or $N < R = M < P < K$.

1. Clearly, in the increasing order of height, Neelam is in third position.

Hence, the answer is (*e*).

2. Clearly, in the descending order of ages, Neelam will be in fourth position (because Monika and Rupa both lie at third position).

Hence, the answer is (*e*).

3. Clearly, only statement (*i*) is not necessary.

Hence, the answer is (*a*).

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EXERCISE 6C

1. Compare the knowledge of persons X, Y, Z, A, B and C in relation to each other :
1. X knows more than A.
2. Y knows as much as B.
3. Z knows less than C.
4. A knows more than Y.

The best knowledgeable person amongst all is : **(S.C.R.A. 1996)**

(*a*) X (*b*) Y (*c*) A (*d*) C

2. Five children were administered psychological tests to know their intellectual levels. In the report, psychologists pointed out that the child A is less intelligent than the child B. The child C is less intelligent than the child D. The child B is less intelligent than the child C and child A is more intelligent than the child E. Which child is the most intelligent ? **(Bank P.O. 1996)**

(*a*) A (*b*) B (*c*) D (*d*) E (*e*) None of these

3. Among five boys, Vineet is taller than Manick, but not as tall as Ravi. Jacob is taller than Dilip but shorter than Manick. Who is the tallest in their group ?

(*a*) Ravi (*b*) Manick (*c*) Vineet
(*d*) Cannot be determined (*e*) None of these **(NABARD, 1994)**

4. If (*i*) P is taller than Q; (*ii*) R is shorter than P; (*iii*) S is taller than T but shorter than Q, then who among them is the tallest ? **(B.S.R.B. 1995)**

(*a*) P (*b*) Q (*c*) S (*d*) T (*e*) Can't be determined

5. Five boys participated in a competition. Rohit was ranked lower than Sanjay. Vikas was ranked higher than Dinesh. Kamal's rank was between Rohit and Vikas. Who was ranked highest ?

(*a*) Sanjay (*b*) Vikas (*c*) Dinesh (*d*) Kamal (*e*) None of these

6. In an examination, Raj got more marks than Mukesh but not as many as Priya. Priya got more marks than Gaurav and Kavita. Gaurav got less marks than Mukesh but his marks are not the lowest in the group. Who is second in the descending order of marks ? **(Bank P.O. 1997)**

(*a*) Priya (*b*) Kavita (*c*) Raj
(*d*) Cannot be determined (*e*) None of these

7. Ashish is heavier than Govind. Mohit is lighter than Jack. Pawan is heavier than Jack but lighter than Govind. Who among them is the heaviest ? **(Bank P.O. 1995)**

(*a*) Govind (*b*) Jack (*c*) Pawan (*d*) Ashish (*e*) Mohit

8. Pune is bigger than Jhansi, Sitapur is bigger than Chittor. Raigarh is not as big as Jhansi, but is bigger than Sitapur. Which is the smallest ? **(Railways, 1994)**

(*a*) Pune (*b*) Jhansi (*c*) Sitapur (*d*) Chittor

9. Rohan is taller than Anand but shorter than Seema. Krishna is taller than Pushpa but shorter than Anand. Dhiraj is taller than Krishna but shorter than Seema. Who among them is the tallest ? **(B.S.R.B. 1996)**

(*a*) Rohan (*b*) Seema (*c*) Krishna
(*d*) Cannot be determined (*e*) None of these

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Directions (*Questions 10-11*) : *Read the following information carefully and answer the questions given below it :* **(Bank P.O. 1994)**

(A) Gopal is shorter than Ashok but taller than Kunal;
(B) Navin is shorter than Kunal;
(C) Jayesh is taller than Navin;
(D) Ashok is taller than Jayesh.

10. Who among them is the tallest ?
(*a*) Gopal (*b*) Ashok (*c*) Kunal (*d*) Navin (*e*) Jayesh

11. Which of the given informations is not necessary to answer the above question ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) None of these

12. B is twice as old as A but twice younger than F.
C is half the age of A but twice the age of D.
Which two persons form the pair of oldest and youngest ?
(*a*) F and A (*b*) F and D (*c*) B and F
(*d*) F and C (*e*) None of these

13. Sudhanshu is as much older than Kokila as he is younger than Praveen. Nitin is as old as Kokila. Which of the following statements is wrong ? **(Bank P.O. 1995)**
(*a*) Kokila is younger than Praveen. (*b*) Nitin is younger than Praveen.
(*c*) Sudhanshu is older than Nitin. (*d*) Praveen is not the oldest.
(*e*) Kokila is younger than Sudhanshu.

14. A is elder to B while C and D are elder to E who lies between A and C. If C be elder to B, which one of the following statements is necessarily true ?
(*a*) A is elder to C (*b*) C is elder to D
(*c*) D is elder to C (*d*) E is elder to B
(Hotel Management, 1995)

15. Hitesh is richer than Jaya whereas Mohan is richer than Pritam. Lalit is as rich as Jaya. Amit is richer than Hitesh. What conclusion can be definitely drawn from the above statements ? **(Hotel Management, 1996; Asstt. Grade, 1995)**
(*a*) Jaya is poorer than Pritam. (*b*) Mohan is richer than Amit.
(*c*) Lalit is poorer than Hitesh. (*d*) Pritam is richer than Lalit.

Directions (*Questions 16 to 21*) : *Study the following information and answer the questions given below it :*

A blacksmith has five iron articles A, B, C, D and E, each having a different weight.

(*i*) A weighs twice as much as B.
(*ii*) B weighs four and a half times as much as C.
(*iii*) C weighs half as much as D.
(*iv*) D weighs half as much as E.
(*v*) E weighs less than A but more than C.

16. Which of the following is the lightest in weight ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

17. E is lighter in weight than which of the other two articles ?
(*a*) A, B (*b*) D, C (*c*) A, C (*d*) D, B (*e*) B, E

18. E is heavier than which of the following two articles ?
(*a*) D, B (*b*) D, C (*c*) A, C (*d*) A, B (*e*) None of these

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19. Which of the following articles is the heaviest in weight ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

20. Which of the following represents the descending order of weights of the articles ?
(*a*) A, B E, D, C (*b*) B, D, E, A, C (*c*) E, C, D, A, B
(*d*) C, A, D, B, E (*e*) A, B, D, E, C

21. Which of the above given statements is not necessary to determine the correct order of articles according to their weights ?
(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4 (*e*) 5

Directions (*Questions 22 to 26*) : *Read the following information and answer the questions given below it :*

(*i*) Seven students P, Q, R, S, T, U and V take a series of tests.
(*ii*) No two students get similar marks.
(*iii*) V always scores more than P.
(*iv*) P always scores more than Q.
(*v*) Each time either R scores the highest and T gets the least, or alternatively S scores the highest and U or Q scores the least.

22. If S is ranked sixth and Q is ranked fifth, which of the following can be true ?
(*a*) V is ranked first or fourth. (*b*) R is ranked second or third.
(*c*) P is ranked second or fifth. (*d*) U is ranked third or fourth.
(*e*) T is ranked fourth or fifth.

23. If R gets most, V should be ranked not lower than :
(*a*) second (*b*) third (*c*) fourth (*d*) fifth (*e*) sixth

24. If R is ranked second and Q is ranked fifth, which of the following must be true ?
(*a*) S is ranked third. (*b*) T is ranked sixth. (*c*) P is ranked sixth.
(*d*) V is ranked fourth. (*e*) U is ranked sixth.

25. If S is ranked second, which of the following can be true ?
(*a*) U gets more than V. (*b*) V gets more than S. (*c*) P gets more than R.
(*d*) P gets more than V. (*e*) T gets more than Q.

26. If V is ranked fifth, which of the following must be true ?
(*a*) S scores the highest. (*b*) R is ranked second. (*c*) T is ranked third.
(*d*) Q is ranked fourth. (*e*) U scores the least.

Directions (*Questions 27 to 31*) : *Study the information given below and answer the questions that follow :* **(Bank P.O. 1995)**

(*i*) A, B, C, D, E and F are six students in a class.
(*ii*) B and C are shorter than F but heavier than A.
(*iii*) D is heavier than B and taller than C.
(*iv*) E is shorter than D but taller than F.
(*v*) F is heavier than D.
(*vi*) A is shorter than E but taller than F.

27. Who among them is the tallest ?
(*a*) A (*b*) B (*c*) D (*d*) E (*e*) None of these

28. Who is third from the top when they are arranged in descending order of height ?
(*a*) A (*b*) B (*c*) C (*d*) E (*e*) None of these

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29. Which of the following groups of friends is shorter than A ?
(*a*) B, C only (*b*) D, B, C only (*c*) E, B, C only
(*d*) F, B, C, only (*e*) None of these

30. Who among them is the lightest ?
(*a*) A (*b*) B (*c*) C (*d*) B or C (*e*) Data inadequate

31. Which of the following statements is true for F as regards height and weight ?
(*a*) He is lighter than E and taller than E.
(*b*) He is heavier than B and taller than E.
(*c*) He is heavier than B and C but shorter than D.
(*d*) He is lighter than E and also shorter than E.
(*e*) He is lighter than B and C but taller than D.

Directions (*Questions 32 to 36*) : *Read the following information carefully and answer the questions given below it :*

(*i*) A, B, C, D and E are five friends.
(*ii*) B is elder to E, but not as tall as C.
(*iii*) C is younger to A, and is taller to D and E.
(*iv*) A is taller to D, but younger to E.
(*v*) D is elder to A but is shortest in the group.

32. Who among the following is the eldest ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) None of these

33. Which of the following pairs of students is elder to D ?
(*a*) BA (*b*) BC (*c*) BE (*d*) EA (*e*) None of these

34. Which of the following statements is correct about B ?
(*i*) B is not the tallest.
(*ii*) B is shorter to E.
(*iii*) When they are asked to stand in ascending order with respect to their heights, B is in the middle.
(*a*) Only (*i*) is correct (*b*) Only (*i*) and (*ii*) are correct (*c*) All are correct
(*d*) All are incorrect (*e*) None of these

35. If F, another friend, is taller than C, how many of them will be between F and E according to their height ?
(*a*) None (*b*) One (*c*) Two (*d*) Three (*e*) None of these

36. If a selection is to be made among them who would be relatively older and also taller, who among them should be chosen ?
(*a*) A (*b*) B (*c*) C (*d*) D (*e*) E

Directions (*Questions 37 to 41*) : *Study the given information carefully and answer the questions that follow :* **(Bank P.O. 1994)**

(*i*) Six friends P, Q, R, S, T and U are members of a club and play a different game of Football, Cricket, Tennis, Basketball, Badminton and Volleyball.
(*ii*) T who is taller than P and S plays Tennis.
(*iii*) The tallest among them plays Basketball.
(*iv*) The shortest among them plays Volleyball.
(*v*) Q and S neither play Volleyball nor Basketball.
(*vi*) R plays Volleyball.
(*vii*) T is between Q who plays Football and P in order of height.

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37. Who among them is taller than R but shorter than P ?
(*a*) Q (*b*) T (*c*) U (*d*) Data inadequate (*e*) None of these

38. Who will be at the third place if they are arranged in the descending order of their height ?
(*a*) Q (*b*) P (*c*) S (*d*) T (*e*) None of these

39. Which of the following statements is not true ?
(*a*) P is shorter than R. (*b*) Q is taller than S. (*c*) S is taller than R.
(*d*) T is taller than R. (*e*) U is taller than Q.

40. Who among them plays Basketball ?
(*a*) Q (*b*) R (*c*) S (*d*) U (*e*) None of these

41. What does S play ?
(*a*) Cricket (*b*) Badminton (*c*) Football
(*d*) Either Cricket or Badminton (*e*) None of these

Directions* (*Questions 42-43*) : *Read the following information carefully and answer the questions given below it : **(Bank P.O. 1997)**

At the end of a cricket series, when five players were arranged in the ascending order of runs scored by them, O was fourth while N was first. When they were arranged in descending order for wickets taken by them, K replaces O while O replaces L. M's position remains unchanged. K has scored more runs than M. L is having first rank in one ranking and fifth in another.

42. Who has scored the highest runs in the series ?
(*a*) K (*b*) L (*c*) M (*d*) Can't be determined (*e*) None of these

43. Who has taken the lowest number of wickets ?
(*a*) L (*b*) M (*c*) P (*d*) Can't be determined (*e*) None of these

Directions (*Questions 44-47*) : *Read the information given below and answer the questions that follow :* **(M.B.A. 1997)**

A * B means A and B are of the same age;
A – B means B is younger than A;
A + B means A is younger than B.

44. Sachin * Madan – Reena means
(*a*) Reena is the youngest. (*b*) Reena is the oldest.
(*c*) Madan is younger than Reena. (*d*) None of these

45. X + Y + Z is same as
(*a*) Y – X – Z (*b*) Z – Y – X (*c*) Z – X – Y (*d*) None of these

46. For an expression Farha – Farida – Arif, which of the following cannot be correct under any circumstances ?
(*a*) Arif is the father of Farha.
(*b*) Arif is the younger brother of Farha.
(*c*) Farha is the mother of both Arif and Farida.
(*d*) None of these

47. Deven – Shashi * Hemant is opposite to
(*i*) Hemant + Shashi + Deven (*ii*) Hemant – Shashi + Deven
(*iii*) Shashi * Hemant + Deven
(*a*) (*i*) only (*b*) (*i*) and (*ii*) only (*c*) (*ii*) and (*iii*) only (*d*) None of these

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ANSWERS

1. (*a*) : Clearly, we have : $A < X$, $Y = B$, $Z < C$, $Z < B$, $Y < A$.
Thus, the sequence becomes :
$X > A > Y = B > C > Z$.
So, X is the best knowledgeable person.

2. (*c*) : We have : $A < B$, $C < D$, $B < C$ and $E < A$.
So, the sequence becomes : $E < A < B < C < D$.
Clearly, child D is the most intelligent.

3. (*a*) : In terms of height, we have :
Manick < Vineet, Vineet < Ravi, Dilip < Jacob, Jacob < Manick.
So, the sequence becomes :
Dilip < Jacob < Manick < Vineet < Ravi.
Clearly, Ravi is the tallest.

4. (*a*) : In terms of height, we have :
$Q < P$, $R < P$, $T < S$, $S < Q$.
So, the sequence becomes : $T < S < Q < R < P$ or $T < S < R < Q < P$.
Whichever may be the case, P is the tallest.

5. (*a*) : In terms of rank, we have :
Rohit < Sanjay, Dinesh < Vikas.
Since Kamal's rank is between Rohit and Vikas, the sequence becomes :
Dinesh < Vikas < Kamal < Rohit < Sanjay.
Clearly, Sanjay was ranked highest.

6. (*c*) : In terms of marks obtained,
Mukesh < Raj, Raj < Priya, Gaurav < Priya, Kavita < Priya, Gaurav < Mukesh.
Since Gaurav's marks are not the lowest, so Kavita's marks are the lowest.
So, the sequence becomes :
Kavita < Gaurav < Mukesh < Raj < Priya.
Clearly, in the descending order, Raj comes second.

7. (*d*) : In terms of weight, we have :
Govind < Ashish, Mohit < Jack, Jack < Pawan, Pawan < Govind.
So, the sequence becomes :
Mohit < Jack < Pawan < Govind < Ashish.
Clearly, Ashish is the heaviest.

8. (*d*) : In terms of size, we have :
Jhansi < Pune, Chittor < Sitapur, Raigarh < Jhansi, Sitapur < Raigarh.
So, the sequence becomes :
Chittor < Sitapur < Raigarh < Jhansi < Pune.
Clearly, Chittor is the smallest.

9. (*b*) : In terms of height, we have :
Anand < Rohan, Rohan < Seema, Pushpa < Krishna, Krishna < Anand.
Krishna < Dhiraj, Dhiraj < Seema.
So, the sequence becomes :
Pushpa < Krishna < Dhiraj < Anand < Rohan < Seema.
Clearly, Seema is the tallest.

10. (*b*) : In terms of height, we have :
Gopal < Ashok, Kunal < Gopal, Navin < Kunal, Navin < Jayesh, Jayesh < Ashok.

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So, the sequence becomes :
Navin < Kunal < Gopal < Jayesh < Ashok.
Clearly, Ashok is the tallest.

11. (*c*) : Clearly, statement C is not necessary.

12. (*b*) : Let A's age be x. Then, B's age is $2x$. B is twice younger than F *i.e.* F is twice older than B. So, F's age is $4x$. C is half the age of A *i.e.* C's age is $\frac{x}{2}$. C is twice the age of D *i.e.* D is half the age of C *i.e.* D's age is $\frac{x}{4}$. So, the descending order of ages is F, B, A, C, D.
Clearly, F is the oldest and D is the youngest.

13. (*d*) : In terms of age, we have :
Kokila < Sudhanshu, Sudhanshu < Praveen, Nitin = Kokila.
So, the sequence becomes :
Nitin = Kokila < Sudhanshu < Praveen.
Clearly, Praveen is the oldest.
Hence, (*d*) is the incorrect statement.

14. (*d*) : In terms of age, we have :
B < A, E < C, E < D, B < C.
Since E lies between A and C, the sequence becomes :
B < A < E < C < D or B < A < E < D < C
Clearly, whichever may be the case, E is elder to B. Hence, (*d*) is necessarily true.

15. (*c*) : In terms of richness, we have :
Jaya < Hitesh, Pritam < Mohan, Lalit = Jaya, Hitesh < Amit.
So, we have : Lalit = Jaya < Hitesh < Amit and Pritam < Mohan.
Clearly, Lalit is poorer than Hitesh.

Questions 16 to 21

Let C's weight be x. Then, D's weight is $2x$. E's weight is $4x$, B's weight is $4.5x$ and A's weight is $9x$.
So, the order of weights can be A > B > E > D > C.

16. (*c*) : C is the lightest in weight.

17. (*a*) : E is lighter in weight than A and B.

18. (*b*) : E is heavier than D and C.

19. (*a*) : Clearly, A is the heaviest in weight.

20. (*a*) : Clearly, the descending order of weights is A, B, E, D, C.

21. (*e*) : Clearly, statement (5) is not required to determine the order of weights.

Questions 22 to 26

In terms of scores, we have :
V > P, P > Q *i.e.* V > P > Q.
If R scores the highest, we have R > ..... > T.
If S scores the highest, we have S > ..... > Q or S > ..... > U.

22. (*d*) : If S is ranked sixth and Q is ranked fifth, we have :
□ > □ > □ > □ > Q > S > □ .
In this case, R will rank the highest and thus T will rank the least. We have :
R > □ > □ > □ > Q > S > T.
Also, the order V > P > Q will be maintained *i.e.*, V and P will have second, third or fourth places. So, statements (*a*), (*b*), (*c*) and (*e*) cannot follow. Thus, (*d*) is the answer.

23. (*c*) : Again, if R ranks most, T ranks lowest and occupies seventh place. Since V always ranks above P and Q, so in the maximum, P and Q will occupy fifth and sixth places. Thus, V will not rank lower than fourth.

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24. (*b*) : If R is ranked second, S will rank first and Q and U lowest. But Q ranks fifth. So, U ranks lowest. Also, the order V > P > Q will be followed.

So, the arrangement will be S > R > V > P > Q > □ > U. Thus, the sixth place will be occupied by T.

25. (*a*) : If S ranks second, R ranks first and T ranks lowest. The order V > P > Q will be followed. So, the arrangement will be R > S > □ > □ > □ > □ > T.

Clearly, statements (*b*), (*c*), (*d*) and (*e*) cannot follow. So, the answer is (*a*).

26. (*a*) : If V ranks fifth, P and Q coming before it will occupy sixth and seventh places respectively *i.e.* Q ranks least. So, S will score the highest.

Questions 27 to 31

In terms of height, we have :

B < F, C < F, C < D, E < D, F < E, A < E, F < A.

So, C < F < E < D, B < F, F < A < E

Thus, the sequence becomes :

B < C < F < A < E < D or C < B < F < A < E < D.

In terms of weight, we have :

A < B, A < C, B < D, D < F.

So, A < B < D < F, A < C.

Thus, the sequence becomes :

A < C < B < D < F or A < B < C < D < F or A < B < D < C < F.

27. (*c*) : Clearly, D is the tallest.

28. (*a*) : The descending order of height is :

D > E > A > F > B > C or D > E > A > F > C > B.

Clearly, A is third from the top.

29. (*d*) : Clearly, F, B and C are shorter than A.

30. (*e*) : Data is inadequate as no clue regarding E's weight is given.

31. (*c*) : Clearly F is heavier than B and C but shorter than D.

Questions 32 to 36

In terms of age, we have :

E < B, C < A, A < E, A < D

So, we have : C < A < E < B, A < D.

In terms of height, we have :

B < C, D < C, E < C, D < A.

32. (*e*) : Either B or D is the eldest.

33. (*e*) : It cannot be determined for sure.

34. (*a*) : B is shorter than C. So, B is not the tallest. Thus, (*i*) is correct.

B and E are shorter than C. So, it cannot be concluded that B is shorter to E. Thus, (*ii*) is incorrect.

A single definite order of heights cannot be obtained from the given information. So, (*iii*) is incorrect.

35. (*e*) : Since no definite order of height can be obtained, so it cannot be determined for sure how many persons lie between F and E.

36. (*b*) : A and C are youngest so they cannot be selected. D is shorter than two persons A and C.

B is shorter than C only and is also relatively older. So, B will be selected.

E is younger than B.

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Questions 37 to 41

In terms of height, we have

T > P, T > S, Q > T > P.

R plays Volleyball, so R is the shortest.

Q plays neither Volleyball nor Basketball.

So, Q is not the tallest. Thus, U is the tallest.

So, the sequence becomes : U > Q > T > P > S > R.

Now, T plays Tennis. U, being tallest, plays Basketball. R plays Volleyball. Q plays Football. Both P and S play either Cricket or Badminton.

37. (*e*) : S is taller than R but shorter than P.

38. (*d*) : The descending order of height is U, Q, T, P, S, R.

Clearly, T is at the third place.

39. (*a*) : Clearly, P is taller than R.

40. (*d*) : U plays Basketball.

41. (*d*) : S plays either Cricket or Badminton.

Questions 42-43

In terms of runs scored, we have the ascending order as N < □ < □ < O < □ .

N has the first rank. So, L will be fifth in this order *i.e.*, N < □ < □ < O < L.

K has scored more runs than M *i.e.*, K > M.

So, the sequence becomes N < M < K < O < L.

In terms of wickets taken, the order becomes : L > M > N > K > O.

42. (*b*) : L has scored the highest runs in the series.

43. (*e*) : O has taken the lowest number of wickets.

44. (*a*) : Sachin * Madan – Reena means Sachin and Madan are of the same age and Reena is younger than Madan. This means that Reena is the youngest.

45. (*b*) : X + Y + Z means X is younger than Y and Y is younger than Z. This can also be written as Z – Y – X.

46. (*a*) : Farha – Farida – Arif means Farida is younger than Farha and Arif is younger than Farida. This means that Arif is younger than Farha. So, Arif cannot be the father of Farha.

47. (*d*) : Deven – Shashi * Hemant means Shashi is younger than Deven, and Shashi and Hemant are of the same age. Thus, Deven is the oldest. Now, the opposite statement would mean : Deven is the youngest.

(*i*) Hemant + Shashi + Deven means Hemant is younger than Shashi, who is younger than Deven. So, Deven is the oldest.

(*ii*) Hemant – Shashi + Deven means Shashi is younger than both Hemant and Deven. Thus, either Hemant or Deven is the oldest, but Deven is not the youngest.

(*iii*) Shashi * Hemant + Deven means Shashi and Hemant are of the same age and Hemant is younger than Deven. So, Deven is the oldest.

TYPE 4 : SEQUENTIAL ORDER OF THINGS

In this type of questions, some clues are given regarding the order of occurrence of certain events. The candidate is required to analyse the given information, frame the right sequence and then answer the questions accordingly.

Ex. 1. Read the following information and answer the questions given below it :

Six plays — A, B, C, D, E and F are to be staged, one on each day from Monday to Saturday. The schedule of the plays is to be in accordance with the following :

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(*i*) A must be staged a day before E.
(*ii*) C must not be staged on Tuesday.
(*iii*) B must be staged on the day following the day on which F is staged.
(*iv*) D must be staged on Friday only and should not be immediately preceded by B.
(*v*) E must not be staged on the last day of the schedule.

1. Which of the following plays immediately follows B ?
(*a*) A (*b*) C (*c*) D (*d*) E (*e*) F

2. Which of the following plays is on Monday ?
(*a*) E (*b*) F (*c*) C (*d*) B (*e*) A

3. Play D is between which of the following pairs of plays ?
(*a*) B and E (*b*) E and F (*c*) A and E (*d*) C and E (*e*) C and F

4. Which of the following is the schedule of plays with the order of their staging from Monday ?
(*a*) E, A, B, F, D, C (*b*) A, F, B, E, D, C (*c*) A, F, B, C, D, E
(*d*) F, A, B, E, D, C (*e*) None of these

5. Play C cannot definitely be staged on which of the following days in addition to Tuesday ?
(*a*) Monday (*b*) Wednesday (*c*) Friday (*d*) Thursday (*e*) Saturday

---

Solution : Clearly, D must be staged on Friday. A must be staged before E *i.e.*, order AE must be followed. But E cannot be staged on last day. Also, B must be staged immediately after F *i.e.*, order FB must be followed. But B cannot precede D. So, F and B can be staged on Monday and Tuesday and A and E on Wednesday and Thursday. C, which cannot be staged on Tuesday shall be staged on Saturday. Thus, the order followed will be :

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
F	B	A	E	D	C

1. Clearly, A immediately follows B. So, the answer is (*a*).
2. F will be played on Monday. So, the answer is (*b*).
3. Play D is between E and C. So, the answer is (*d*).
4. Clearly, order of staging of plays is F, B, A, E, D, C. So, the answer is (*e*).
5. C cannot be staged on Friday as well because D has to be staged on that day.

Ex. 2. Read the following information carefully and answer the questions given below it : **(S.B.I.P.O. 1997)**

(*i*) Eight doctors P, Q, R, S, T, U, V and W visit a charitable dispensary every day except on a holiday *i.e.* Monday.
(*ii*) Each doctor visits for one hour from Tuesday to Sunday except Saturday. The timings are 9 a.m. to 1 p.m. and 2 p.m. to 6 p.m.; 1 p.m. to 2 p.m. is lunch break.
(*iii*) On Saturday, it is open only in the morning *i.e.* 9 a.m. to 1 p.m. and each doctor visits for only half an hour.
(*iv*) No other doctor visits the dispensary before doctor Q and after doctor U.
(*v*) Doctor W comes immediately after lunch break and is followed by R.
(*vi*) S comes in the same order as P in the afternoon session.

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9. LOGICAL VENN DIAGRAMS

This section deals with questions which aim at analysing a candidate's ability to relate a certain given group of items and illustrate it diagrammatically.

Here are a few different types of Venn diagrams with their implications made clear.

Suppose you are given a group of three items. Then,

1. if the items evidently belong to three different groups, the Venn diagram representing it would be as shown alongside.

   Ex. Doctors, Engineers, Lawyers

   These three items bear no relationship to each other. So, they are represented by 3 disjoint figures as shown in Fig. 1.

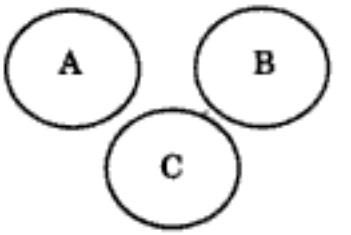


Fig. 1

2. if one item belongs to the class of the second and the second belongs to the class of third, then the representation is in the form of three concentric circles, as shown in Fig. 2.

   Ex. Seconds, Minutes, Hours

   Clearly, seconds are a part of minutes and minutes are a part of hours. So, the Venn diagram would be as shown in the adjoining figure with circle A representing *Seconds*, circle B representing *Minutes* and circle C representing *Hours*.

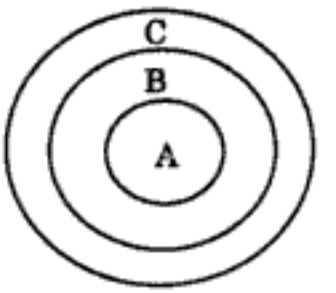


Fig. 2

3. if two separate items belong to the class of the third, they are represented by two disjoint circles inside a bigger circle as shown in Fig. 3.

   Ex. Table, Chair, Furniture

   Clearly, table and chair are separate items but both are items of furniture. So, they would be represented as in the adjoining figure with circle A representing *Table*, circle B representing *Chair* and circle C representing *Furniture*.

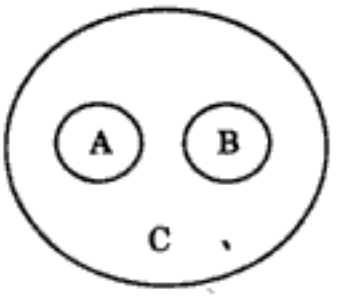


Fig. 3

4. if two items belong to the class of the third such that some items of each of these two groups are common in relationship, then they are represented by two intersecting circles enclosed within a bigger circle.

   Ex. Males, Fathers, Brothers

   Clearly, some fathers may be brothers. So, fathers and brothers would be represented by two intersecting circles. Also both fathers and brothers are males. So, the diagrammatic representation would be as shown in Fig. 4, with circle A representing *Fathers*, circle B representing *Brothers* and circle C representing *Males*.

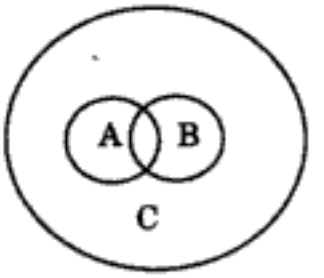


Fig. 4

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5. if two items are partly related to the third, and are themselves independent of each other they are represented by three intersecting circles in a line.

Ex. Dogs, Pets, Cats

Clearly, some dogs and some cats are pets. But, all the pets are not dogs or cats. Also, dogs and cats are not related to each other. So, the given items would be represented as shown in Fig. 5 with circle A representing *Dogs*, circle B representing *Pets* and circle C representing *Cats*.

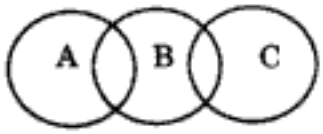


Fig. 5

6. if the three items are partly related to each other, they are represented as shown in the adjoining figure.

Ex. Clerks, Government Employees, Educated Persons

Clearly, some clerks may be government employees and some may be educated. Similarly, some government employees may be clerks and some may be educated. Also, some educated persons may be clerks and some may be government employees. So, the given items may be represented as shown in Fig. 6 with three different circles denoting the three classes.

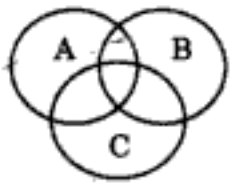


Fig. 6

7. if one item belongs to the class of second while third item is entirely different from the two, then they may be represented by the adjoining diagram.

Ex. Engineers, Human Beings, Rats

Clearly, all engineers are human beings. This would be represented by two concentric circles. But the class of rats is entirely different from these two. Thus, these items would be represented as shown in Fig. 7 with circle A representing *Engineers*, circle B representing *Human Beings* and circle C representing *Rats*.

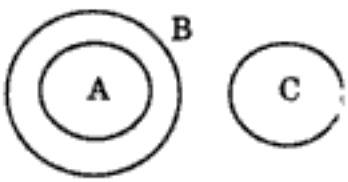


Fig. 7

8. if one item belongs to the class of second and the third item is partly related to these two, they are represented as shown alongside.

Ex. Females, Mothers, Doctors

Clearly, all mothers are females. This would be represented by two concentric circles. But, some females and some mothers can be doctors. So, the circle representing doctors would intersect the two concentric circles. Thus, the diagram becomes as shown in Fig. 8 with circle A representing *Mothers*, circle B representing *Females* and circle C representing *Doctors*.

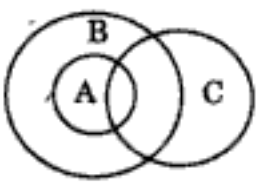


Fig. 8

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9. if one item belongs to the class of second and the third item is partly related to the second, they are represented as shown alongside.

Ex. Grass-eating Animals, Cows, Flesh-eating Animals

Clearly, cows are grass-eating animals. So, they would be represented by two concentric circles. But some grass-eating animals are flesh-eating also. Thus, the Venn diagram is as shown in Fig. 9 with circle A representing *Cows*, circle B representing *Grass-eating Animals* and circle C representing *Flesh-eating Animals*.

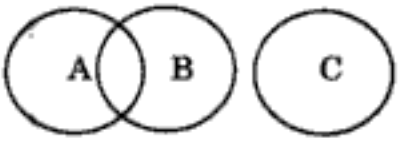


Fig. 9

EXERCISE 9A

1. Which of the following diagrams correctly represents *Elephants, Wolves, Animals* ? **(I.A.S. 1992)**



(*a*) (*b*) (*c*) (*d*)

2. Which one of the following Venn diagrams correctly illustrates the relationship among the classes : *Carrot, Food, Vegetable* ? **(I.A.S. 1996)**



(*a*) (*b*) (*c*) (*d*)

Directions (*Questions 3 to 7*) : *In the following questions, three classes are given. Out of the four figures that follow, you are to indicate which figure will best represent the relationship amongst the three classes.*



(*a*) (*b*) (*c*) (*d*)

3. Women, Mothers, Widows

4. Authors, Teachers, Men

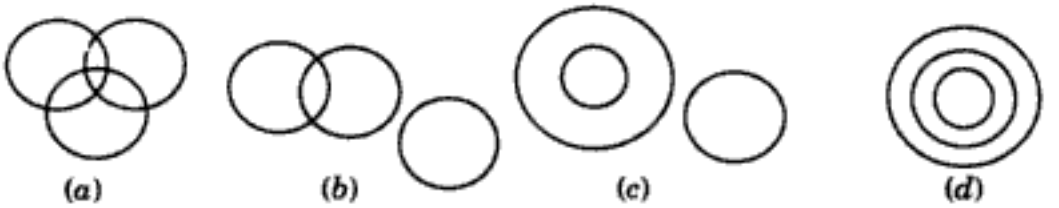
5. Sparrows, Birds, Mice

6. Tea, Coffee, Beverages

7. Boys, Students, Athletes

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8. Select from the given diagrams, the one that illustrates the relationship among the given three classes : *Judge, Thief, Criminal.* **(S.C.R.A. 1994)**



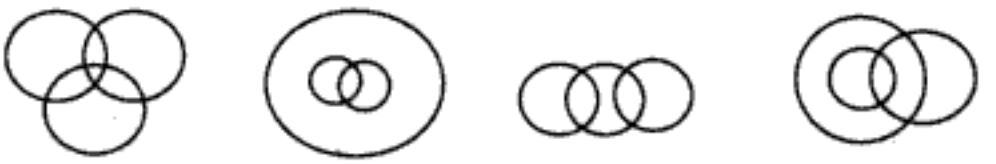
(*a*) (*b*) (*c*) (*d*)

9. Choose from the four diagrams given below, the one that illustrates the relationship among *Languages, French, German.*



(*a*) (*b*) (*c*) (*d*)

10. Which one of the following diagrams correctly represents the relationship among the classes : *Tennis fans, Cricket players, Students* ? **(I.A.S. 1990)**



(*a*) (*b*) (*c*) (*d*)

11. Which one of the following Venn diagrams best illustrates the three classes : *Rhombus, Quadrilaterals, Polygons* ?



(*a*) (*b*) (*c*) (*d*)

Directions (*Questions 12 to 15*) : *Each of these questions below contains three groups of things. You are to choose from the following five numbered diagrams, the diagram that depicts the correct relationship among the three groups of things in each question.* **(S.B.I.P.O. 1995)**



(*a*) (*b*) (*c*) (*d*) (*e*)

12. Tables, Chairs, Furniture

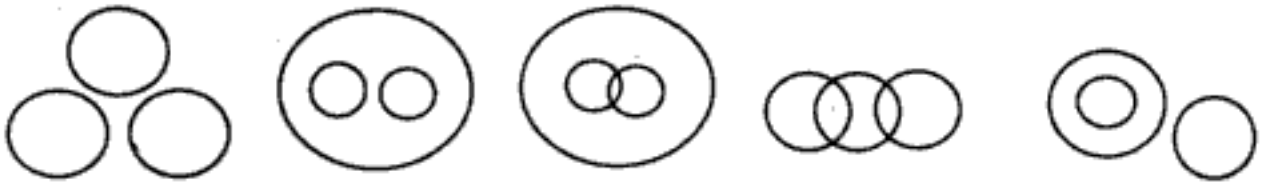
13. Tie, Shirt, Pantaloon

14. Dogs, Pets, Cats

15. Brinjal, Meat, Vegetables

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Directions (*Questions 16 to 28*) : *Each one of the following questions contains three items. Using the relationship between these items, match each question with the most suitable diagram. Your answer is the letter denoting that diagram.*



(*a*) (*b*) (*c*) (*d*) (*e*)

16. Deer, Rabbit, Mammal

17. Human beings, Teachers, Graduates **(Central Excise, 1995)**

18. Whales, Fishes, Crocodiles

19. Plums, Tomatoes, Fruits

20. Mountains, Forests, Earth

21. Tiger, Fox, Carnivores

22. Grams, Beans, Legumes

23. Flowers, Clothes, White

24. Uncles, Parents, Friends

25. Rohtak, Haryana, Punjab

26. Engineer, Doctor, People

27. Thieves, Lawyers, Criminals

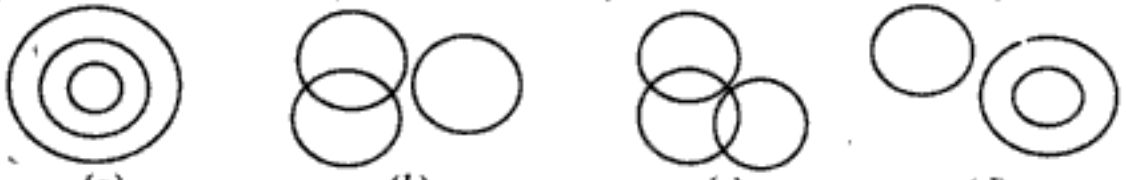
28. Sea, Island, Mountain

29. Which is the most suitable Venn diagram among the following, which represents interrelationship among **Antisocial elements, Pick pockets and Black mailers ?**
(M.B.A. 1997)



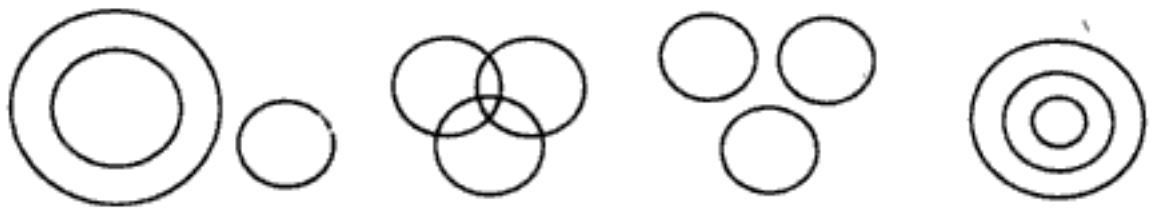
(*a*) (*b*) (*c*) (*d*)

30. Which one of the following four logical diagrams represents correctly the relationship between : *Musicians, Instrumentalists, Violinists* ? **(I.A.S. 1994)**



(*a*) (*b*) (*c*) (*d*)

31. Select from the four alternative diagrams, the one that best illustrates the relationship among the three classes : *Pigeons, Birds, Dogs.* **(Assistant Grade, 1993)**



(*a*) (*b*) (*c*) (*d*)

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Directions (*Questions 32 to 37*) : *In each of the following questions, choose the Venn diagram which best illustrates the relationship among three given items ?*



(*a*) (*b*) (*c*) (*d*) (*e*)

32. Diseases, Leprosy, Scurvy **(Hotel Management, 1991)**

33. Hockey, Cricket, Games

34. Yak, Zebra, Bear **(Hotel Management, 1991)**

35. Sun, Moon, Stars

36. Animals, Men, Plants

37. Mercury, Mars, Planets **(Hotel Management, 1991)**

38. Which of the following figures correctly represents the relation between : *Doctors, Lawyers, Professionals* ?



(*a*) (*b*) (*c*) (*d*)

39. Which one of the following sets is best represented in the adjoining diagram ? **(S.C.R.A. 1994)**



(*a*) Animals, Insects, Cockroaches

(*b*) Country, States, Districts

(*c*) Animals, Males, Females and Hermaphrodites

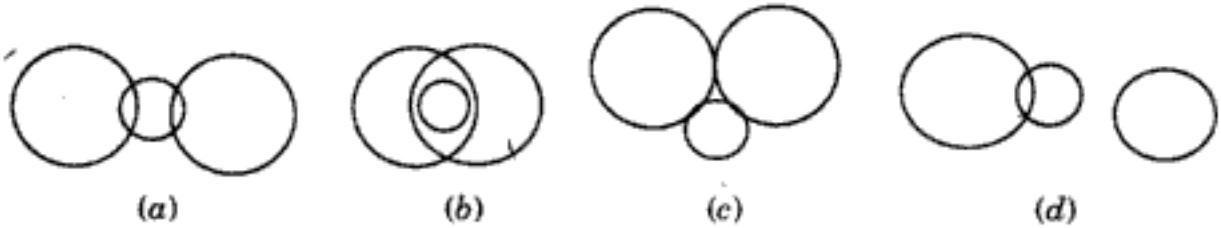
(*d*) States, Districts, Union Territory

40. Which of the following gives the proper relation of *Tall men, Black haired people, Indians* ?



(*a*) (*b*) (*c*) (*d*)

41. If animals that live on land and the animals that live in water are represented by two big circles and animals that live in water and on land are represented by small circle, the combination of these three can be best represented as



(*a*) (*b*) (*c*) (*d*)

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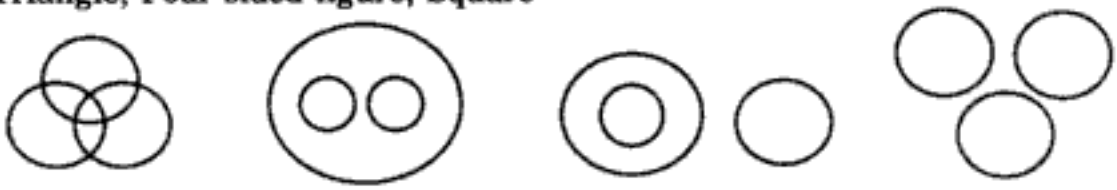
Directions (*Questions 42 to 44*) : *In each of the following questions, find out which of the alternatives* (*a*), (*b*) (*c*) *or* (*d*) *indicates the correct relationship between the three given words* ? **(I. Tax & Central Excise, 1995)**

42. Elected house, M.P., M.L.A.



(*a*) (*b*) (*c*) (*d*)

43. Triangle, Four-sided figure, Square



(*a*) (*b*) (*c*) (*d*)

44. Doctor, Nurse, Human being



(*a*) (*b*) (*c*) (*d*)

Questions 45 to 53 :

(*a*)



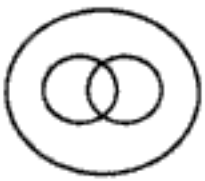
Indicates that one class is completely contained in the other but not the third.

(*b*)



Indicates that two classes are completely contained in the third.

(*c*)



Indicates that neither class is completely contained in the other but the two have common members, forming one entity.

(*d*)



Indicates that two classes are interrelated and third one is not.

Directions : *Choose the Venn diagram which best illustrates the three given classes in each question.*

45. Protons, Electrons, Atoms

46. Sun, Planets, Earth **(Railways, 1990)**

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47. Dog, Animal, Pet

48. Science, Physics, Chemistry

49. Atmosphere, Hydrogen, Oxygen **(Railways, 1990)**

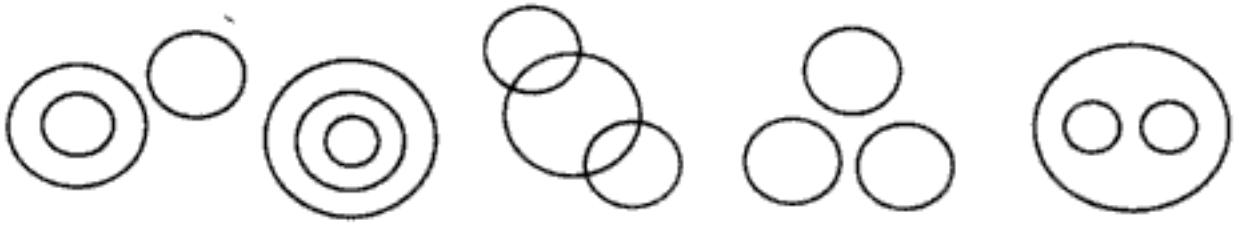
50. Wheat, Grains, Maize

51. Machine, Lathe, Mathematics **(C.B.I., 1990)**

52. Biology, Botany, Zoology **(Railways, 1990)**

53. Citizens, Educated, Men

Directions (*Questions 54 to 58*) : *Each of the questions below contains three elements. These elements may or may not have some inter se linkage. Each group of the elements may fit into one of the diagrams at* (*a*), (*b*), (*c*), (*d*) *and* (*e*). *You have to indicate the group of elements which correctly fits into the diagrams.* **(S.B.I.P.O. 1997)**



(*a*) (*b*) (*c*) (*d*) (*e*)

54. Pencil, Stationery, Jeep

55. Factory, Machinery, Product

56. Vegetable, Brinjal, Cauliflower

57. Honesty, Intelligence, Aptitude

58. Truck, Ship, Goods

Directions (*Questions 59 to 68*) : *Of the four alternatives in each of the following questions, three alternatives are such that the three words in each are related among themselves in one of the five ways represented by* (*a*), (*b*), (*c*), (*d*) *and* (*e*) *below while none of these relationships is applicable to the remaining alternative. That is your answer.* **(L.I.C.A.A.O. 1988)**



(*a*) (*b*) (*c*) (*d*) (*e*)

59. (*a*) Army, General, Colonel
(*b*) Boy, Student, Player
(*c*) Painter, Scholar, Table
(*d*) Man, Typist, Peon

60. (*a*) Hen, Dog, Cat
(*b*) Body, Ear, Mouth
(*c*) Bed, Ward, Nurse
(*d*) Tiger, Animal, Carnivorous

61. (*a*) Atmosphere, Air, Oxygen
(*b*) Boy, Girl, Student
(*c*) Man, Worker, Garden
(*d*) Animal, Dog, Cat

62. (*a*) Animal, Mammal, Cow
(*b*) Colour, Cloth, Merchant
(*c*) Colour, Red, Blue
(*d*) Male, Horse, Mare

63. (*a*) Body, Hand, Finger
(*b*) Mammal, Nurse, Woman
(*c*) Cereal, Wheat, Rice
(*d*) Males, Cousins, Nephews

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64. (*a*) Bed, Ward, Hospital
(*b*) Boy, Girl, Player
(*c*) Copper, Zinc, Iron
(*d*) Book, Page, Paragraph

65. (*a*) Star, Moon, Mars
(*b*) Professor, Scholar, Politician
(*c*) Nurse, Doctor, Woman
(*d*) Swimmer, Carpenter, Singer

66. (*a*) Periodical, Weekly, Book
(*b*) Mineral, Copper, Wood
(*c*) Doctors, Human beings, Married people
(*d*) Army, Doctor, Engineer

67. (*a*) Director, Engineer, Musician
(*b*) Apple, Orange, Mango
(*c*) Fruit, Mango, Grass
(*d*) Oxygen, Air, Water

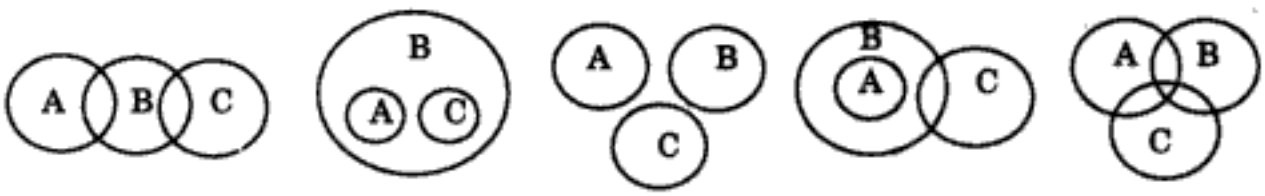
68. (*a*) Mineral, Iron, Copper
(*b*) Dean, Painter, Singer
(*c*) Seed, Leaf, Root
(*d*) Piston, Engine, Wheel

Directions (*Questions 69 to 74*) : *In each of the following questions, there are three words which are related in some way. The relationship in each case is indicated by one of the four alternatives (a), (b), (c) and (d) given below. The alternative which best states the relationship is your answer.* **(C.A.T. 1997)**

(*a*) P includes part of Q and part of R but Q and R are independent of each other.
(*b*) P includes Q and part of R but Q is independent of R.
(*c*) P, Q and R include parts of one another.
(*d*) P includes both Q and R.

69. Wheat, Loaf, Barley
70. Singer, Writer, Actor
71. Soldier, Army, Engineer
72. Wood, Steel, Furniture
73. Researcher, Historian, Scholar
74. Tiger, Elephant, Quadruped

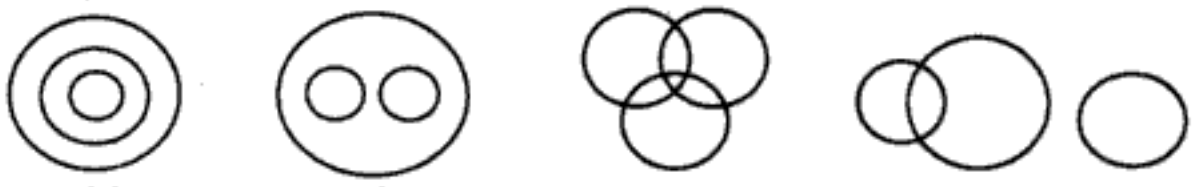
Directions (*Questions 75 to 78*) : *Given below are five patterns represented by circles A, B and C which indicate the logical relationship between and among the respective descriptions. On the basis of description given for A, B and C respectively in the questions, decide which of the given patterns (a), (b), (c), (d) or (e) best indicates the logical relationship.*



(*a*) (*b*) (*c*) (*d*) (*e*)

75. (A) Doctor (B) Male (C) Actor
76. (A) Rose (B) Flower (C) Lotus
77. (A) Father (B) Mother (C) Child
78. (A) Gold (B) Ornament (C) Silver

Directions (*Questions 79 to 83*) : *Choose the Venn diagram which best illustrates the three given classes in each question :* **(Hotel Management, 1997)**



(*a*) (*b*) (*c*) (*d*)

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79. Girl, Athlete, Singer
80. Window, Room, Wall
81. State, Country, City
82. Copper, Paper, Wire
83. Teacher, Graduate, Player

Directions (*Questions 84-85*) : *Choose the Venn diagram which best illustrates the three given classes in each of the following questions :*

(M.B.A. 1998)



(*a*) (*b*) (*c*) (*d*) (*e*)

84. Vertebrates, Non-vertebrates, Living beings
85. Women, Teachers, Doctors
86. In a dinner party both fish and meat were served. Some took only fish and some only meat. There were some vegetarians who did not accept either. The rest accepted both fish and meat. Which of the following logic diagrams correctly reflects this situation ? **(I.A.S. 1998)**



(*a*) (*b*) (*c*) (*d*)

ANSWERS

1. (*a*) : *Elephants and Wolves* bear no relationship to each other. But, both of them are *animals*. (Fig. 1)

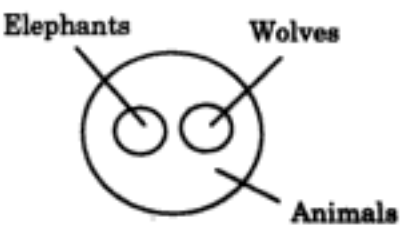


Fig. 1

2. (*a*) : All *carrots* are *vegetables*. All *vegetables* are *foods*. (Fig. 2)

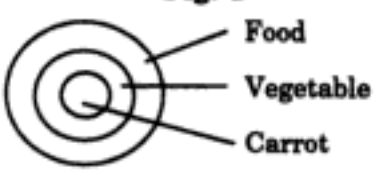


Fig. 2

3. (*d*) : All *mothers* are *women*. Some *mothers* and some *women* can be widows. (Fig. 3)



Fig. 3

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4. (*a*) : Some *authors* can be *teachers*. Some *teachers* can be *men*. Some *authors* can be *men*. So, the given items are partly related to each other. (Fig. 4)

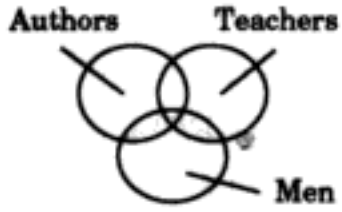


Fig. 4

5. (*b*) : All *sparrows* are *birds*. But, *mice* is entirely different. (Fig. 5)

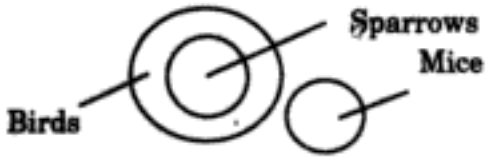


Fig. 5

6. (*c*) : *Tea* and *Coffee* are two separate unrelated items. But, both of them are *beverages*. (Fig. 6)

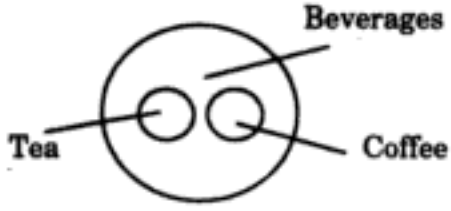


Fig. 6

7. (*a*) : Some *boys* are *students*. Some *students* are *athletes*. Some *boys* are *athletes*. So, the given items are partly related to each other. (Fig. 7)

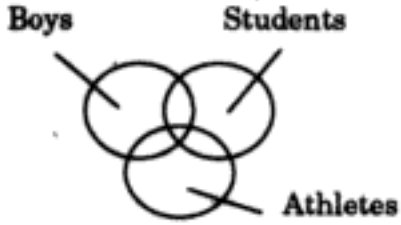


Fig. 7

8. (*c*) : All *thieves* are *criminals*. But *judge* is entirely different. (Fig. 8)

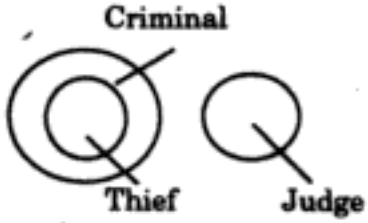


Fig. 8

9. (*c*) : Both *French* and *German* are *languages*. But, both of them are different from each other. (Fig. 9)

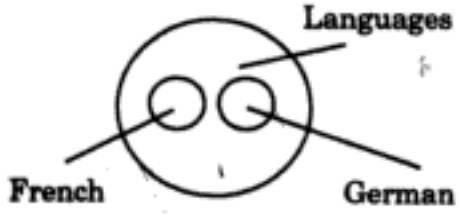


Fig. 9

10. (*a*) : Some *students* can be *cricket players*. Some *cricket players* can be *tennis fans*.
Some *students* can be *tennis fans*. So, the given items are partly related to each other. (Fig. 10).

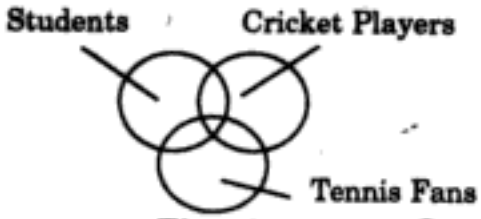


Fig. 10

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11. (*a*) : All *rhombus* are *quadrilaterals*. All *quadrilaterals* are *polygons*. (Fig. 11)

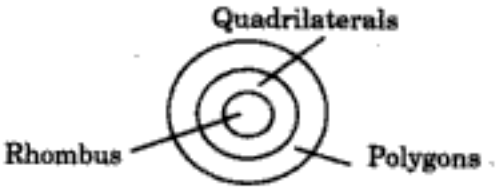


Fig. 11

12. (*d*) : *Tables* and *Chairs* are unrelated items. But, both are items of *Furniture*. (Fig. 12)



Fig. 12

13. (*c*) : *Tie, Shirt* and *Pantaloon* are separate items, entirely different from each other. (Fig. 13)

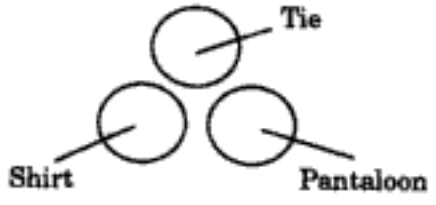


Fig. 13

14. (*d*) : *Dogs* and *Cats* are entirely different from each other. But, both are *pet* animals. (Fig. 14)

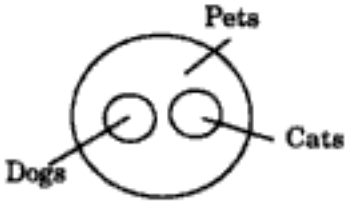


Fig. 14

15. (*e*) : *Brinjal* is a *vegetable*. But *Meat* is entirely different. (Fig. 15)

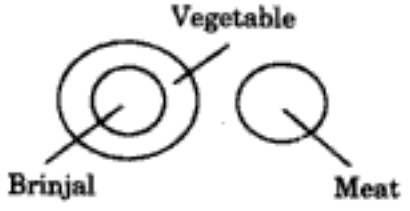


Fig. 15

16. (*b*) : *Deer* and *Rabbit* are unrelated items. But, both are *mammals*. (Fig. 16)

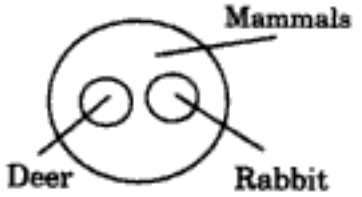


Fig. 16

17. (*c*) : All *teachers* and *graduates* are *human beings*. But, some *teachers* can be *graduates* and some *graduates* can be *teachers*. (Fig. 17)



Fig. 17

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18. (*a*) : *Whales, Fishes* and *Crocodiles* are all separate items, entirely different from each other.
So, they would be represented by three disjoint circles. (Fig. 18)

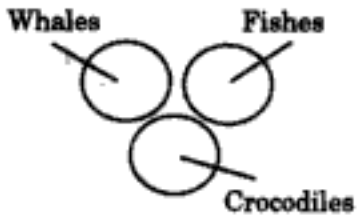


Fig. 18

19. (*e*) : All *plums* are *fruits*. But, *tomatoes* are entirely different. (Fig. 19)

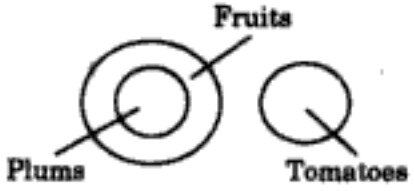


Fig. 19

20. (*c*) : *Mountains* and *Forests* are parts of *earth*. But, some *mountains* are *forested* and some *forests* are *mountainous*. (Fig. 20)

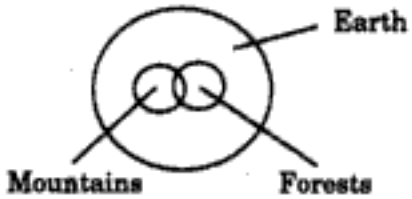


Fig. 20

21. (*b*) : *Tiger* and *Fox* are unrelated and entirely different. But, both are *carnivores* or flesh-eating animals. (Fig. 21)

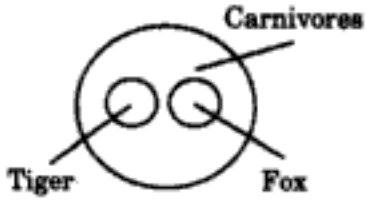


Fig. 21

22. (*b*) : *Grams* and *Beans* are entirely different from each other. But, both are *legumes*. (Fig. 22)

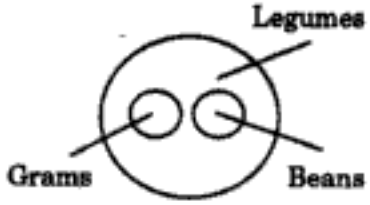


Fig. 22

23. (*d*) : Some *flowers* are *white*. Some *clothes* are *white*. But, all *white* things are not *flowers* or *clothes*. (Fig. 23)

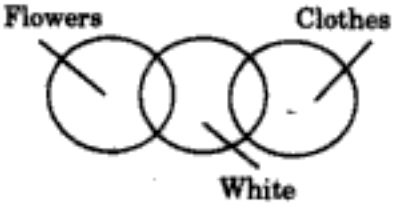


Fig. 23

24. (*a*) : *Uncles, Parents* and *Friends* are entirely different from each other. (Fig. 24)



Fig. 24

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25. (*e*) : *Rohtak* is a part of *Haryana*. *Punjab* is a separate state. (Fig. 25)

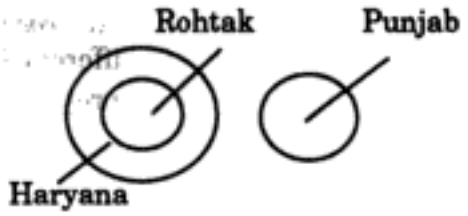


Fig. 25

26. (*b*) : Both *Engineer* and *Doctor* are *people*. But, both of them are different from each other. (Fig. 26)



Fig. 26

27. (*e*) : All *thieves* are *criminals*. But, *lawyers* are entirely different. (Fig. 27)



Fig. 27

28. (*e*) : *Island* is a part of *Sea*. But, *Mountain* is entirely different. (Fig. 28)

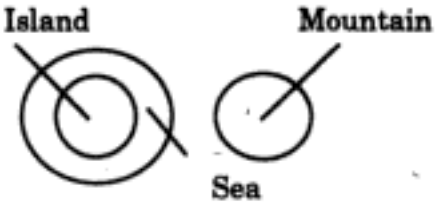


Fig. 28

29. (*c*) : Both *pickpockets* and *blackmailers* are *anti-social elements*. But, some *pickpockets* can be *blackmailers* and vice-versa. (Fig. 29)

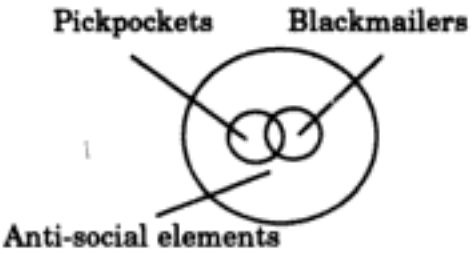


Fig. 29

30. (*a*) : All *violinists* are *instrumentalists*. All *instrumentalists* are *musicians*. (Fig. 30)

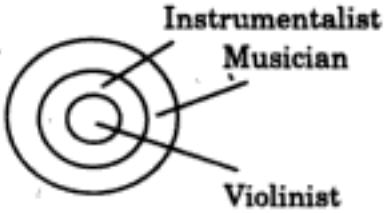


Fig. 30

31. (*a*) : All *pigeons* are *birds*. But, *dogs* are entirely different. (Fig. 31)

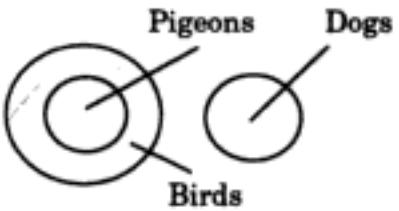


Fig. 31

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32. (*a*) : Both *Leprosy* and *Scurvy* are *Diseases*. But, both are entirely different from each other. (Fig. 32)



Fig. 32

33. (*a*) : Both *Hockey* and *Cricket* are *Games*. But, both are entirely different from each other. (Fig. 33)

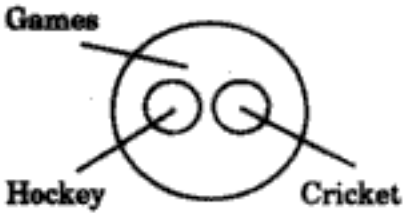


Fig. 33

34. (*b*) : *Yak*, *Zebra* and *Bear* are all different from each other. (Fig. 34)

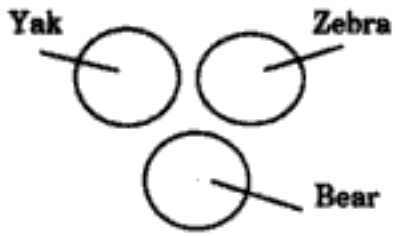


Fig. 34

35. (*c*) : *Sun* is a *star*. *Moon* is entirely different from the two. (Fig. 35)



Fig. 35

36. (*c*) : *Men* belong to the class of *animals*. *Plants* are entirely different from the two. (Fig. 36)

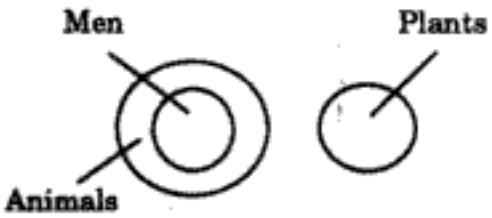


Fig. 36

37. (*a*) : *Mercury* and *Mars* are entirely different from each other. But, both are *planets*. (Fig. 37)



Fig. 37

38. (*c*) : Both *Doctors* and *Lawyers* are *Professionals*. But, both are entirely different from each other. (Fig. 38)



Fig. 38

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39. (*d*) : *Districts* form part of the *State*. But, *Union Territory* is entirely different. (Fig. 39)

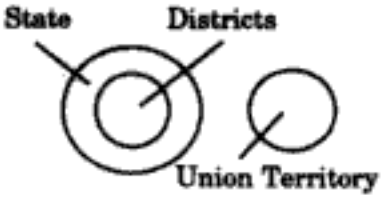


Fig. 39

40. (*d*) : Some *tall men* can be *black haired*. Some *black haired* persons can be *Indians*.
Some *tall men* can be *Indians*.
So, all the three items are partly interrelated. (Fig. 40)

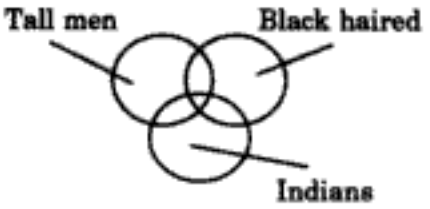


Fig. 40

41. (*b*) : Clearly, each one of the animals that live in water and on land, lies in both the other two categories. Also, some of the animals that live on land also live in water.

42. (*b*) : Clearly, no M.P. can be M.L.A. Also, all M.P.s and M.L.A.s belong to the elected house.

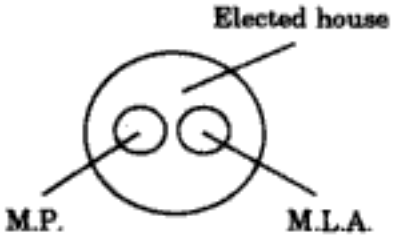


Fig. 41

43. (*c*) : *Square* is a *four-sided figure*. But, *triangle* is entirely different. (Fig. 42)

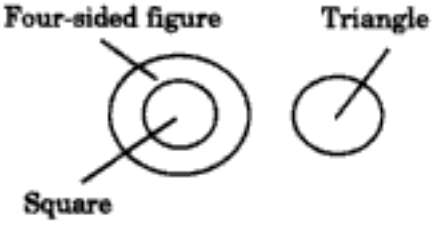


Fig. 42

44. (*a*) : *Doctor* and *Nurse* are entirely different. But, both are *human beings*. (Fig. 43)

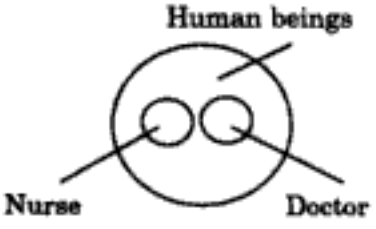


Fig. 43

45. (*b*) : *Protons* and *Electrons* are entirely different from each other. But, both are parts of *atoms*. (Fig. 44)

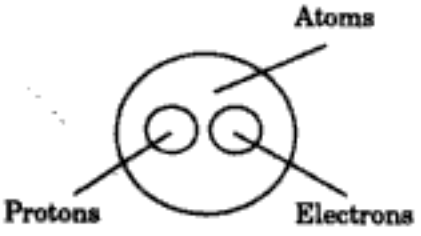


Fig. 44

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46. (*a*) : *Earth* belongs to the class of *Planets*. But, *Sun* is entirely different from the two. (Fig. 45)

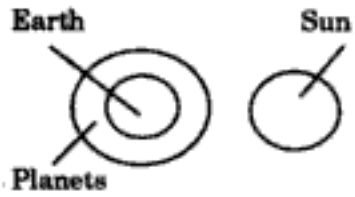


Fig. 45

47. (*c*) : Some *dogs* are *pets* and some *pets* are *dogs*. Both, *dog* and *pets* are *animals*. (Fig. 46)

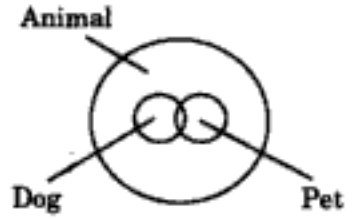


Fig. 46

48. (*b*) : *Physics* and *Chemistry* are entirely different from each other. But, both belong to the class of *Science*. (Fig. 47)

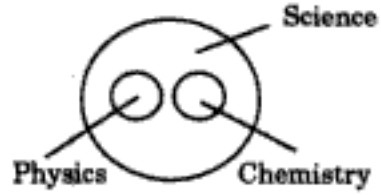


Fig. 47

49. (*b*) : *Hydrogen* and *Oxygen* are entirely different from each other.
But, both are parts of *atmosphere*. (Fig. 48)



Fig. 48

50. (*b*) : *Wheat* and *Maize* are two different items. But, both belong to the class of *Grains*. (Fig. 49)



Fig. 49

51. (*a*) : *Lathe* is a type of *machine*. But, *Mathematics* is entirely different from the two. (Fig. 50)

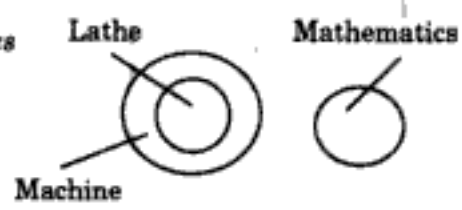


Fig. 50

52. (*b*) : *Botany* and *Zoology* are entirely different from each other. But, both are branches of *Biology*. (Fig. 51)

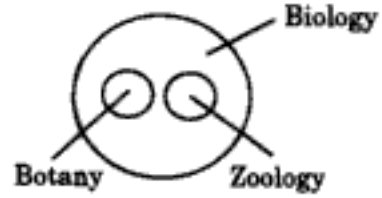


Fig. 51

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53. (*c*) : Some *educated* are *citizens*. Some *citizens* are *educated*. But, both *educated* and *citizens* are *men*. (Fig. 52)

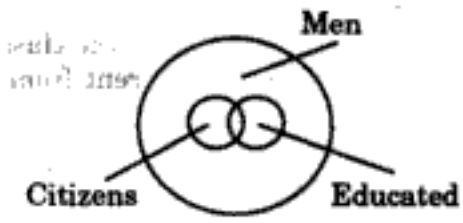


Fig. 52

54. (*a*) : *Pencil* is an item of *Stationery*. But, *Jeep* is entirely different. (Fig. 53)

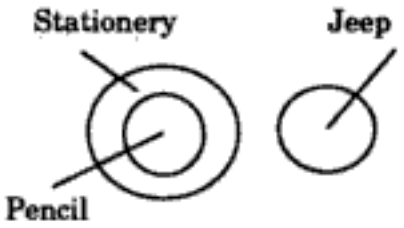


Fig. 53

55. (*e*) : *Machinery* and *product* are entirely different. But, both are present in a *factory*. (Fig. 54)

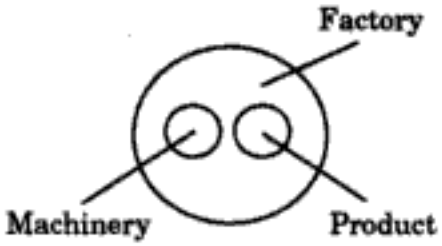


Fig. 54

56. (*e*) : *Brinjal* and *Cauliflower* are entirely different. But, both are *vegetables*. (Fig. 55)

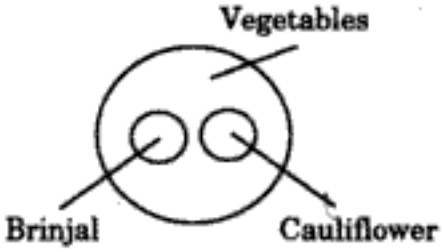


Fig. 55

57. (*d*) : Aptitude, intelligence and honesty are entirely different.

58. (*c*) : *Truck* and *Ship* are entirely different. But, some *goods* are carried by some *trucks* and some *goods* are carried by some *ships*. (Fig. 56)

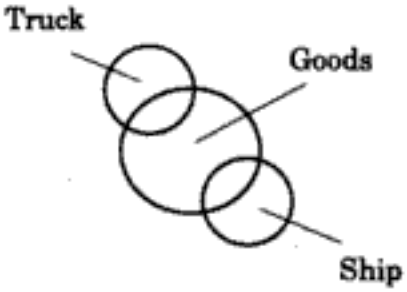


Fig. 56

59. (*c*) : This group of items can be represented as in Fig. 57.
Since there is no such diagram in the question, so (*c*) is the answer.

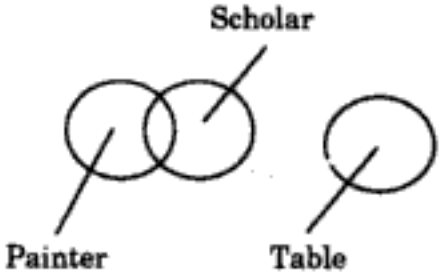


Fig. 57

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60. (*a*) : This group of items can be represented as in Fig. 58.
Since there is no such diagram in the question, so (*a*) is the answer.

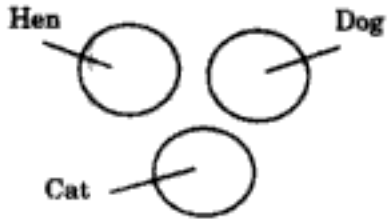


Fig. 58

61. (*c*) : This group of items can be represented as in Fig. 59.
Since there is no such diagram in the question, so (*c*) is the answer.

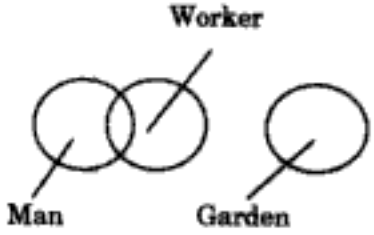


Fig. 59

62. (*b*) : This group of items can be represented as in Fig. 60.
Since there is no such diagram in the question, so (*b*) is the answer.

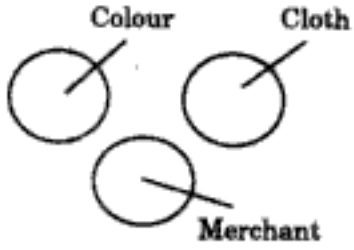


Fig. 60

63. (*d*) : This group of items can be represented as in Fig. 61.
Since there is no such diagram in the question, so (*d*) is the answer.

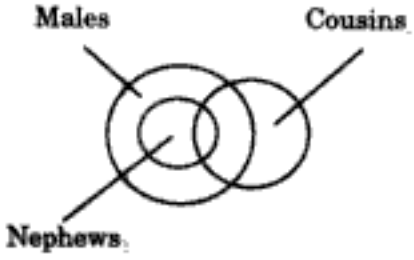


Fig. 61

64. (*c*) : This group of items can be represented as in Fig. 62.
Since there is no such diagram in the question, so (*c*) is the answer.



Fig. 62

65. (*a*) : This group of items can be represented as in Fig. 63.
Since there is no such diagram in the question, so (*a*) is the answer.



Fig. 63

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66. (*c*) : This group of items can be represented as in Fig. 64.
Since there is no such diagram in the question, so (*c*) is the answer.

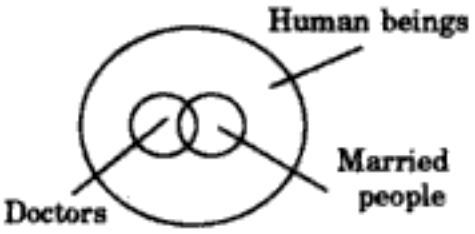


Fig. 64

67. (*b*) : This group of items can be represented as in Fig. 65.
Since there is no such diagram in the question, so (*b*) is the answer.

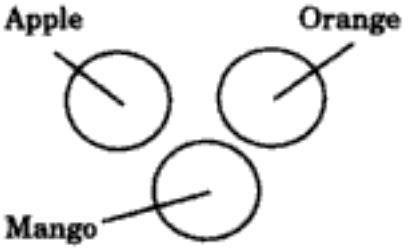


Fig. 65

68. (*c*) : This group of items can be represented as in Fig. 66.
Since there is no such diagram in the question, so (*c*) is the answer.

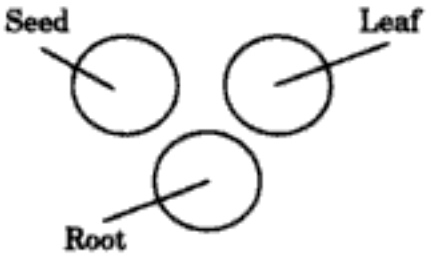


Fig. 66

69. (*a*) : Here P is Loaf, Q is Wheat and R is Barley.
70. (*c*) : Here P is Singer, Q is Writer and R is Actor.
71. (*a*) : Here P is Army, Q is Soldier and R is Engineer.
72. (*a*) : Here P is Furniture, Q is Wood and R is Steel.
73. (*c*) : Here P is Researcher, Q is Historian and R is Scholar.
74. (*d*) : Here P is Quadruped, Q is Tiger and R is Elephant.
75. (*a*) : Some *doctors* and some *actors* are *males*. But, *doctor* and *actor* are entirely different.
76. (*b*) : Both *Rose* and *Lotus* are *flowers*. But, *Rose* and *Lotus* are entirely different.
77. (*c*) : *Father, Mother* and *Child* are entirely different.
78. (*a*) : Some *ornaments* are made of *gold* and some of *silver*. *Gold* and *Silver* are entirely different.

79. (*c*) : Some *girls* can be *athletes*.
Some *athletes* can be *singers*.
Some *girls* can be *singers*.

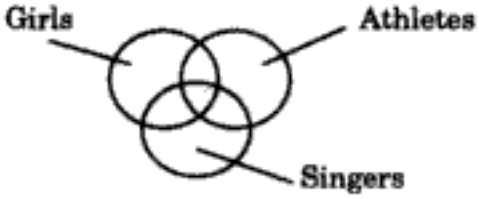


Fig. 67

80. (*b*) : Both *wall* and *window* are parts of a *room*.
But, *wall* and *window* are entirely different.

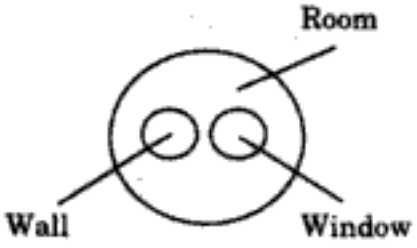


Fig. 68

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81. (*a*) : A *city* lies within a *state,* which lies within a *country.*

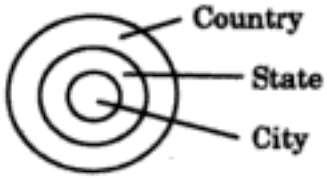


Fig. 69

82. (*d*) : Some *wires* are made of *copper.* But, *Paper* is entirely different.

83. (*c*) : All three items are partly related to each other.

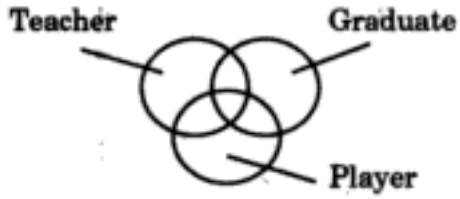


Fig. 70

84. (*b*) : Both *vertebrates* and *non-vertebrates* are entirely different, but both are *living beings.*

85. (*d*) : Some *teachers* and some *doctors* are *women.* But, *Teacher* and *Doctor* are entirely different.

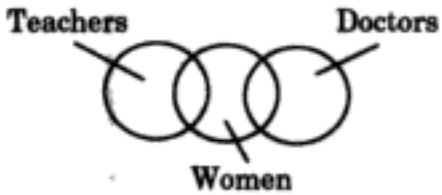


Fig. 71

86. (*a*) : The given situation can be represented as under :

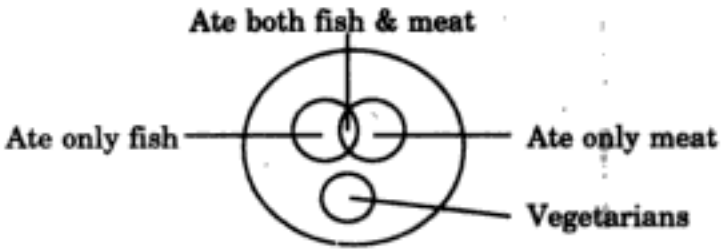
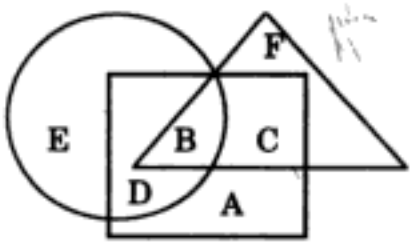


Fig. 72

TYPE-2

In this type of questions, generally a Venn diagram is given. Each geometrical figure in the diagram represents a certain class. The candidate is required to study and analyse the figure carefully and then answer certain questions regarding the given data.

Example 1 : ***In the following diagram, three classes of population are represented by three figures. The triangle represents the school teachers, the square represents the married persons and the circle represents the persons living in joint families.***



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1. Married persons living in joint families but not working as school teachers are represented by

(*a*) C (*b*) F (*c*) D (*d*) A

Sol. Married persons living in joint families are represented by the region common to the square and the circle *i.e.*, D and B. But, according to the given conditions, the persons should not be school teachers. So, B is to be excluded. Hence, the required condition is denoted by region D. So, the answer is (*c*).

2. Persons who live in joint families, are unmarried and who do not work as school teachers are represented by

(*a*) C (*b*) B (*c*) E (*d*) D

Sol. Persons living in joint families are represented by the circle. According to the given conditions, the persons should be unmarried and not working as school teachers. So, the region should not be a part of either the square or the triangle. Thus, the given conditions are satisfied by the region E. So, the answer is (*c*).

3. Married teachers living in joint families are represented by

(*a*) C (*b*) B (*c*) D (*d*) A

Sol. Married teachers are represented by the region common to the square and the triangle *i.e.*, B and C. But, according to the given conditions, the persons should be living in joint families. So, the required region should be a part of the circle. Since B lies inside the circle, so the given conditions are satisfied by the persons denoted by the region B. Hence, the answer is (*b*).

4. School teachers who are married but do not live in joint families are represented by

(*a*) C (*b*) F (*c*) A (*d*) D

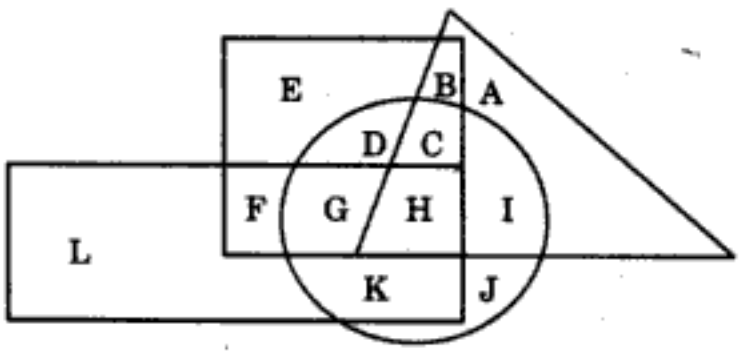
Sol. As in the above question, married teachers are represented by B and C. But, here, the given conditions lay down that the persons should not be living in joint families. So, the required region should lie outside the circle. Since C lies outside the circle, so the given conditions are satisfied by the persons denoted by the region C. Hence, the answer is (*a*).

5. School teachers who are neither married nor do live in joint families are represented by

(*a*) F (*b*) C (*c*) B (*d*) A

Sol. School teachers are represented by the triangle. But according to the given conditions, persons are neither married nor do they live in joint families. So, the region should not be a part of either the square or the circle. Such a region is F. Hence, the answer is (*a*).

Example 2 : ***In the following diagram, the square represents girls, the circle tall persons, the triangle is for tennis players and the rectangle stands for the swimmers.***



On the basis of the above diagram, answer the following questions.

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1. Which letter represents tall girls who are swimmers but don't play tennis ?

(*a*) C (*b*) D (*c*) G (*d*) H

Sol. Tall girls, who are swimmers are represented by the region common to the square, circle and the rectangle *i.e.*, G and H. But, according to the given conditions, the girls shouldn't be tennis players. So, the required region should not be a part of the triangle *i.e.*, H should be excluded. Thus, the region representing the persons satisfying the given conditions is G. Hence, the answer is (*c*).

2. Which letter represents girls who are swimmers, play tennis but are not tall ?

(*a*) B (*b*) E (*c*) F (*d*) None of these

Sol. Girls who are swimmers and play tennis are represented by the region common to the square, triangle and rectangle *i.e.*, H. But, it is given that the girls shouldn't be tall. So, the required region should not be a part of the circle. Since H is a part of the circle, so the answer is (*d*).

3. Which letter represents tall girls who do not play tennis and are not swimmers ?

(*a*) C (*b*) D (*c*) E (*d*) G

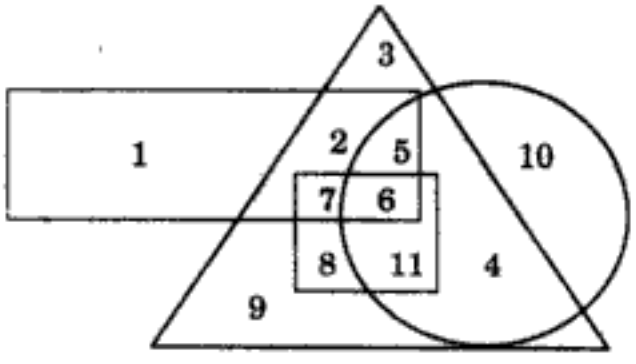
Sol. Tall girls are represented by the region common to the square and the circle *i.e.*, D, C, G and H. But, according to the given conditions, the girls are neither tennis players nor swimmers. So, the required region should be neither a part of the rectangle nor the triangle. G lies inside the rectangle, C inside the triangle and H is common to both. So, the answer is (*b*).

4. Which letter represents tall persons who are gents and swimmers but do not play tennis ?

(*a*) I (*b*) J (*c*) K (*d*) L

Sol. The tall persons are represented by regions inside the circle *i.e.*, C, D, G, H, I, J and K. Since the persons are not girls and do not play tennis, so the region should not be a part of either the square or the triangle. Thus, C, D, G, H should be excluded. Also, according to the given conditions, the persons should be swimmers. So, the required region should be a part of the rectangle and such a region is K. Hence, the answer is (*c*).

Example 3 : ***The following questions are based on the diagram given below :***



(1) The rectangle represents government employees.

(2) The triangle represents urban people.

(3) The circle represents graduates.

(4) The square represents clerks.

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1. Which of the following statements is true ?

(*a*) All government employees are clerks.

(*b*) Some government employees are graduates as well as clerks.

(*c*) All government employees are graduates.

(*d*) All clerks are government employees but not graduates.

Sol. The above cases may be considered as under :

For statement (*a*) to be true, the rectangle should lie inside the square. This is not true. Hence, (*a*) is false.

For statement (*b*) to be true, there should be a region common to the rectangle, circle and the square. Such a region is 6. Hence, (*b*) is true.

Further, for statement (*c*) to be true, the rectangle should lie inside the circle. So, (*c*) is false.

For statement (*d*) to be true, square should lie wholly inside the rectangle, with no region common to the circle. This is not true. So, (*d*) is false.

2. Which of the following statements is true ?

(*a*) All urban people are graduates.

(*b*) Some clerks are government employees but not urban.

(*c*) All government employees are clerks.

(*d*) Some urban people are not graduates.

Sol. For the validity of condition (*a*), the triangle should lie inside the circle. This is not true. So, (*a*) is false.

For the validity of statement (*b*), there should be a region which is common to the square and the rectangle but is not a part of the triangle. Since no such region exists, (*b*) is false.

For the validity of statement (*c*), the rectangle should lie inside the square. This is not true. So, (*c*) is false.

For the validity of statement (*d*), some region of the triangle should lie outside the circle. Since this is true, so, (*d*) is true.

3. Choose the correct statement :

(*a*) Some clerks are government employees.

(*b*) No clerk is urban.

(*c*) All graduates are urban.

(*d*) All graduates are government employees.

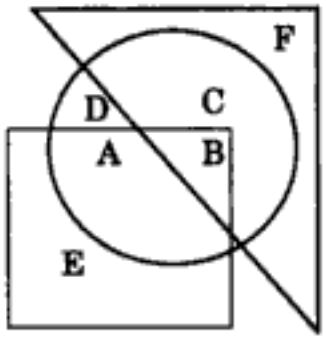
Sol. For the validity of statement (*a*), there should be a region common to the square and rectangle. Such regions are 6 and 7. So, (*a*) is true.

Further, for statement (*b*) to be true, there should be no region common to the square and the triangle. But since square lies wholly inside the triangle, (*b*) is false. For statement (*c*) to be true, circle should lie inside the triangle. Clearly, (*c*) is false. For the validity of statement (*d*), the circle should lie inside the rectangle. Clearly, (*d*) is false.

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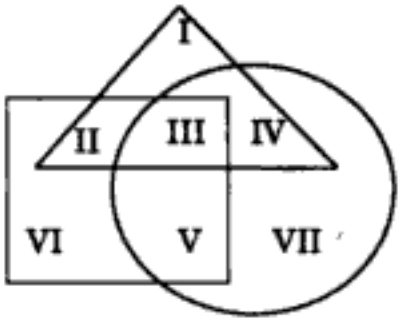
EXERCISE 9B

1. Which one of the following statements is correct with regard to the given figure ? **(S.C.R.A. 1994)**



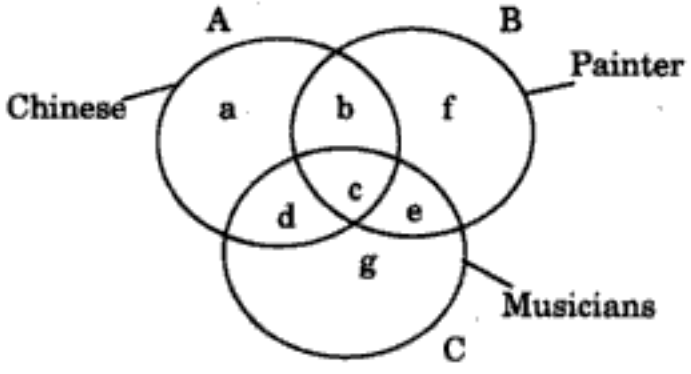
(*a*) A and B are in all the three shapes.
(*b*) E, A, B, C are in all the three shapes.
(*c*) F, C, D, B, A are in all the three shapes.
(*d*) Only B is in all the three shapes.

2. The triangle, square and circle shown below respectively represent the urban, hard working and educated people. Which one of the areas marked I-VII is represented by the urban educated people who are not hard working ?
(Civil Services, 1992)



(*a*) II (*b*) I (*c*) IV (*d*) III

Directions (*Questions 3 to 6*) : *In the figure given below, there are three intersecting circles each representing certain section of people. Different regions are marked a—g. Read the statements in each of the following questions and choose the letter of the region which correctly represents the statement.*



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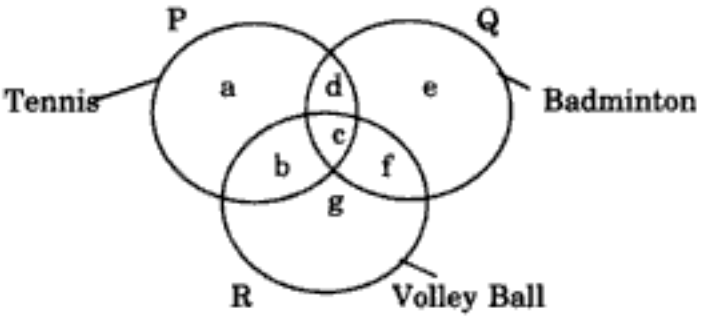
3. Chinese who are painters but not musicians.
(*a*) b (*b*) c (*c*) d (*d*) g

4. Painters who are neither Chinese nor musicians.
(*a*) b (*b*) c (*c*) f (*d*) g

5. Chinese who are musicians but not painters.
(*a*) d (*b*) c (*c*) b (*d*) a

6. Chinese who are painters as well as musicians.
(*a*) a (*b*) b (*c*) c (*d*) d

Directions (*Questions 7 to 10*) : *The figure given below consists of three intersecting circles which represent sets of students who play Tennis, Badminton and Volley Ball. Each region in the figure is represented by a small letter.*



On the basis of the above figure, answer the questions given below.

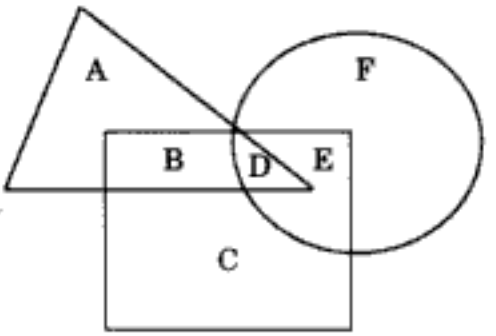
7. Which letter represents the set of persons who play all the three games ?
(*a*) b (*b*) c (*c*) f (*d*) g

8. Which letter represents the set of persons who play Tennis and Volley Ball but not Badminton ?
(*a*) g (*b*) e (*c*) c (*d*) b

9. Which letter represents the set of persons who play Tennis but neither Badminton nor Volley Ball ?
(*a*) a (*b*) b (*c*) c (*d*) d

10. Which letter represents the set of persons who play Tennis and Badminton but not Volley Ball ?
(*a*) b (*b*) c (*c*) d (*d*) f

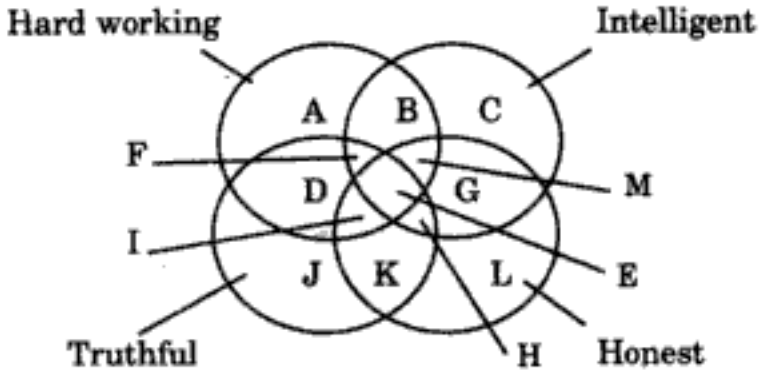
11. In the given figure, the triangle represents girls, the square represents sports persons and the circle represents coaches. The portion in the figure which represents girls, who are sports persons but not coaches is the one labelled
(I.A.S. 1996)



(*a*) A (*b*) B (*c*) D (*d*) E

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Directions (*Questions 12 to 16*) : *Below is given a figure with four intersecting circles, each representing a group of persons having the quality written against it. Study the figure carefully and answer the questions that follow.*



12. The region which represents the people who are intelligent, honest and truthful but not hard working is denoted by
(*a*) E (*b*) F (*c*) H (*d*) I

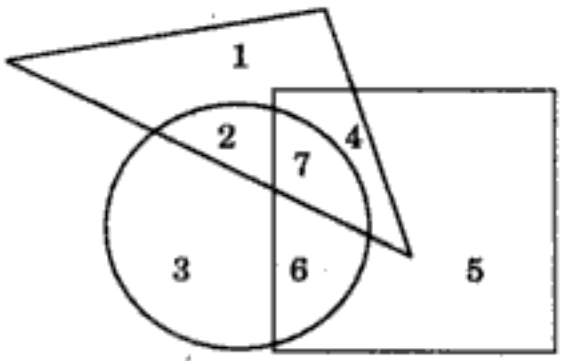
13. The people possessing all the qualities are represented by
(*a*) I (*b*) H (*c*) F (*d*) E

14. The region which represents people who are not honest but possess all other three qualities, is denoted by
(*a*) B (*b*) D (*c*) F (*d*) I

15. People who are not hard working, intelligent and truthful are represented by
(*a*) G (*b*) H (*c*) K (*d*) L

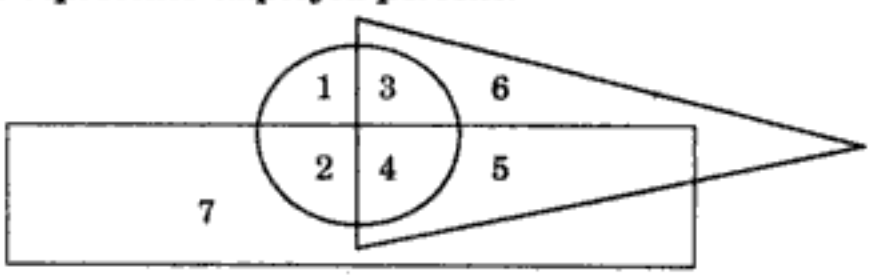
16. People who are not honest and truthful but are hard working and intelligent both, are represented by
(*a*) E (*b*) B (*c*) M (*d*) I

17. If Tall is equivalent to circle, Armymen to triangle and Strong to square, indicate which number will represent strong armymen ? **(I.A.S. 1982)**



(*a*) 3 (*b*) 4 (*c*) 5 (*d*) 6

Directions (*Questions 18 to 21*) : *In the figure given below, the circle represents young persons, the triangle represents uneducated persons and the rectangle represents employed persons.*



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Study the figure carefully and answer the questions given below.

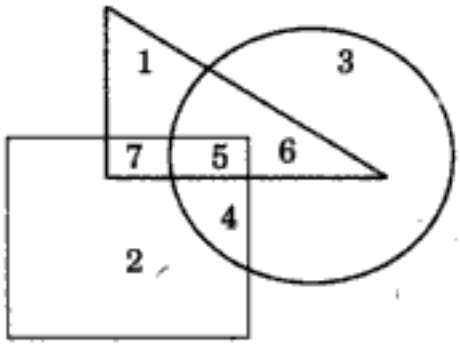
18. Which region represents young, uneducated and employed persons ?
(*a*) 6 (*b*) 5 (*c*) 4 (*d*) 3 (*e*) 2

19. The region which represents educated, employed young persons, is denoted by
(*a*) 2 (*b*) 3 (*c*) 4 (*d*) 5 (*e*) 6

20. Which region represents young, educated and unemployed persons ?
(*a*) 7 (*b*) 4 (*c*) 1 (*d*) 5 (*e*) 3

21. Which region represents young, uneducated and unemployed persons ?
(*a*) 1 (*b*) 2 (*c*) 6 (*d*) 3 (*e*) 5

Directions (*Questions 22 to 24*) : *The following questions are based on the diagram given below. In the diagram, the triangle stands for graduates, square stands for membership of professional organisations and the circle stands for membership of social organisations. Read each statement and find out the appropriate number(s) to represent the people covered by the given statement.* **(Assistant Grade, 1994)**

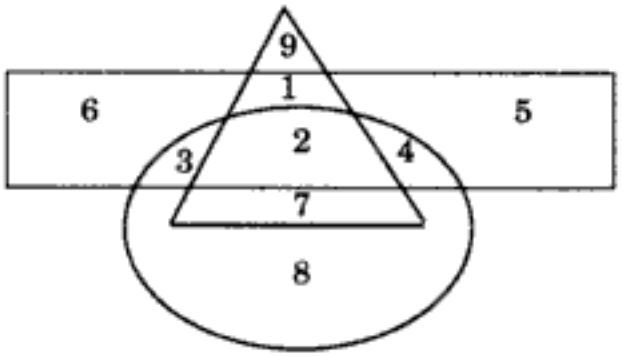


22. Number of graduates in social organisations
(*a*) 1 (*b*) 5 (*c*) 6 (*d*) 5 and 6

23. Number of graduates in social organisations only
(*a*) 3 (*b*) 4 (*c*) 5 (*d*) 6

24. Number of graduates in professional organisations
(*a*) 5 and 7 (*b*) 5, 6 and 7 (*c*) 6 and 7 (*d*) 4, 5 and 6

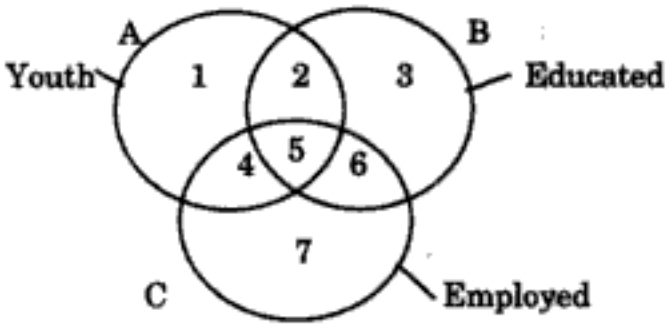
25. In the figure given below, triangle represents the women, rectangle represents the employed and circle represents the doctors, find out the area of the figure which represents women doctors who are not employed. **(C.B.I. 1993)**



(*a*) 1 (*b*) 3 (*c*) 7 (*d*) 8

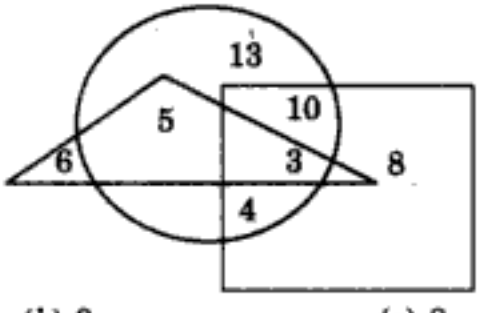
26. Study the diagram below and identify the region representing youth who are employed but not educated. **(S.S.C. 1995)**

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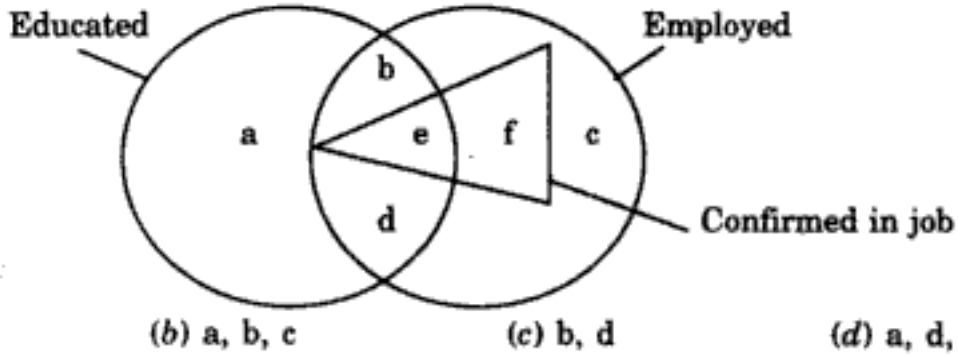
(*a*) 4 only (*b*) 1, 4, 7 (*c*) 4, 7 (*d*) 4, 5, 6

27. In the following diagram, parallelogram represents women, triangle represents sub-inspectors of police and circle represents graduates. Which numbered area represents women graduate sub-inspectors of police ? **(C.B.I. 1994)**



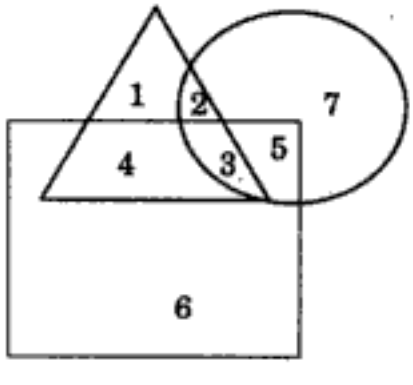
(*a*) 5 (*b*) 3 (*c*) 8 (*d*) 13

28. Read the figure and find the region representing persons who are educated and employed but not confirmed. **(Assistant Grade, 1993)**



(*a*) a, c (*b*) a, b, c (*c*) b, d (*d*) a, d, c

29. In the figure given below, triangle represents the healthy, square represents the old and circle represents the men. Find out the area of the figure which represents the men who are healthy but not old ? **(C.B.I. 1993)**



(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 7

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$\Rightarrow \quad cx > acy \Rightarrow cx > abz \quad (\because cy = bz)$

$\Rightarrow \quad cx \neq abz \ i.e. \ cx \ \delta \ abz.$

44. (*a*) : $a^2x \ \alpha \ byz, czx \ \alpha \ b^2y$ and $c^2z \ \alpha \ axy \Rightarrow a^2x = byz, czx = b^2y, \ c^2z = axy.$

$czx = b^2y \Rightarrow c^2zx = cb^2y \Rightarrow axyx = cb^2y \quad (\because c^2z = axy)$

$\Rightarrow ax^2 = cb^2.$

Now, $a^2x = byz \Rightarrow a^2x^2 = bxyz \Rightarrow a.ax^2 = bxyz$

$\Rightarrow acb^2 = bxyz \quad (\because ax^2 = cb^2)$

$\Rightarrow abc = xyz \ i.e. \ abc \ \alpha \ xyz.$

45. (*a*) : $A + B > C + D, \ C + D > B + E, \ B + E = 2C$

$\Rightarrow A + B > B + E, \ B + E = 2C \Rightarrow A + B > 2C.$

46. (*d*) : $A + D > C + E \Rightarrow A + D > (2B - D) + E \quad (\because C + D = 2B)$

$\Rightarrow A + D > (B + E) + (B - D)$

$\Rightarrow A + D > (C + D) + (B - D)$

$\Rightarrow A + D > B + C.$

47. (*b*) : Clearly, time taken $= \dfrac{\text{sum of lengths of two trains}}{\text{total speed of two trains}}$

$= \dfrac{L_1 + L_2}{V_1 + V_2} = (L_1 \text{"} L_2) \ @ \ (V_1 \text{"} V_2).$

48. (*b*) : Total fare = B + 15% of B + 2% of B + 200

$= B + \dfrac{B \times 15}{100} + \dfrac{B \times 2}{100} + 200$

$= B \text{"} (B \bullet 15) \ @ \ 100 \text{"} (B \bullet 2) \ @ \ 100 \text{"} 200.$

49. (*c*) : Profit percentage $= \dfrac{S - (C + L + T)}{C + L + T} \times 100$

$= \{S' \ (C \text{"} L \text{"} \ T)\} \ @ \ (C \text{"} L' \ T) \bullet 100$

50. (*e*) : Clearly, total marks $= (T - 2) \times 2 + \dfrac{4K}{3} + 5 \times 2$

$= (T' 2) \bullet 2 \text{"} 4 \bullet K \ @ \ 3 \text{"} 5 \bullet 2.$

51. (*b*) : Marks out of 150 in first periodical = P.

Marks out of 100 in first periodical $= \left(\dfrac{P}{150} \times 100\right).$

Marks out of 180 in second periodical = T.

Marks out of 100 in second periodical $= \left(\dfrac{T}{180} \times 100\right).$

Marks out of 400 in final examination = M.

Marks out of 100 in final examination $= \left(\dfrac{M}{400} \times 100\right).$

$\therefore$ Total marks

$= \left[10\% \text{ of } \left(\dfrac{P}{150} \times 100\right)\right] + \left[15\% \text{ of } \left(\dfrac{T}{180} \times 100\right)\right] + \left[75\% \text{ of } \left(\dfrac{M}{400} \times 100\right)\right]$

$= \left[\dfrac{10}{100} \text{ of } \left(\dfrac{P}{150} \times 100\right)\right] + \left[\dfrac{15}{100} \text{ of } \left(\dfrac{T}{180} \times 100\right)\right] + \left[\dfrac{75}{100} \text{ of } \left(\dfrac{M}{400} \times 100\right)\right]$

$= \left(\dfrac{P}{150} \times 10\right) + \left(\dfrac{T}{180} \times 15\right) + \left(\dfrac{M}{400} \times 75\right)$

$= (P \ @ \ 150 \bullet 10) \text{"} (T \ @ \ 180 \bullet 15) \text{"} (M \ @ \ 400 \bullet 75).$

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13. LOGICAL SEQUENCE OF WORDS

In this type of questions, a group of words is given. The candidate is required to arrange these words in a meaningful order such as the sequence of occurrence of events, sequence from a part to the whole, sequence of increasing/decreasing size, value, intensity etc., and then choose the correct sequence accordingly.

Ex. 1. ***Arrange the following in a meaningful sequence :***

1. Consultation 2. Illness 3. Doctor
4. Treatment 5. Recovery

(*a*) 2,3,1,4,5 (*b*) 2,3,4,1,5 (*c*) 4,3,1,2,5 (*d*) 5,1,4,3,2

Sol. We know that **illness** occurs first. One then goes to the **doctor** and after **consultation** with him, undergoes **treatment** to finally attain **recovery.**

Thus, the correct order is 2, 3, 1, 4, 5.

Hence, the answer is (*a*).

Ex. 2. ***Arrange the following in a logical order :***

1. Euphoria 2. Happiness 3. Ambivalence
4. Ecstasy 5. Pleasure

(*a*) 4,1,3,2,5 (*b*) 3,2,5,1,4 (*c*) 2,1,3,4,5 (*d*) 1,4,2,5,3

Sol. All the given words stand for 'Joy', but the intensity increases in the order — Ambivalence, Happiness, Pleasure, Euphoria, Ecstasy.

Thus, the correct order is 3, 2, 5, 1, 4.

Hence, the answer is (*b*).

Ex. 3. ***Arrange the following in a meaningful order, from particular to general :***

1. Family 2. Community 3. Member
4. Locality 5. Country

(*a*) 3,1,2,4,5 (*b*) 3,1,2,5,4 (*c*) 3,1,4,2,5 (*d*) 3,1,4,5,2

(Asstt. Grade, 1996)

Sol. Clearly, a **member** is a part of a **family,** which in turn is a part of **community.** The community lives in a **locality** which lies within a **country.**

Thus, the correct order is 3, 1, 2, 4, 5.

Hence, the answer is (*a*).

EXERCISE 13

Directions (*Questions 1 to 33*) : ***In each of the following questions, arrange the given words in a meaningful sequence and then choose the most appropriate sequence from amongst the alternatives provided below each question.***

1. 1. Birth 2. Death 3. Funeral
4. Marriage 5. Education

(Asstt. Grade, 1995)

(*a*) 4,5,3,1,2 (*b*) 2,3,4,5,1 (*c*) 1,5,4,2,3 (*d*) 1,3,4,5,2

2. 1. Site 2. Plan 3. Rent
4. Money 5. Building

(Central Excise, 1996)

(*a*) 4,1,2,5,3 (*b*) 3,4,2,5,1 (*c*) 2,3,5,1,4 (*d*) 1,2,3,5,4

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3. 1. Table 2. Tree 3. Wood
4. Seed 5. Plant **(C.B.I. 1994)**
(*a*) 4,5,3,2,1 (*b*) 4,5,2,3,1 (*c*) 1,3,2,4,5 (*d*) 1,2,3,4,5

4. 1. College 2. Child 3. Salary
4. School 5. Employment **(Central Excise, 1992)**
(*a*) 1,2,4,3,5 (*b*) 2,4,1,5,3 (*c*) 4,1,3,5,2 (*d*) 5,3,2,1,4

5. 1. Reading 2. Composing 3. Writing 4. Printing
(*a*) 1,3,2,4 (*b*) 2,3,4,1 (*c*) 3,1,2,4 (*d*) 3,2,4,1
(Railways, 1995)

6. 1. Cutting 2. Dish 3. Vegetable
4. Market 5. Cooking **(Asstt. Grade, 1994)**
(*a*) 1,2,4,5,3 (*b*) 3,2,5,1,4 (*c*) 4,3,1,5,2 (*d*) 5,3,2,1,4

7. 1. Income 2. Status 3. Education
4. Well-being 5. Job **(S.S.C. 1993)**
(*a*) 1,3,2,5,4 (*b*) 1,2,5,3,4 (*c*) 3,1,5,2,4 (*d*) 3,5,1,2,4

8. 1. Milky way 2. Sun 3. Moon
4. Earth 5. Stars **(C.B.I. 1993)**
(*a*) 4,3,2,5,1 (*b*) 3,4,2,5,1 (*c*) 2,3,4,5,1 (*d*) 1,4,3,2,5

9. 1. Sea 2. Rivulet 3. Ocean
4. River 5. Glacier **(I. Tax & Central Excise, 1995)**
(*a*) 5,4,3,2, 1 (*b*) 5,4,2,3,1 (*c*) 5,2,4,1,3 (*d*) 5,2,1,3,4

10. 1. Poverty 2. Population 3. Death
4. Unemployment 5. Disease **(S.S.C. 1996)**
(*a*) 3,4,2,5,1 (*b*) 2,4,1,5,3 (*c*) 2,3,4,5,1 (*d*) 1,2,3,4,5

11. 1. Yarn 2. Plant 3. Saree
4. Cotton 5. Cloth **(Asstt. Grade, 1996)**
(*a*) 2,4,1,5,3 (*b*) 2,4,3,5,1 (*c*) 2,4,5,1,3 (*d*) 2,4,5,3,1

12. 1. Puberty 2. Adulthood 3. Childhood
4. Infancy 5. Senescence **(U.D.C. 1995)**
(*a*) 5,2,3,4,1 (*b*) 4,3,2,1,5 (*c*) 4,3,1,2,5 (*d*) 2,4,3,1,5

13. 1. Windows 2. Walls 3. Floor
4. Foundation 5. Roof 6. Room **(C.B.I. 1995)**
(*a*) 4,5,3,2,1,6 (*b*) 4,3,5,6,2,1 (*c*) 4,2,1,5,3,6 (*d*) 4,1,5,6,2,3

14. 1. Post-box 2. Letter 3. Envelope
4. Delivery 5. Clearance
(*a*) 3,2,4,5,1 (*b*) 3,2,1,5,4 (*c*) 3,2,1,4,5 (*d*) 2,3,1,4,5

15. 1. Key 2. Door 3. Lock
4. Room 5. Switch on **(Asstt. Grade, 1995)**
(*a*) 5,1,2,4,3 (*b*) 4,2,1,5,3 (*c*) 1,2,3,5,4 (*d*) 1,3,2,4,5

16. 1. Gold 2. Iron 3. Sand
4. Platinum 5. Diamond **(Asstt. Grade, 1997)**
(*a*) 2,4,3,5,1 (*b*) 3,2,1,5,4 (*c*) 4,5,1,3,2 (*d*) 5,4,3,2,1

17. 1. Cut 2. Put on 3. Mark
4. Measure 5. Tailor **(Central Excise, 1994)**
(*a*) 4,3,1,5,2 (*b*) 3,1,5,4,2 (*c*) 2,4,3,1,5 (*d*) 1,3,2,4,5

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18. 1. Rainbow 2. Rain 3. Sun
4. Happy 5. Child **(S.S.C. 1993)**
(*a*) 2,1,4,3,5 (*b*) 2,3,1,5,4 (*c*) 4,2,3,5,1 (*d*) 4,5,1,2,3

19. 1. Study 2. Job 3. Examination
4. Earn 5. Apply **(I. Tax & Central Excise, 1992)**
(*a*) 1,2,3,4,5 (*b*) 1,3,2,5,4 (*c*) 1,3,5,4,2 (*d*) 1,3,5,2,4

20. 1. Shoulder 2. Wrist 3. Elbow
4. Palm 5. Finger
(*a*) 5,4,2,3,1 (*b*) 3,4,5,2,1 (*c*) 3,1,4,2,5 (*d*) 2,4,5,3,1

21. 1. Frog 2. Eagle 3. Grasshopper
4. Snake 5. Grass **(S.S.C. 1996)**
(*a*) 5,3,4,2,1 (*b*) 5,3,1,4,2 (*c*) 3,4,2,5,1 (*d*) 1,3,5,2,4

22. 1. Punishment 2. Prison 3. Arrest
4. Crime 5. Judgement **(Asstt. Grade, 1996)**
(*a*) 5,1,2,3,4 (*b*) 4,3,5,2,1 (*c*) 4,3,5,1,2 (*d*) 2,3,1,4,5

23. 1. Child 2. Job 3. Marriage
4. Infant 5. Education **(S.S.C. 1993)**
(*a*) 1,3,5,2,4 (*b*) 3,5,2,1,4 (*c*) 4,1,3,5,2 (*d*) 4,1,5,2,3

24. 1. Mother 2. Child 3. Milk
4. Cry 5. Smile **(Asstt. Grade, 1995)**
(*a*) 1,5,2,4,3 (*b*) 2,4,1,3,5 (*c*) 2,4,3,1,5 (*d*) 3,2,1,5,4

25. 1. Travel 2. Destination 3. Payment
4. Berth/Seat number 5. Reservation
6. Availability of berth/seat for reservation
(*a*) 6,2,5,4,3,1 (*b*) 5,3,4,1,6,2 (*c*) 2,6,3,5,4,1 (*d*) 1,2,5,4,3,6
(Central Excise, 1994)

26. 1. Curd 2. Grass 3. Butter
4. Milk 5. Cow **(Asstt. Grade, 1994)**
(*a*) 5,2,4,1,3 (*b*) 5,2,3,4,1 (*c*) 4,2,5,3,1 (*d*) 2,5,4,3,1

27. 1. Elephant 2. Cat 3. Mosquito
4. Tiger 5. Whale **(Asstt. Grade, 1996)**
(*a*) 1,3,5,4,2 (*b*) 2,5,1,4,3 (*c*) 3,2,4,1,5 (*d*) 5,3,1,2,4

28. 1. Probation 2. Interview 3. Selection
4. Appointment 5. Advertisement 6. Application
(*a*) 5,6,2,3,4,1 (*b*) 5,6,3,2,4,1 (*c*) 5,6,4,2,3,1 (*d*) 6,5,4,2,3,1

29. 1. District 2. Village 3. State 4. Town
(*a*) 2,1,4,3 (*b*) 2,3,4,1 (*c*) 2,4,1,3 (*d*) 3,2,1,4
(S.S.C. 1995)

30. 1. Index 2. Contents 3. Title
4. Chapters 5. Introduction
(*a*) 3,2,5,1,4 (*b*) 2,3,4,5,1 (*c*) 5,1,4,2,3 (*d*) 3,2,5,4,1

31. 1. Country 2. Furniture 3. Forest
4. Wood 5. Trees **(S.S.C. 1993)**
(*a*) 1,3,5,4,2 (*b*) 1,4,3,2,5 (*c*) 2,4,3,1,5 (*d*) 5,2,3,1,4

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32. 1. Protect 2. Pressure 3. Relief
4. Rain 5. Flood **(I. Tax & Central Excise, 1994)**
(*a*) 2,4,3,1,5 (*b*) 2,4,5,1,3 (*c*) 2,5,4,1,3 (*d*) 3,2,4,5,1

33. 1. Andhra Pradesh 2. Universe 3. Tirupathi
4. World 5. India **(Asstt. Grade, 1995)**
(*a*) 1,5,3,2,4 (*b*) 2,1,3,5,4 (*c*) 3,1,5,4,2 (*d*) 5,4,2,1,3

34. Arrange the following in a meaningful order : **(C.B.I. 1997)**
Doctor, Fever, Medicine, Medical shop
(*a*) Medicine, Doctor, Medical shop, Fever
(*b*) Doctor, Medical shop, Medicine, Fever
(*c*) Fever, Doctor, Medical shop, Medicine
(*d*) Medical shop, Medicine, Fever, Doctor

35. Which would be the proper order of the following (in ascending order) ?
1. Trillion 2. Thousand 3. Billion
4. Hundred 5. Million **(S.S.C. 1993)**
(*a*) 1,2,4,3,5 (*b*) 1,5,3,2,4 (*c*) 4,2,3,5,1 (*d*) 4,2,5,3,1

36. Which of the following number sequences represents a correct sequence from a part to the whole ?
1. Caste 2. Family 3. Newly married couple
4. Clan 5. Species **(C.B.I. 1993)**
(*a*) 2,3,1,4,5 (*b*) 3,2,1,4,5 (*c*) 3,4,5,1,2 (*d*) 4,5,3,2,1

37. Arrange the following items from general to particular :
1. Animal 2. Feline 3. Leopard
4. Mammal 5. Vertebrate 6. Cat
(*a*) 1,5,4,2,3,6 (*b*) 1,4,3,2,5,6 (*c*) 1,3,5,4,2,6 (*d*) 1,2,3,4,5,6

38. Which number sequence of the following represents a correct sequence from small to big ?
1. Bungalow 2. Flat 3. Cottage
4. House 5. Palace 6. Mansion
(*a*) 3,2,1,4,6,5 (*b*) 3,2,4,1,5,6 (*c*) 3,2,4,1,6,5 (*d*) 5,6,4,1,2,3

ANSWERS

1. (*c*)	**2.** (*a*)	**3.** (*b*)	**4.** (*b*)	**5.** (*d*)	**6.** (*c*)	**7.** (*d*)	**8.** (*b*)	**9.** (*c*)	**10.** (*b*)
11. (*a*)	**12.** (*c*)	**13.** (*c*)	**14.** (*b*)	**15.** (*d*)	**16.** (*b*)	**17.** (*a*)	**18.** (*b*)	**19.** (*d*)	**20.** (*a*)
21. (*b*)	**22.** (*c*)	**23.** (*d*)	**24.** (*b*)	**25.** (*c*)	**26.** (*a*)	**27.** (*c*)	**28.** (*a*)	**29.** (*c*)	**30.** (*d*)
31. (*a*)	**32.** (*b*)	**33.** (*c*)	**34.** (*c*)	**35.** (*d*)	**36.** (*b*)	**37.** (*a*)	**38.** (*c*)		

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14. ARITHMETICAL REASONING

TYPE 1 : CALCULATION-BASED PROBLEMS

Ex. 1. In a chess tournament each of six players will play every other player exactly once. How many matches will be played during the tournament ?

(*a*) 12 (*b*) 15 (*c*) 30 (*d*) 36

(C.B.I. 1995)

Sol. Clearly, we will consider the following matches :

(*i*) matches of first player with other 5 players;

(*ii*) matches of second player with 4 players other than the first player;

(*iii*) matches of third player with 3 players other than the first and second players;

(*iv*) matches of fourth player with 2 players other than the first three players; and

(*v*) match of fifth player with the sixth one.

So, number of matches played during the tournament = 5 + 4 + 3 + 2 + 1 = 15.

Ex. 2. A man has a certain number of small boxes to pack into parcels. If he packs 3, 4, 5 or 6 in a parcel, he is left with one over; if he packs 7 in a parcel, none is left over. What is the number of boxes, he may have to pack ?

(*a*) 106 (*b*) 301 (*c*) 309 (*d*) 400

Sol. Clearly, the required number would be such that it leaves a remainder of 1 when divided by 3, 4, 5 or 6 and no remainder when divided by 7. Such a number is 301.

Hence, the answer is (*b*).

Ex. 3. A, B, C and D play a game of cards. A says to B, "If I give you 8 cards, you will have as many as C has and I shall have 3 less than what C has. Also, if I take 6 cards from C, I shall have twice as many as D has." If B and D together have 50 cards, how many cards has A got ? **(Hotel Management, 1997)**

(*a*) 40 (*b*) 37 (*c*) 27 (*d*) 23

Sol. Clearly, we have :

$B + 8 = C$ ...(*i*) $A - 8 = C - 3$ ...(*ii*)

$A + 6 = 2D$ ...(*iii*) $B + D = 50$ ...(*iv*)

Putting $C = A - 5$ from (*ii*) into (*i*), we have :

$B + 8 = A - 5$ or $A - B = 13$ ...(*v*)

Putting $D = 50 - B$ from (*iv*) into (*iii*), we have :

$A + 6 = 100 - 2B$ or $A + 2B = 94$ ...(*vi*)

Solving (*v*) and (*vi*), we get $B = 27$ and $A = 40$.

$\therefore$ A has 40 cards.

Hence, the answer is (*a*).

Ex. 4. In a group of cows and hens, the number of legs are 14 more than twice the number of heads. The number of cows is

(*a*) 5 (*b*) 7 (*c*) 10 (*d*) 12

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Sol. Let the number of cows be x and the number of hens be y. Then,
number of legs in the group $= 4x + 2y$.
number of heads in the group $= x + y$.
So, $4x + 2y = 2(x + y) + 14$ or $4x + 2y = 2x + 2y + 14$ or $2x = 14$ or $x = 7$.
$\therefore$ Number of cows = 7.
Hence, the answer is (*b*).

Ex. 5. A worker may claim Rs 15 for each km which he travels by taxi and Rs 5 for each km which he drives his own car. If in one week he claimed Rs 500 for travelling 80 km how many kms did he travel by taxi ?
(*a*) 10 (*b*) 20 (*c*) 30 (*d*) 40

Sol. Let the distance covered by taxi be x km.
Then, distance covered by car $= (80 - x)$ km.
$15x + 5(80 - x) = 500$ or $15x + 400 - 5x = 500$ or $10x = 100$ or $x = 10$.
$\therefore$ Distance covered by taxi = 10 km.
Hence, the answer is (*a*).

TYPE 2 : DATA-BASED QUESTIONS

Ex. 6. The following questions are based on the given data for an examination.

(A)	Candidates appeared	10500
(B)	Passed in all the five subjects	5685
(C)	Passed in three subjects only	1498
(D)	Passed in two subjects only	1250
(E)	Passed in one subject only	835
(F)	Failed in English only	78
(G)	Failed in Maths only	275
(H)	Failed in Physics only	149
(I)	Failed in Chemistry only	147
(J)	Failed in Biology only	221

Q. 1. How many candidates failed in all the subjects ?
(*a*) 4815 (*b*) 3317 (*c*) 2867 (*d*) 362

Sol. Clearly, candidates failed in all the subjects
= (Candidates appeared) – (Candidates failed in 1, 2, 3 or 5 subjects + Candidates failed in 1 subject only)
= 10500 – (5685 + 1498 + 1250 + 835 + 78 + 275 + 149 + 147 + 221)
= 10500 – 10138 = 362.
Hence, the answer is (*d*).

Q. 2. How many candidates passed at least in four subjects ?
(*a*) 6555 (*b*) 5685 (*c*) 1705 (*d*) 870

Sol. Candidates passed at least in four subjects
= (Candidates passed in 4 subjects) + (Candidates passed in all 5 subjects)
= (Candidates failed in only 1 subject) + (Candidates passed in all 5 subjects)
= (78 + 275 + 149 + 147 + 221) + 5685
= 870 + 5685 = 6555.
Hence, the answer is (*a*).

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Q. 3. How many candidates failed because of having failed in four or less subjects ?

(*a*) 4815 (*b*) 4453 (*c*) 3618 (*d*) 2368

Sol. Candidates failed in four or less subjects

= (Candidates failed in only 1 subject) + (Candidates failed in only 2 subjects) + (Candidates failed in only 3 subjects) + (Candidates failed in only 4 subjects)

= (Candidates failed in only 1 subject) + (Candidates passed in only 3 subjects) + (Candidates passed in only 2 subjects) + (Candidates passed in only 1 subject)

$= (78 + 275 + 149 + 147 + 221) + 1498 + 1250 + 835$

$= 4453.$

Hence, the answer is (*b*).

TYPE 3 : PROBLEMS ON AGES

Ex. 7. Reena is twice as old as Sunita. Three years ago, she was three times as old as Sunita. How old is Reena now ? **(I. Tax & Central Excise, 1995)**

(*a*) 6 years (*b*) 7 years (*c*) 8 years (*d*) 12 years

Sol. Let Sunita's present age be x years.

Then, Reena's present age = $2x$ years.

Three years ago, Sunita's age = $(x - 3)$ and Reena's age = $(2x - 3)$.

So, $(2x - 3) = 3(x - 3)$ or $2x - 3 = 3x - 9$ or $x = 6$.

$\therefore$ Reena's present age = $2x = 12$ years.

Hence, the answer is (*d*).

Ex. 8. The age of a father is twice that of the elder son. Ten years hence the age of the father will be three times that of the younger son. If the difference of ages of the two sons is 15 years, the age of the father is

(*a*) 50 years (*b*) 55 years (*c*) 60 years (*d*) 70 years

Sol. Let the age of the elder son be x.

Then, age of younger son = $(x - 15)$; and

age of the father = $2x$.

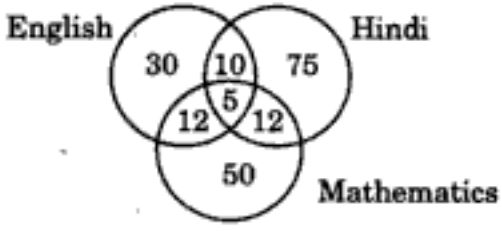
So, $2x + 10 = 3(x - 15 + 10)$ or $2x + 10 = 3x - 15$ or $x = 25$.

$\therefore$ Father's age = $2x = 50$ years.

Hence, the answer is (*a*).

TYPE 4 : VENN-DIAGRAM BASED QUESTIONS

Ex. 9. Consider the diagram given below : **(I.A.S. 1994)**



Five hundred candidates appeared in an examination comprising of tests in English, Hindi and Mathematics. The diagram gives the number of candidates who failed in different tests. What is the percentage of candidates who failed in at least two subjects ?

(*a*) 0.078 (*b*) 1.0 (*c*) 6.8 (*d*) 7.8

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Sol. Clearly, number of candidates who failed in at least two subjects
= number of candidates who failed in two or more subjects
= (10 + 12 + 12 + 5) = 39.

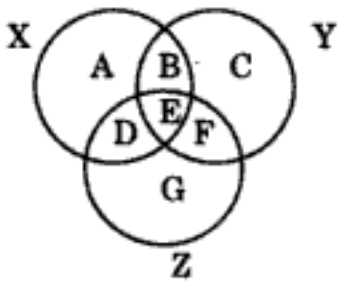
$\therefore$ Required percentage $= \left(\frac{39}{500} \times 100\right)\% = 7.8\%$.

Hence, the answer is (*d*).

Ex. 10. In a group of persons travelling in a bus, 6 persons can speak Tamil, 15 can speak Hindi and 6 can speak Gujarati. In that group, none can speak any other language. If 2 persons in the group can speak two languages and one person can speak all the three languages, then how many persons are there in the group ? **(I.A.S. 1997)**

(*a*) 21 (*b*) 22 (*c*) 23 (*d*) 24

Sol. Let circles X, Y, and Z represent persons who can speak Tamil, Hindi and Gujarati respectively.



Tamil-speaking persons = A + B + D + E = 6 ...(*i*)
Hindi-speaking persons = B + C + E + F = 15 ...(*ii*)
Gujarati-speaking persons = D + E + F + G = 6 ...(*iii*)
Persons speaking 2 languages = B + D + F = 2 ...(*iv*)
Persons speaking all 3 languages = E = 1 ...(*v*)
Clearly, we have : A + B + D = 5 ...(*vi*)
B + C + F = 14 ...(*vii*)
D + F + G = 5 ...(*viii*)
B + D + F = 2 ...(*ix*)
Subtracting (*ix*) from (*vi*), we get :
A − F = 3 ...(*x*)
Adding (*vii*) and (*viii*), we get :
B + C + D + 2F + G = 19 ...(*xi*)
Adding (*x*) and (*xi*), we get :
A + B + C + D + F + G = 22
or A + B + C + D + E + F + G = 23. ($\because$ E = 1)
$\therefore$ Total number of persons = 23.
Hence, the answer is (*c*).

EXERCISE 14

1. A shepherd had 17 sheep. All but nine died. How many was he left with ?

(*a*) Nil (*b*) 8 (*c*) 9 (*d*) 17

(Railways, 1995)

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2. A bird shooter was asked how many birds he had in the bag. He replied that there were all sparrows but six, all pigeons but six, and all docks but six. How many birds had he in all ?

(*a*) 9 (*b*) 18 (*c*) 27 (*d*) 36

3. What is the smallest number of ducks that could swim in this formation — two ducks in front of a duck, two ducks behind a duck and a duck between two ducks ?

(*a*) 3 (*b*) 5 (*c*) 7 (*d*) 9

4. A group of 1200 persons consisting of captains and soldiers is travelling in a train. For every 15 soldiers there is one captain. The number of captains in the group is **(Hotel Management, 1992)**

(*a*) 85 (*b*) 80 (*c*) 75 (*d*) 70

5. Aruna cut a cake into two halves and cuts one half into smaller pieces of equal size. Each of the small pieces is twenty grams in weight. If she has seven pieces of the cake in all with her, how heavy was the original cake ? **(L.I.C. 1994)**

(*a*) 120 grams (*b*) 140 grams (*c*) 240 grams
(*d*) 280 grams (*e*) None of these

6. First bunch of bananas has $\frac{1}{4}$ again as many bananas as a second bunch. If the second bunch has 3 bananas less than the first bunch, then the number of bananas in the first bunch are **(S.C.R.A. 1996)**

(*a*) 9 (*b*) 10 (*c*) 12 (*d*) 15

7. At the end of a business conference the ten people present all shake hands with each other once. How many handshakes will there be altogether ? **(M.B.A. 1997)**

(*a*) 20 (*b*) 45 (*c*) 55 (*d*) 90

8. A student got twice as many sums wrong as he got right. If he attempted 48 sums in all, how many did he solve correctly ? **(M.B.A. 1994)**

(*a*) 12 (*b*) 16 (*c*) 24 (*d*) 18

9. The number of boys in a class is three times the number of girls. Which one of the following numbers cannot represent the total number of children in the class ? **(S.C.R.A. 1993)**

(*a*) 48 (*b*) 44 (*c*) 42 (*d*) 40

10. A placed three sheets with two carbons to get two extra copies of the original. Then he decided to get more carbon copies and folded the paper in such a way that the upper half of the sheets were on top of the lower half. Then he typed. How many carbon copies did he get ?

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

11. A motorist knows four different routes from Bristol to Birmingham. From Birmingham to Sheffield he knows three different routes and from Sheffield to Carlisle he knows two different routes. How many routes does he know from Bristol to Carlisle ?

(*a*) 4 (*b*) 8 (*c*) 12 (*d*) 24

12. In a class, there are 18 boys who are over 160 cm tall. If these constitute three-fourths of the boys and the total number of boys is two-thirds of the total number of students in the class, what is the number of girls in the class ? **(I.A.S. 1992)**

(*a*) 6 (*b*) 12 (*c*) 18 (*d*) 24

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13. A bus starts from city X. The number of women in the bus is half of the number of men. In city Y, 10 men leave the bus and five women enter. Now, number of men and women is equal. In the beginning, how many passengers entered the bus ? **(I. Tax & Central Excise, 1995)**

(*a*) 15 (*b*) 30 (*c*) 36 (*d*) 45

14. In a class, 20% of the members own only two cars each, 40% of the remaining own three cars each and the remaining members own only one car each. Which of the following statements is definitely true from the given statements ?

(*a*) Only 20% of the total members own three cars each.

(*b*) 48% of the total members own only one car each.

(*c*) 60% of the total members own at least two cars each.

(*d*) 80% of the total members own at least one car.

(*e*) None of these **(Bank P.O. 1998)**

15. Between two book-ends in your study are displayed your five favourite puzzle books. If you decide to arrange the five books in every possible combination and moved just one book every minute, how long would it take you ?

(*a*) 1 hour (*b*) 2 hours (*c*) 3 hours (*d*) 4 hours

16. I have a few sweets to be distributed. If I keep 2, 3 or 4 in a pack, I am left with one sweet. If I keep 5 in a pack, I am left with none. What is the minimum number of sweets I can have to pack and distribute ? **(Assistant Grade, 1992)**

(*a*) 25 (*b*) 37 (*c*) 54 (*d*) 65

17. Mr. Johnson was to earn £ 300 and a free holiday for seven weeks' work. He worked for only 4 weeks and earned £ 30 and a free holiday. What was the value of the holiday ?

(*a*) £ 300 (*b*) £ 330 (*c*) £ 360 (*d*) £ 420

18. In a cricket match, five batsmen A, B, C, D and E scored an average of 36 runs. D scored 5 more than E; E scored 8 fewer than A; B scored as many as D and E combined; and B and C scored 107 between them. How many runs did E score ?

(*a*) 62 (*b*) 45 (*c*) 28 (*d*) 20

19. Mac has £ 3 more than Ken, but then Ken wins on the horses and trebles his money, so that he now has £ 2 more than the original amount of money that the two boys had between them. How much money did Mac and Ken have between them before Ken's win ?

(*a*) £ 9 (*b*) £ 11 (*c*) £ 13 (*d*) £ 15

20. Robin says, "If Jai gives me Rs 40, he will have half as much as Atul, but if Atul gives me Rs 40, then the three of us will all have the same amount." What is the total amount of money that Robin, Jai and Atul have between them ?

(*a*) Rs 240 (*b*) Rs 320 (*c*) Rs 360 (*d*) Rs 420

21. A, B, C, D and E play a game of cards. A says to B, "If you give me three cards, you will have as many as E has and if I give you three cards, you will have as many as D has." A and B together have 10 cards more than what D and E together have. If B has two cards more than what C has and the total number of cards be 133, how many cards does B have ? **(Hotel Management, 1995)**

(*a*) 22 (*b*) 23 (*c*) 25 (*d*) 35

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22. A, B, C, D and E play a game of cards. A says to B, "If you give me 3 cards, you will have as many as I have at this moment while if D takes 5 cards from you, he will have as many as E has." A and C together have twice as many cards as E has. B and D together also have the same number of cards as A and C taken together. If together they have 150 cards, how many cards has C got ?

(*a*) 28 (*b*) 29 (*c*) 31 (*d*) 35

(C.A.T. 1997)

23. In a caravan in addition to 50 hens, there are 45 goats and 8 camels with some keepers. If the total number of feet be 224 more than the number of heads in the caravan, the number of keepers is **(Hotel Management, 1995)**

(*a*) 5 (*b*) 8 (*c*) 10 (*d*) 15

24. A certain number of horses and an equal number of men are going somewhere. Half of the owners are on their horses' back while the remaining ones are walking along leading their horses. If the number of legs walking on the ground is 70, how many horses are there ? **(Railways, 1998)**

(*a*) 10 (*b*) 12 (*c*) 14 (*d*) 16

25. At a farm, there are hens, cows and bullocks, and keepers to look after them. There are 69 heads less than legs; the number of cows is double of that of the bullocks; the number of cows and hens is the same and there is one keeper per ten birds and cattle. The total number of hens plus cows and bullocks and their keepers does not exceed 50. How many cows are there ?

(*a*) 10 (*b*) 12 (*c*) 14 (*d*) 16

26. In a certain office, $\frac{1}{3}$ of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ of the married women have children. If $\frac{3}{4}$ of the men are married and $\frac{2}{3}$ of the married men have children, what part of workers are without children ?

(*a*) $\frac{5}{18}$ (*b*) $\frac{4}{9}$ (*c*) $\frac{11}{18}$ (*d*) $\frac{17}{36}$

27. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 75 questions and secures 125 marks, the number of questions he attempts correctly, is

(*a*) 35 (*b*) 40 (*c*) 42 (*d*) 46

Directions (*Questions 28 to 32*) : *The following questions are based on the information given below :* **(Hotel Management, 1997)**

Data on 450 candidates, who took an examination in Social Sciences, Mathematics and Science is given below :

Passed in all the subjects	167
Failed in all the subjects	60
Failed in Social Sciences	175
Failed in Mathematics	199
Failed in Science	191
Passed in Social Sciences only	62
Passed in Mathematics only	48
Passed in Science only	52

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28. How many failed in Social Sciences only ?
(*a*) 15 (*b*) 21 (*c*) 30 (*d*) 42

29. How many failed in one subject only ?
(*a*) 152 (*b*) 144 (*c*) 61 (*d*) 56

30. How many passed in Mathematics and at least one more subject ?
(*a*) 210 (*b*) 203 (*c*) 170 (*d*) 94

31. How many failed in two subjects only ?
(*a*) 56 (*b*) 61 (*c*) 152 (*d*) 162

32. How many passed at least in one subject ?
(*a*) 450 (*b*) 390 (*c*) 304 (*d*) 167

33. A is three times as old as B. C was twice as old as A four years ago. In four years' time, A will be 31. What is the present age of B and C ?
(*a*) 9, 46 (*b*) 9, 50 (*c*) 10, 46 (*d*) 10, 50
(Assistant Grade, 1997)

34. A father tells his son, "I was of your present age when you were born." If the father is 36 now, how old was the boy 5 years back ? **(Railways, 1994)**
(*a*) 13 (*b*) 15 (*c*) 17 (*d*) 20

35. In a family, a couple has a son and a daughter. The age of the father is three times that of his daughter and the age of the son is half of his mother. The wife is 9 years younger to her husband and the brother is seven years older than his sister. What is the age of the mother ? **(I.A.S. 1998)**
(*a*) 40 years (*b*) 45 years (*c*) 50 years (*d*) 60 years

36. A father is now three times as old as his son. Five years back, he was four times as old as his son. The age of the son is **(I.A.S. 1993)**
(*a*) 12 (*b*) 15 (*c*) 18 (*d*) 20

37. Ravi's brother is 3 years senior to him. His father was 28 years of age when his sister was born while his mother was 26 years of age when he was born. If his sister was 4 years of age when his brother was born, what was the age of Ravi's father and mother respectively when his brother was born ?
(*a*) 32 years, 23 years (*b*) 32 years, 29 years
(*c*) 35 years, 29 years (*d*) 35 years, 33 years
(Hotel Management, 1995)

38. When Rahul was born, his father was 32 years older than his brother and his mother was 25 years older than his sister. If Rahul's brother is 6 years older than him and his mother is 3 years younger than his father, how old was Rahul's sister when he was born ? **(Hotel Management, 1997)**
(*a*) 7 years (*b*) 10 years (*c*) 14 years (*d*) 19 years

39. In a town, 65% people watched the news on television, 40% read a newspaper and 25% read a newspaper and watched the news on television also. What percent of the people neither watched the news on television nor read a newspaper ?
(*a*) 5 (*b*) 10 (*c*) 15 (*d*) 20

40. In a group of 15 people, 7 read French, 8 read English while 3 of them read none of these two. How many of them read French and English both ?
(*a*) 0 (*b*) 3 (*c*) 4 (*d*) 5
(I. Tax & Central Excise, 1995)

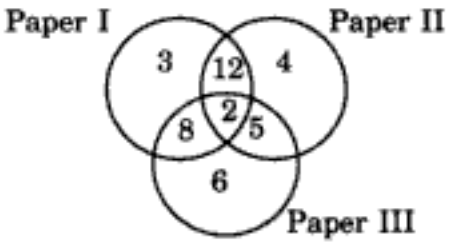
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41. There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. Ten students can speak both English and Hindi. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi and how many can speak only English ? **(I.A.S. 1998)**

(*a*) 39, 29 and 11 respectively (*b*) 37, 27 and 13 respectively

(*c*) 28, 18 and 22 respectively (*d*) 21, 11 and 29 respectively

42. Consider the Venn diagram given below : **(I.A.S. 1993)**



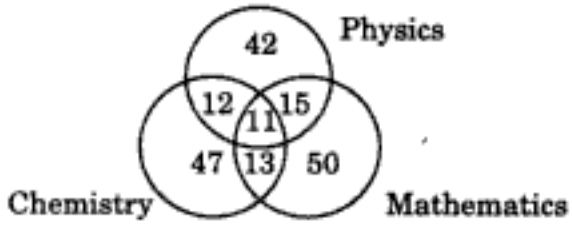
The number in the Venn diagram indicates the number of persons reading the newspapers. The diagram is drawn after surveying 50 persons. In a population of 10,000, how many can be expected to read at least two newspapers ?

(*a*) 5000 (*b*) 5400 (*c*) 6000 (*d*) 6250

43. Out of a total of 120 musicians in a club, 5% can play all the three instruments — guitar, violin and flute. It so happens that the number of musicians who can play any two and only two of the above instruments is 30. The number of musicians who can play the guitar alone is 40. What is the total number of those who can play violin alone or flute alone ? **(I.A.S. 1995)**

(*a*) 30 (*b*) 38 (*c*) 44 (*d*) 45

Directions (*Questions 44 to 46*) : *The diagram given below shows the number of students who got distinction in three subjects out of 500 students. Study the diagram carefully and answer the questions that follow.*



44. What is the percentage of students who got distinction in two subjects ?

(*a*) 8% (*b*) 9% (*c*) 10% (*d*) 12%

45. What is the percentage of students who got distinction ?

(*a*) 28% (*b*) 35% (*c*) 38% (*d*) 40%

46. The percentage of students with distinction marks in Mathematics is

(*a*) 17.8% (*b*) 18.6% (*c*) 19.2% (*d*) 20.6%

Directions (*Questions 47 to 49*) : *Study the information given below and answer the questions that follow :*

A publishing firm publishes newspapers A, B and C. In an effort to persuade advertisers to insert advertisements in these newspapers, the firm sends out the following statement to possible advertisers :

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A survey of representative sample of the whole population shows that —
Newspaper A is read by 26%;
Newspaper B is read by 25%;
Newspaper C is read by 14%;
Newspaper A and B are read by 11%;
Newspaper B and C are read by 10%;
Newspaper C and A are read by 9%;
Newspaper C only is read by 0%.

47. The percentage of readers who read all the three newspapers is
(*a*) 1 (*b*) 4 (*c*) 5 (*d*) 6

48. The percentage of readers who read A and B but not C, is
(*a*) 2 (*b*) 4 (*c*) 5 (*d*) 6

49. The percentage of readers who read at least one of the three newspapers is
(*a*) 40 (*b*) 50 (*c*) 60 (*d*) 65

50. A number of friends decided to go on a picnic and planned to spend Rs 96 on eatables. Four of them, however, did not turn up. As a consequence, the remaining ones had to contribute Rs 4 each extra. The number of those who attended the picnic was
(*a*) 8 (*b*) 12 (*c*) 16 (*d*) 24

ANSWERS

1. (*c*) : 'All but nine died' means 'All except nine died' *i.e.* nine sheep remained alive.

2. (*a*) : 'There were all sparrows but six' means that six birds were not sparrows but only pigeons and docks.
Similarly, number of sparrows + number of docks = 6
and number of sparrows + number of pigeons = 6.
This is possible when there are 3 sparrows, 3 pigeons and 3 docks *i.e.* 9 birds in all.

3. (*a*) : Clearly, the smallest such number is 3.
Three ducks can be arranged as shown along side to satisfy all the three given conditions.

D
D
D

4. (*c*) : Clearly, out of every 16 persons, there is one captain.

So, number of captains $= \frac{1200}{16} = 75$.

5. (*c*) : The seven pieces consist of 6 smaller equal pieces and one half cake piece.
Weight of each small piece = 20 g.
So, total weight of the cake = 2 × (20 × 6) = 240 g.

6. (*d*) : Let the number of bananas in the second bunch be x.

Then, number of bananas in the first bunch $= x + \frac{1}{4}x = \frac{5}{4}x$.

So, $\frac{5}{4}x - x = 3 \Rightarrow 5x - 4x = 12 \Rightarrow x = 12$.

$\therefore$ Number of bananas in first bunch $= \left(\frac{5}{4} \times 12\right) = 15$.

7. (*b*) : Clearly, total number of handshakes $= (9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1) = 45$.

8. (*b*) : Suppose the boy got x sums right and $2x$ sums wrong. Then,

$x + 2x = 48$ or $3x = 48$ or $x = 16$.

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3. **a)** The product rule applies, since the student will perform each of 10 tasks, one after the other. There are 4 ways to do each task. Therefore there are $4 \cdot 4 \cdots 4 = 4^{10} = 1{,}048{,}576$ ways to answer the questions on the test.

b) This is identical to part **(a)**, except that now there are 5 ways to answer each question—give any of the 4 answers or give no answer at all. Therefore there are $5^{10} = 9{,}765{,}625$ ways to answer the questions on the test.

5. The product rule applies here, since a flight is determined by choosing an airline for the flight from New York to Denver (which can be done in 6 ways) and then choosing an airline for the flight from Denver to San Francisco (which can be done in 7 ways). Therefore there are $6 \cdot 7 = 42$ different possibilities for the entire flight.

7. Three-letter initials are determined by specifying the first initial (26 ways), then the second initial (26 ways), and then the third initial (26 ways). Therefore by the product rule there are $26 \cdot 26 \cdot 26 = 26^3 = 17{,}576$ possible three-letter initials.

9. There is only one way to specify the first initial, but as in Exercise 7, there are 26 ways to specify each of the other initials. Therefore there are, by the product rule, $1 \cdot 26 \cdot 26 = 26^2 = 676$ possible three-letter initials beginning with A.

11. A bit string is determined by choosing the bits in the string, one after another, so the product rule applies. We want to count the number of bit strings of length 10, except that we are not free to choose either the first bit or the last bit (they are mandated to be 1's). Therefore there are 8 choices to make, and each choice can be made in 2 ways (the bit can be either a 1 or a 0). Thus the product rule tells us that there are $2^8 = 256$ such strings.

13. This is a trick question, since it is easier than one might expect. Since the string is given to consist entirely of 1's, there is nothing to choose except the length. Since there are $n + 1$ possible lengths not exceeding n (if we include the empty string, of length 0), the answer is simply $n + 1$. Note that the empty string consists—vacuously—entirely of 1's.

15. By the sum rule we can count the number of strings of length 4 or less by counting the number of strings of length i, for $0 \leq i \leq 4$, and then adding the results. Now there are 26 letters to choose from, and a string of length i is specified by choosing its characters, one after another. Therefore, by the product rule there are 26^i strings of length i. The answer to the question is thus $\sum_{i=0}^{4} 26^i = 1 + 26 + 676 + 17576 + 456976 = 475{,}255$.

17. The easiest way to count this is to find the total number of ASCII strings of length five and then subtract off the number of such strings that do not contain the @ character. Since there are 128 characters to choose from in each location in the string, the answer is $128^5 - 127^5 = 34{,}359{,}738{,}368 - 33{,}038{,}369{,}407 = 1{,}321{,}368{,}961$.

19. Recall that an RNA sequence is a sequence of letters, each of which is one of A, C, G, or U. Thus by the product rule there are 4^6 RNA sequences of length six if we impose no restrictions.

a) If U is excluded, then each position can be chosen from among three letters, rather than four. Therefore the answer is $3^6 = 729$.

b) If the last two letters are specified, then we get to choose only four letters, rather than six, so the answer is $4^4 = 256$.

c) If the first letter is specified, then we get to choose only five letters, rather than six, so the answer is $4^5 = 1024$.

d) If only A or U is allowed in each position, then there are just two choices at each of six stages, so the answer is $2^6 = 64$.

21. Because neither 100 nor 50 is divisible by either 7 or 11, whether the ranges are meant to be inclusive or exclusive of their endpoints is moot.

a) There are $\lfloor 100/7 \rfloor = 14$ integers less than 100 that are divisible by 7, and $\lfloor 50/7 \rfloor = 7$ of them are less than 50 as well. This leaves $14 - 7 = 7$ numbers between 50 and 100 that are divisible by 7. They are 56, 63, 70, 77, 84, 91, and 98.

b) There are $\lfloor 100/11 \rfloor = 9$ integers less than 100 that are divisible by 11, and $\lfloor 50/11 \rfloor = 4$ of them are less than 50 as well. This leaves $9 - 4 = 5$ numbers between 50 and 100 that are divisible by 11. They are 55, 66, 77, 88, and 99.

c) A number is divisible by both 7 and 11 if and only if it is divisible by their least common multiple, which is 77. Obviously there is only one such number between 50 and 100, namely 77. We could also work this out as we did in the previous parts: $\lfloor 100/77 \rfloor - \lfloor 50/77 \rfloor = 1 - 0 = 1$. Note also that the intersection of the sets we found in the previous two parts is precisely what we are looking for here.

23. This problem deals with the set of positive integers between 100 and 999, inclusive. Note that there are exactly $999 - 100 + 1 = 900$ such numbers. A second way to see this is to note that to specify a three-digit number, we need to choose the first digit to be nonzero (which can be done in 9 ways) and then the second and third digits (which can each be done in 10 ways), for a total of $9 \cdot 10 \cdot 10 = 900$ ways, by the product rule. A third way to see this (perhaps most relevant for this problem) is to note that a number of the desired form is a number less than or equal to 999 (and there are 999 such numbers) but not less than or equal to 99 (and there are 99 such numbers); therefore there are $999 - 99 = 900$ numbers in the desired range.

a) Every seventh number—7, 14, and so on—is divisible by 7. Therefore the number of positive integers less than or equal to n and divisible by 7 is $\lfloor n/7 \rfloor$ (the floor function is used—we have to round down—because the first six positive integers are not multiples of 7; for example there are only $\lfloor 20/7 \rfloor = 2$ multiples of 7 less than or equal to 20). So we find that there are $\lfloor 999/7 \rfloor = 142$ multiples of 7 not exceeding 999, of which $\lfloor 99/7 \rfloor = 14$ do not exceed 99. Therefore there are exactly $142 - 14 = 128$ numbers in the desired range divisible by 7.

b) This is similar to part **(a)**, with 7 replaced by 2, but with the added twist that we want to count the numbers *not* divisible by 2. Mimicking the analysis in part **(a)**, we see that there are $\lfloor 999/2 \rfloor = 499$ even numbers not exceeding 999, and therefore $999 - 499 = 500$ odd ones; there are similarly $99 - \lfloor 99/2 \rfloor = 50$ odd numbers less than or equal to 99. Therefore there are $500 - 50 = 450$ odd numbers between 100 and 999 inclusive.

c) There are just 9 possible digits that a three-digit number can start with. If all of its digits are to be the same, then there is no choice after the leading digit has been specified. Therefore there are 9 such numbers.

d) This is similar to part **(b)**, except that 2 is replaced by 4. Following the analysis there, we find that there are $999 - \lfloor 999/4 \rfloor = 750$ positive integers less than or equal to 999 not divisible by 4, and $99 - \lfloor 99/4 \rfloor = 75$ such positive integers less than or equal to 99. Therefore there are $750 - 75 = 675$ three-digit integers not divisible by 4.

e) The method is similar to that used in the earlier parts. There are $\lfloor 999/3 \rfloor - \lfloor 99/3 \rfloor = 300$ three-digit numbers divisible by 3, and $\lfloor 999/4 \rfloor - \lfloor 99/4 \rfloor = 225$ three-digit numbers divisible by 4. Moreover there are $\lfloor 999/12 \rfloor - \lfloor 99/12 \rfloor = 75$ numbers divisible by both 3 and 4, i.e., divisible by 12. In order to count each number divisible by 3 or 4 once and only once, we need to add the number of numbers divisible by 3 to the number of numbers divisible by 4, and then subtract the number of numbers divisible by both 3 and 4 so as not to count them twice. Therefore the answer is $300 + 225 - 75 = 450$.

f) In part **(e)** we found that there were 450 three-digit integers that are divisible by either 3 or 4. The

others, of course, are not. Therefore there are $900 - 450 = 450$ three-digit integers that are not divisible by either 3 or 4.

g) We saw in part **(e)** that there are 300 three-digit numbers divisible by 3 and that 75 of them are also divisible by 4. The remainder of those 300 numbers, therefore, are not divisible by 4. Thus the answer is $300 - 75 = 225$.

h) We saw in part **(e)** that there are 75 three-digit numbers divisible by both 3 and 4.

25. This problem involves 1000 possible strings, since there is a choice of 10 digits for each of the three positions in the string.

a) This is most easily done by subtracting from the total number of strings the number of strings that violate the condition. Clearly there are 10 strings that consist of the same digit three times (000, 111, ..., 999). Therefore there are $1000 - 10 = 990$ strings that do not.

b) If we must begin our string with an odd digit, then we have only 5 choices for this digit. We still have 10 choices for each of the remaining digits. Therefore there are $5 \cdot 10 \cdot 10 = 500$ such strings. Alternatively, we note that by symmetry exactly half the strings begin with an odd digit (there being the same number of odd digits as even ones). Therefore half of the 1000 strings, or 500 of them, begin with an odd digit.

c) Here we need to choose the position of the digit that is not a 4 (3 ways) and choose that digit (9 ways). Therefore there are $3 \cdot 9 = 27$ such strings.

27. There are 50 choices to make, each of which can be done in 3 ways, namely by choosing the governor, choosing the senior senator, or choosing the junior senator. By the product rule the answer is therefore $3^{50} \approx 7.2 \times 10^{23}$.

29. By the sum rule we need to add the number of license plates of the first type and the number of license plates of the second type. By the product rule there are $26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 \cdot 10 = 6{,}760{,}000$ license plates consisting of 2 letters followed by 4 digits; and there are $10 \cdot 10 \cdot 26 \cdot 26 \cdot 26 \cdot 26 = 45{,}697{,}600$ license plates consisting of 2 digits followed by 4 letters. Therefore the answer is $6{,}760{,}000 + 45{,}697{,}600 = 52{,}457{,}600$.

31. First let us compute the number of ways to choose the letters. By the sum rule this is the sum of the number of ways to use two letters and the number of ways to use three letters. By the product rule there are 26^2 ways to choose two letters and 26^3 ways to choose three letters. Therefore there are $26^2 + 26^3$ ways to choose the letters. By similar reasoning there are $10^2 + 10^3$ ways to choose the digits. Thus the answer to the question is $(26^2 + 26^3)(10^2 + 10^3) = 18252 \cdot 1100 = 20{,}077{,}200$.

33. We take as known that there are 26 letters including 5 vowels in the English alphabet.

a) There are 8 slots, each of which can be filled with one of the $26 - 5 = 21$ nonvowels (consonants), so by the product rule the answer is $21^8 = 37{,}822{,}859{,}361$.

b) There are 21 choices for the first slot in our string, but only 20 choices for the second slot, 19 for the third, and so on. So the answer is $21 \cdot 20 \cdot 19 \cdot 18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 = 8{,}204{,}716{,}800$.

c) There are 26 choices for each slot except the first, for which there are 5 choices, so the answer is $5 \cdot 26^7 = 40{,}159{,}050{,}880$.

d) This is similar to **(b)**, except that there are only five choices in the first slot, and we are free to choose from all the letters not used so far, rather than just the consonants. Thus the answer is $5 \cdot 25 \cdot 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20 \cdot 19 = 12{,}113{,}640{,}000$.

e) We subtract from the total number of strings (26^8) the number that do not contain at least one vowel (21^8, the answer to **(a)**), obtaining the answer $26^8 - 21^8 = 208{,}827{,}064{,}576 - 37{,}822{,}859{,}361 = 171{,}004{,}205{,}215$.

f) The best way to do this is first to decide where the vowel goes (8 choices), then to decide what the vowel is to be (A, E, I, O, or U—5 choices), and then to fill the remaining slots with any consonants (21^7 choices, since one slot has already been filled). Therefore the answer is $8 \cdot 5 \cdot 21^7 = 72{,}043{,}541{,}640$.

g) We can ignore the first slot, since there is no choice. Now the problem is almost identical to **(e)**, except that there are only 7 slots to fill. So the answer is $26^7 - 21^7 = 8{,}031{,}810{,}176 - 1{,}801{,}088{,}541 = 6{,}230{,}721{,}635$.

h) The problem is almost identical to **(g)**, except that there are only 6 slots to fill. So the answer is $26^6 - 21^6 = 308{,}915{,}776 - 85{,}766{,}121 = 223{,}149{,}655$.

35. For each part of this problem, we need to find the number of one-to-one functions from a set with 5 elements to a set with k elements. To specify such a function, we need to make 5 choices, in succession, namely the values of the function at each of the 5 elements in its domain. Therefore the product rule applies. The first choice can be made in k ways, since any element of the codomain can be the image of the first element of the domain. After that choice has been made, there are only $k-1$ elements of the codomain available to be the image of the second element of the domain, since images must be distinct for the function to be one-to-one. Similarly, for the third element of the domain, there are $k-2$ possible choices for a function value. Continuing in this way, and applying the product rule, we see that there are $k(k-1)(k-2)(k-3)(k-4)$ one-to-one functions from a set with 5 elements to a set with k elements.

a) By the analysis above the answer is $4 \cdot 3 \cdot 2 \cdot 1 \cdot 0 = 0$, what we would expect since there are no one-to-one functions from a set to a strictly smaller set.

b) By the analysis above the answer is $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.

c) By the analysis above the answer is $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$.

d) By the analysis above the answer is $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 2520$.

37. a) There can clearly be no one-to-one function from $\{1, 2, \ldots, n\}$ to $\{0, 1\}$ if $n > 2$. If $n = 1$, then there are 2 such functions, the one that sends 1 to 0, and the one that sends 1 to 1. If $n = 2$, then there are again 2 such functions, since once it is determined where 1 is sent, the function must send 2 to the other value in the codomain.

b) If the function assigns 0 to both 1 and n, then there are $n-2$ function values free to be chosen. Each can be chosen in 2 ways. Therefore, by the product rule (since we have to choose values for all the elements of the domain) there are 2^{n-2} such functions, as long as $n > 1$. If $n = 1$, then clearly there is just one such function.

c) If $n = 1$, then there are no such functions, since there are no positive integers less than n. So assume $n > 1$. In order to specify such a function, we have to decide which of the numbers from 1 to $n-1$, inclusive, will get sent to 1. There are $n-1$ ways to make this choice. There is no choice for the remaining numbers from 1 to $n-1$, inclusive, since they all must get sent to 0. Finally, we are free to specify the value of the function at n, and this may be done in 2 ways. Hence, by the product rule the final answer is $2(n-1)$.

39. The easiest way to view a partial function in terms of counting is to add an additional element to the codomain of the function—let's call it u for "undefined"— and then imagine that the function assigns a value to *all* elements of the domain. If the original function f had previously been undefined at x, we now say that $f(x) = u$. Thus all we have done is to increase the size of the codomain from n elements to $n+1$ elements. By Example 6 we conclude that there are $(n+1)^m$ such partial functions.

41. The trick here is to realize that a palindrome of length n is completely determined by its first $\lceil n/2 \rceil$ bits. This is true because once these bits are specified, the remaining bits, read from right to left, must be identical to the first $\lfloor n/2 \rfloor$ bits, read from left to right. Furthermore, these first $\lceil n/2 \rceil$ bits can be specified at will, and by the product rule there are $2^{\lceil n/2 \rceil}$ ways to do so.

43. Recall that an RNA sequence is a sequence of letters, each of which is one of A, C, G, or U. Thus by the product rule there are 4^4 RNA sequences of length four if we impose no restrictions.

a) There are 3^4 RNA sequences that use only letters other than U, so the answer is $4^4 - 3^4 = 175$.

b) We "contain the sequence" to mean that those letters must be consecutive. We compute the number of sequences that *do* contain CUG. There are two cases: these are the first three bases, or these are the last three. In either case, there are four ways to fill in the remaining base, so there are $4 + 4 = 8$ such sequences. Therefore there are $4^4 - 8 = 248$ sequences that do not.

c) We first count the number of sequences that *do* contain all four bases. All that matters is the order of the bases. There are four choices for which base to put first, three choices for which base to put second, then two choices for the third position, and one choice for the last position. By the product rule, the answer to this subproblem is $4 \cdot 3 \cdot 2 \cdot 1 = 24$. Therefore the answer to the given question is $4^4 - 24 = 232$.

d) We need to choose two the two bases to be included, and this can be done in six ways (AU, AC, AG, UC, UG, CG). Once we make this choice (say AU), we can count the sequences that use those two bases. There are $2^4 = 16$ ways to choose which base goes at which position (two choices at each of the four positions), but two of these in fact use only one base, so there are 14 sequences that use exactly those two bases. Therefore the answer is $6 \cdot 14 = 84$.

45. We can use the division rule here; see Example 20. If we imagine the places at the table as fixed, then there are $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 6!$ ways to seat the six people—choose one of the six people for seat #1, choose one of the five remaining people for seat #2, and so on. However, this overcounts by a factor of 6, corresponding to rotating the table (which does not change a person's neighbors), and by a factor of 2, because any given arrangement is the "same" under the stated rules as the arrangement that reverses the order of the people (changing clockwise to counterclockwise). Therefore the answer is $6!/(6 \cdot 2) = 60$. An alternative way to look at it is as follows. Without loss of generality, have person A take a seat. Choose his two neighbors in one of $5 \cdot 4/2 = 10$ ways and have them sit on either side of him. Now place the other three people in one of the 3! ways possible in the remaining seats. The number of ways, then, is $10 \cdot 6 = 60$.

47. **a)** Here is a good way (but certainly not the only way) to approach this problem. Since the bride and groom must stand next to each other, let us treat them as one unit. Then the question asks for the number of ways to arrange five units in a row (the bride-and-groom unit and the four other people). We can think of filling five positions one at a time, so by the product rule there are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ ways to make these choices. This is not quite the answer, however, since there are also two ways to decide on which side of the groom the bride will stand. Therefore the final answer is $120 \cdot 2 = 240$.

b) There are clearly $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$ arrangements in all. We just determined in part **(a)** that 240 of them involve the bride standing next to the groom. Therefore there are $720 - 240 = 480$ ways to arrange the people with the bride not standing next to the groom.

c) Of the 720 arrangements of these people (see part **(b)**), exactly half must have the bride somewhere to the left of the groom. (We are invoking **symmetry** here—a useful tool for solving some combinatorial problems.) Therefore the answer is $720/2 = 360$.

49. There are 2^7 bit strings of length 10 that begin with three 0's, since each of the remaining seven bits has two possible values. Similarly, there are 2^8 bit strings of length 10 that end with two 0's. Furthermore, there are 2^5 bit strings of length 10 that both begin with three 0's and end with two 0's, since each of the five "middle" bits can be specified in two ways. Using the principle of inclusion–exclusion, we conclude that there are $2^7 + 2^8 - 2^5 = 352$ such strings. The idea behind this principle here is that the strings that both begin with three 0's and end with two 0's were counted twice when we added 2^7 and 2^8, so we need to subtract for the overcounting. It is definitely *not* the case that we are subtracting because we do not want to count such strings at all.

51. First let us count the number of 8-bit strings that contain three consecutive 0's. We will organize our count by looking at the leftmost bit that contains a 0 followed by at least two more 0's. If this is the first bit,

then the second and third bits are determined (namely, they are both 0), but bits 4 through 8 are free to be specified, so there are $2^5 = 32$ such strings. If it is the second, third, or fourth bits, then the bit preceding it must be a 1 and the two bits after it must be 0's, but the other four bits are free. Therefore there are $2^4 = 16$ such strings in each of these three cases, or 48 in all. Next let us suppose that the substring of three or more 0's starts at bit 5. Then bit 4 must be a 1. Bits 1 through 3 can be anything other than three 0's (if it were three 0's, then we already counted this string); thus there are $2^3 - 1 = 7$ ways to specify them. Bit 8 is free. Therefore there are $7 \cdot 2 = 14$ such strings. Finally, suppose that the substring 000 starts in bit 6, so that bit 5 is a 1. There are $2^4 = 16$ possibilities for the first four bits, but three of them contain three consecutive 0's (0000, 0001, and 1000); therefore there are only 13 such strings. Adding up all the cases we have discussed, we obtain the final answer of $32 + 48 + 14 + 13 = 107$ for the number of 8-bit strings that contain three consecutive 0's.

Next we need to compute the number of 8-bit strings that contain four consecutive 1's. The analysis is very similar to what we have just done. There are $2^4 = 16$ such strings starting with 1111; there are $2^3 = 8$ starting 01111; there are $2^3 = 8$ starting x01111 (where x is either 0 or 1); and there are similarly 8 starting each of xy01111 and xyz01111. This gives us a total of $16 + 4 \cdot 8 = 48$ strings containing four consecutive 1's.

Finally, we need to count the number of strings that contain both three consecutive 0's and four consecutive 1's. It is easy enough to just list them all: 00001111, 00011111, 00011110, 10001111, 11110000, 11111000, 01111000, and 11110001, eight in all. Now applying the principle of inclusion–exclusion to what we have calculated above, we obtain the answer to the entire problem: $107 + 48 - 8 = 147$. There are only 256 bit strings of length eight altogether, so this answer is somewhat surprising, in that more than half of them satisfy the stated condition.

53. We can solve this problem by computing the number of positive integers not exceeding 100 that are divisible by 4, the number of them divisible by 6, and the number divisible by both 4 and 6; and then applying the principle of inclusion–exclusion. There are clearly $100/4 = 25$ multiples of 4 in this range, since every fourth number is divisible by 4. The number of integers in this range divisible by 6 is $\lfloor 100/6 \rfloor = 16$; we needed to round down because the multiples of 6 occur at the end of each consecutive block of 6 integers (6, 12, 18, etc.). Furthermore a number is divisible by both 4 and 6 if and only if it is divisible by their least common multiple, namely 12. Therefore there are $\lfloor 100/12 \rfloor = 8$ numbers divisible by both 4 and 6 in this range. Finally, applying inclusion–exclusion, we see that the number of positive integers not exceeding 100 that are divisible either by 4 or by 6 is $25 + 16 - 8 = 33$.

55. **a)** We are told that there are $26 + 26 + 10 + 6 = 68$ available characters. A password of length k using these characters can be formed in 68^k ways. Therefore the number of passwords with the specified length restriction is $68^8 + 68^9 + 68^{10} + 68^{11} + 68^{12} = 9{,}920{,}671{,}339{,}261{,}325{,}541{,}376$, which is about 9.9×10^{21} or about ten sextillion.

b) For a password *not* to contain one of the special characters, it must be constructed from the other 62 characters. There are $62^8 + 62^9 + 62^{10} + 62^{11} + 62^{12} = 3{,}279{,}156{,}377{,}874{,}257{,}103{,}616$ of these. Thus there are $6{,}641{,}514{,}961{,}387{,}068{,}437{,}760 \approx 6.6 \times 10^{21}$ (about seven sextillion) passwords that do contain at least one occurrence of one of the special symbols.

c) Assuming no restrictions, it will take one nanosecond (one billionth of a second, or 10^{-9} sec) for each password. We just multiply this by our answer in part **(a)** to find the number of seconds the hacker will require. We can convert this to years by dividing by $60 \cdot 60 \cdot 24 \cdot 365.2425$ (the average number of seconds in a year). It will take about 314,374 years.

57. Suppose the name has length k, where $1 \leq k \leq 65535$. There are $26 + 26 + 1 + 1 + 10 = 64$ choices for each character, except that the first character cannot be a digit, so there are only 54 choices for the first character.

By the product rule there are $54 \cdot 64^{k-1}$ choices for such a string. To get the final answer, we need to sum this over all lengths, for a total of

$$\sum_{k=1}^{65535} 54 \cdot 64^{k-1},$$

Applying the formula for the sum of a geometric series gives $54(64^{65536}-1)/63$, which is about 3.5×10^{118369}. A programmer will never run out of variable names!

59. By the result of Example 8, there are $C = 6{,}400{,}000{,}000$ possible numbers of the form *NXX-NXX-XXXX*. To determine the number of different telephone numbers worldwide, then, we need to determine how many country codes there are and multiply by C. There are clearly 10 country codes of length 1, 100 country codes of length 2, and 1000 country codes of length 3. Thus there are $10 + 100 + 1000 = 1110$ country codes in all, by the sum rule. Our final answer is $1110 \cdot C = 7{,}104{,}000{,}000{,}000$.

61. There are 16 hexadecimal digits. We just apply the sum rule and the product rule to obtain the answer $16^{10} + 16^{26} + 16^{58} \approx 6.9 \times 10^{69}$. Almost all of this comes from the last term (16^{58}).

63. Every fourth number is divisible by 4, and every sixth number is divisible by 6. Therefore $\lfloor 999{,}999/4 \rfloor = 249{,}999$ numbers in the given range are divisible by 4, and $\lfloor 999{,}999/6 \rfloor = 166{,}666$ numbers in the given range are divisible by 6. We want to exclude these from the 999,999 numbers we have. However, numbers can be divisible by both 4 and 6; namely those divisible by 12, the least common multiple of 4 and 6. There are $\lfloor 999{,}999/12 \rfloor = 83{,}333$ such numbers. Using the principle of inclusion–exclusion, we must add this number back in to avoid overcounting. Therefore our answer is $999{,}999 - 249{,}999 - 166{,}000 + 83{,}000 = 666{,}667$.

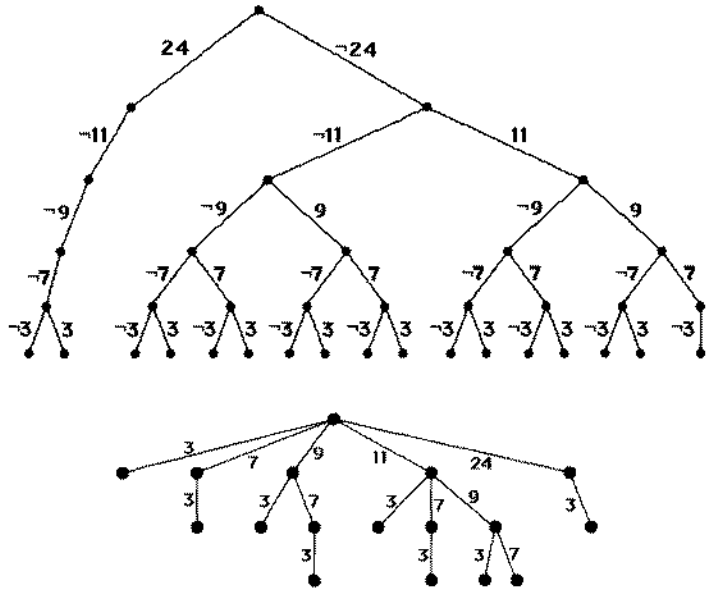
65. We assume that what is intended is that each of the 4 letters is to be used exactly once. There are at least two ways to do this problem. First let us break it into two cases, depending on whether the a comes at the end of the string or not. If a is not at the end, then there are 3 places to put it. After we have placed the a, there are only 2 places to put the b, since it cannot go into the position occupied by the a and it cannot go into the position following the a. Then there are 2 positions in which the c can go, and only 1 position for the d. Therefore, there are by the product rule $3 \cdot 2 \cdot 2 \cdot 1 = 12$ allowable strings in which the a does not come last. Second, if the a comes last, then there are $3 \cdot 2 \cdot 1 = 6$ ways to arrange the letters b, c, and d in the first three positions. The answer, by the sum rule, is therefore $12 + 6 = 18$.

Here is another approach. Ignore for a moment the restriction that a b cannot follow an a. Then we need to choose the letter that comes first (which can be done in 4 ways), then the letter that comes second (which can be done in 3 ways, since one letter has already been used), then the letter that comes third (which can be done in 2 ways, since two of the letters have already been used), and finally the letter that comes last (which can only be done in 1 way, since there is only one unused letter at that point). Therefore there are, by the product rule, $4 \cdot 3 \cdot 2 \cdot 1 = 24$ such strings. Now we need to subtract from this total the number of strings in which the a is immediately followed by the b. To count these, let us imagine the a and b glued together into one superletter, ab. (This **gluing** technique often comes in handy.) Now there are 3 things to arrange. We can choose any of them (the letters c or d or the superletter ab) to come first, and that can be done in 3 ways. We can choose either of the other two to come second (which can be done in 2 ways), and we are forced to choose the remaining one to come third. By the product rule there are $3 \cdot 2 \cdot 1 = 6$ ways to make these choices. Therefore our final answer is $24 - 6 = 18$.

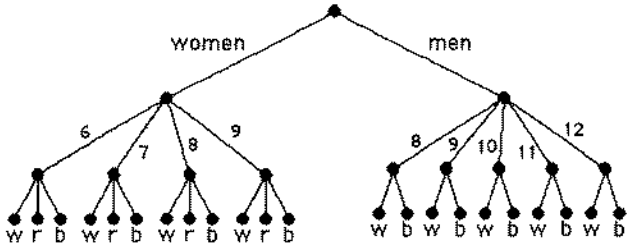
67. There are at least two approaches that are effective here. In our first tree, we let each branching point represent a decision as to whether to include the next element in the set (starting with the largest element). At the top of the tree, for example, we can either choose to include 24 or to exclude it (denoted $\neg 24$). We branch one

way for each possibility. In the first figure below, the entire subtree to the right represents those sets that do not include 24, and the subtree to the left represents those that do. At the point below and to the left of the 24, we have only one branch, -11, since after we have included 24 in our set, we cannot include 11, because the sum would not be less than 28 if we did. At the point below and to the right of the -24, however, we again branch twice, since we can choose either to include 11 or to exclude it. To answer the question, we look at the points in the last row of the tree. Each represents a set whose sum is less than 28. For example, the sixth point from the right represents the set $\{11, 3\}$. Since there are 17 such points, the answer to the problem is 17.

Our other solution is more compact. In the tree below we show branches from a point only for the inclusion of new numbers in the set. The set formed by including no more numbers is represented by the point itself. This time every point represents a set. For example, the point at the top represents the empty set, the point below and to the right of the number 11 represents the set $\{11\}$, and the left-most point on the bottom row represents the set $\{3, 7, 9\}$. In general the set that a point represents is the set of numbers found on the path up the tree from that point to the root of the tree (the point at the top). We only included branches when the sum would be less than 28. Since there are 17 points altogether in this figure, the answer to the problem is 17.



69. **a)** The tree shown here enumerates the possible outcomes. First we branch on gender, then on size, and finally on color. There are 22 ends, so the answer to the question is 22.



b) First we apply the sum rule: the number of shoes is the sum of the number of men's shoes and the number of women's shoes. Next we apply the product rule. For a woman's shoe we need to specify size (4 choices) and then for each choice of size, we need to specify color (3 choices). Therefore there are $4 \cdot 3 = 12$ possible women's models. Similarly, there are $5 \cdot 2 = 10$ men's models. Therefore the answer is $12 + 10 = 22$.

71. We want to prove $P(m)$, the sum rule for m tasks, which says that if tasks T_1, T_2, $\ldots$, T_m can be done in n_1, n_2, $\ldots$, n_m ways, respectively, and no two of them can be done at the same time, then there are $n_1 + n_2 + \cdots + n_m$ ways to do one of the tasks. The basis step of our proof by mathematical induction is $m = 2$, and that has already been given. Now assume that $P(m)$ is true, and we want to prove $P(m+1)$. There are $m+1$ tasks, no two of which can be done at the same time; we want to do one of them. Either we choose one from among the first m, or we choose the task T_{m+1}. By the sum rule for two tasks, the number of ways we can do this is $n + n_{m+1}$, where n is the number of ways we can do one of the tasks among the first m. But by the inductive hypothesis $n = n_1 + n_2 + \cdots + n_m$. Therefore the number of ways we can do one of the $m+1$ tasks is $n_1 + n_2 + \cdots + n_m + n_{m+1}$, as desired.

73. A diagonal joins two vertices of the polygon, but they must be vertices that are not already joined by a side of the polygon. Thus there are $n-3$ diagonals emanating from each vertex of the polygon (we've excluded two of the $n-1$ other vertices as possible targets for diagonals). If we multiply $n-3$ by n, the number of vertices, we will have counted each diagonal exactly twice—once for each endpoint. We compensate for this overcounting by dividing by 2. Therefore there are $n(n-3)/2$ diagonals. (Note that the convexity of the polygon had nothing to do with the problem—we were counting the diagonals, whether or not we could be sure that they all lay inside the polygon.)

SECTION 6.2 The Pigeonhole Principle

The pigeonhole principle seems so trivial that it is difficult to realize how powerful it is in solving some mathematical problems. As usual with combinatorial problems, the trick is to look at things the right way, which usually means coming up with the clever insight, perhaps after hours of agonizing and frustrating exploration with a problem.

Try to solve these problems by invoking the pigeonhole principle explicitly, even if you can see other ways of doing them; you will gain some insights by formulating the problem and your solution in this way. The trick, of course, is to figure out what should be the pigeons and what should be the pigeonholes. Many of the the hints of Section 6.1 apply here, as well as general problem-solving techniques, especially the willingness to play with a problem for a long time before giving up.

Many of the elegant applications are quite subtle and difficult, and there are even more subtle and difficult applications not touched on here. Not every problem, of course, fits into the model of one of the examples in the text. In particular, Exercise 42 looks deceptively like a problem amenable to the technique discussed in Example 10. Keep in mind that the process of grappling with problems such as these is worthwhile and educational in itself, even if you never find the solution.

1. There are six classes: these are the pigeons. There are five days on which classes may meet (Monday through Friday): these are the pigeonholes. Each class must meet on a day (each pigeon must occupy a pigeonhole). By the pigeonhole principle at least one day must contain at least two classes.

3. **a)** There are two colors: these are the pigeonholes. We want to know the least number of pigeons needed to insure that at least one of the pigeonholes contains two pigeons. By the pigeonhole principle the answer is 3. If three socks are taken from the drawer, at least two must have the same color. On the other hand two socks are not enough, because one might be brown and the other black. Note that the number of socks was irrelevant (assuming that it was at least 3).

b) He needs to take out 14 socks in order to insure at least two black socks. If he does so, then at most 12 of them are brown, so at least two are black. On the other hand, if he removes 13 or fewer socks, then 12 of them could be brown, and he might not get his pair of black socks. This time the number of socks did matter.

5. There are four possible remainders when an integer is divided by 4 (these are the pigeonholes here): 0, 1, 2, or 3. Therefore, by the pigeonhole principle at least two of the five given remainders (these are the pigeons) must the same.

7. Let the n consecutive integers be denoted $x+1$, $x+2$, ..., $x+n$, where x is some integer. We want to show that exactly one of these is divisible by n. There are n possible remainders when an integer is divided by n, namely 0, 1, 2, ..., $n-1$. There are two possibilities for the remainders of our collection of n numbers: either they cover all the possible remainders (in which case exactly one of our numbers has a remainder of 0 and is therefore divisible by n), or they do not. If they do not, then by the pigeonhole principle, since there are then fewer than n pigeonholes (remainders) for n pigeons (the numbers in our collection), at least one remainder must occur twice. In other words, it must be the case that $x+i$ and $x+j$ have the same remainder when divided by n for some pair of numbers i and j with $0 < i < j \leq n$. Since $x+i$ and $x+j$ have the same remainder when divided by n, if we subtract $x+i$ from $x+j$, then we will get a number divisible by n. This means that $j-i$ is divisible by n. But this is impossible, since $j-i$ is a positive integer strictly less than n. Therefore the first possibility must hold, that exactly one of the numbers in our collection is divisible by n.

9. The generalized pigeonhole principle applies here. The pigeons are the students (no slur intended), and the pigeonholes are the states, 50 in number. By the generalized pigeonhole principle if we want there to be at least 100 pigeons in at least one of the pigeonholes, then we need to have a total of N pigeons so that $\lceil N/50 \rceil \geq 100$. This will be the case as long as $N \geq 99 \cdot 50 + 1 = 4951$. Therefore we need at least 4951 students to guarantee that at least 100 come from a single state.

11. We must recall from analytic geometry that the midpoint of the segment whose endpoints are (a, b, c) and (d, e, f) is $((a+d)/2, (b+e)/2, (c+f)/2)$. We are concerned only with integer values of the original coordinates. Clearly the coordinates of these fractions will be integers as well if and only if a and d have the same parity (both odd or both even), b and e have the same parity, and c and f have the same parity. Thus what matters in this problem is the parities of the coordinates. There are eight possible triples of parities: (odd, odd, odd), (odd, odd, even), (odd, even, odd), ..., (even, even, even). Since we are given nine points, the pigeonhole principle guarantees that at least two of them will have the same triple of parities. The midpoint of the segment joining these two points will therefore have integer coordinates.

13. **a)** We can group the first eight positive integers into four subsets of two integers each, each subset adding up to 9: $\{1, 8\}$, $\{2, 7\}$, $\{3, 6\}$, and $\{4, 5\}$. If we select five integers from this set, then by the pigeonhole principle (at least) two of them must come from the same subset. These two integers have a sum of 9, as desired.

b) No. If we select one element from each of the subsets specified in part **(a)**, then no sum will be 9. For example, we can select 1, 2, 3, and 4.

15. We can apply the pigeonhole principle by grouping the numbers cleverly into pairs (subsets) that add up to 7, namely $\{1, 6\}$, $\{2, 5\}$, and $\{3, 4\}$. If we select four numbers from the set $\{1, 2, 3, 4, 5, 6\}$, then at least two of them must fall within the same subset, since there are only three subsets. Two numbers in the same subset are the desired pair that add up to 7. We also need to point out that choosing three numbers is not enough, since we could choose $\{1, 2, 3\}$, and no pair of them add up to more than 5.

17. The given information tells us that there are $50 \cdot 85 \cdot 5 = 21{,}250$ bins. If we had this many products, then each could be stored in a separate bin. By the pigeonhole principle, however, if there are at least 21,251 products, then at least two of them must be stored in the same bin. This is the number the problem is asking us for.

19. **a)** If this statement were not true, then there would be at most 8 from each class standing, for a total of at most 24 students. This contradicts the fact that there are 25 students in the class.

b) If this statement were not true, then there would be at most 2 freshmen, at most 18 sophomores, and at most 4 juniors, for a total of at most 24 students. This contradicts the fact that there are 25 students in the class.

21. One way to do this is to have the sequence contain four groups of four numbers each, so that the numbers within each group are decreasing, and so that the numbers between groups are increasing. For example, we could take the sequence to be 4, 3, 2, 1; 8, 7, 6, 5; 12, 11, 10, 9; 16, 15, 14, 13. There can be no increasing subsequence of five terms, because any increasing subsequence can have only one element from each of the four groups. There can be no decreasing subsequence of five terms, because any decreasing subsequence cannot have elements from more than one group.

23. The key here is that 25 is an odd number. If there were an even number of boys and the same even number of girls, then we could position them around the table in the order BBGGBBGG..., and never have a person both of whose neighbors are boys. But with an odd number of each sex, that cannot happen. Here is one nice way to see why when there are 25 of each. Number the seats around the table from 1 to 50, and think of seat 50 as being adjacent to seat 1. There are 25 seats with odd numbers and 25 seats with even numbers. If no more than 12 boys occupied the odd-numbered seats, then at least 13 boys would occupy the even-numbered seats, and vice versa. Without loss of generality, assume that at least 13 boys occupy the 25 odd-numbered seats. Then at least two of those boys must be in consecutive odd-numbered seats. The person sitting between those two boys will have boys as both of his or her neighbors.

25. This is actually a fairly hard problem, in terms of what we need to keep track of. Call the given sequence $a_1, a_2, \ldots, a_n$. We will keep track of the lengths of long increasing or decreasing subsequences by assigning values i_k and d_k for each k from 1 to n, indicating the length of the longest increasing subsequence ending with a_k and the length of the longest decreasing subsequence ending with a_k, respectively. Thus $i_1 = d_1 = 1$, since a_1 is both an increasing and a decreasing subsequence of length 1. If $a_2 < a_1$, then $i_2 = 1$ and $d_2 = 2$, since the longest increasing subsequence ending at a_2 is just a_2, but the longest decreasing subsequence ending at a_2 is a_1, a_2. If $a_2 > a_1$, then it is the other way around: $i_2 = 2$ and $d_2 = 1$. In general, we can determine i_k in the following manner (the determination of d_k is similar, with the roles of i and d, and the roles of $<$ and $>$, reversed). We look at the numbers $a_1, a_2, \ldots, a_{k-1}$. For each a_j that is less than a_k, we know that the value of i_k is at least $i_j + 1$, since the increasing subsequence of length i_j ending at a_j can be extended by following it by a_k, resulting in an increasing subsequence of length $i_j + 1$, ending at a_k. Furthermore, there is no other way of producing an increasing subsequence ending at a_k, other than the subsequence of length 1. Thus we set i_k equal to either 1 or the maximum of the numbers $i_j + 1$ for those values of $j < k$ for which $a_j < a_k$. Finally, once we have determined all the values i_k and d_k, we choose the largest of these $2n$ numbers as our answer.

The procedure just described, however, does not keep track of what the longest subsequence is, so we need to use two more sets of variables, $iprev_k$ and $dprev_k$. These will point back to the terms in the sequence that caused the values of i_k and d_k to be what they are. To retrieve the longest increasing or decreasing subsequence, once we know which type it is and where it ends, we follow these pointers, thereby exhibiting the subsequence backwards. We will not write the pseudocode for this final phase of the algorithm. (See the answer in the back of the textbook for an alternative procedure, which explicitly computes the sequence.)

```
procedure long subsequence(a_1, a_2, ..., a_n : distinct integers)
for k := 1 to n
        i_k := 1; d_k := 1
        iprev_k := k; dprev_k := k
        for j := 1 to k − 1
                if a_j < a_k and i_j + 1 > i_k then
                        i_k := i_j + 1
                        iprev_k := j
                if a_j > a_k and d_j + 1 > d_k then
                        d_k := d_j + 1
                        dprev_k := j
{at this point correct values of i_k and d_k have all been assigned}
longest := 1
for k := 2 to n
        if i_k > longest then longest := i_k
        if d_k > longest then longest := d_k
{ longest is the length of the longest increasing or decreasing subsequence}
```

27. We can prove these statements using both the result and the method of Example 13. First note that the role of "mutual friend" and "mutual enemy" is symmetric, so it is really enough to prove one of these statements; the other will follow by interchanging the roles. So let us prove that in every group of 10 people, either there are 3 mutual friends or 4 mutual enemies. Consider one person; call this person A. Of the 9 other people, either there must be 6 enemies of A, or there must be 4 friends of A (if there were 5 or fewer enemies and 3 or fewer friends, that would only account for 8 people). We need to consider the two cases separately. First suppose that A has 6 enemies. Apply the result of Example 13 to these 6 people: among them either there are 3 mutual friends or there are 3 mutual enemies. If there are 3 mutual friends, then we are done. If there are 3 mutual enemies, then these 3 people, together with A, form a group of 4 mutual enemies, and again we are done. That finishes the first case. The second case was that A had 4 friends. If some pair of these people are friends, then they, together with A, form the desired group of 3 mutual friends. Otherwise, these 4 people are the desired group of 4 mutual enemies. Thus in either case we have found either 3 mutual friends or 4 mutual enemies.

29. We need to show two things: that if we have a group of n people, then among them we must find either a pair of friends or a subset of n of them all of whom are mutual enemies; and that there exists a group of $n-1$ people for which this is not possible. For the first statement, if there is any pair of friends, then the condition is satisfied, and if not, then every pair of people are enemies, so the second condition is satisfied. For the second statement, if we have a group of $n-1$ people all of whom are enemies of each other, then there is neither a pair of friends nor a subset of n of them all of whom are mutual enemies.

31. First we need to figure out how many distinct combinations of initials and birthdays there are. The product rule tells us that since there are 26 ways to choose each of the 3 initials and 366 ways to choose the birthday, there are $26 \cdot 26 \cdot 26 \cdot 366 = 6{,}432{,}816$ such combinations. By the generalized pigeonhole principle, with these combinations as the pigeonholes and the 37 million people as the pigeons, there must be at least $\lceil 37{,}000{,}000/6{,}432{,}816 \rceil = 6$ people with the same combination.

33. The numbers from 1 to 200,000 are the pigeonholes, and the inhabitants of Paris are the pigeons, which number at least 800,001. Therefore by Theorem 1 there are at least two Parisians with the same number of hairs on their heads; and by Theorem 2 there are at least $\lceil 800{,}001/200{,}000 \rceil = 5$ Parisians with the same number of hairs on their heads.

35. The 38 time periods are the pigeonholes, and the 677 classes are the pigeons. By the generalized pigeonhole principle there is some time period in which at least $\lceil 677/38 \rceil = 18$ classes are meeting. Since each class must meet in a different room, we need 18 rooms.

37. Let c_i be the number of computers that the ith computer is connected to. Each of these integers is between 0 and 5, inclusive. The pigeonhole principle does not allow us to conclude immediately that two of these numbers must be the same, since there are six numbers (pigeons) and six possible values (pigeonholes). However, if not all of the values are used, then the pigeonhole principle would allow us to draw the desired conclusion. Let us therefore show that not all of the numbers can be used. The only way that the value 5 can appear as one of the c_i's is if one computer is connected to each of the others. In that case, the number 0 cannot appear, since no computer could be connected to none of the others. Thus not both 5 and 0 can appear in our list, and the above argument is valid.

39. This is similar to Example 9. Label the computers C_1 through C_{100}, and label the printers P_1 through P_{20}. If we connect C_k to P_k for $k = 1, 2, \ldots, 20$ and connect each of the computers C_{21} though C_{100} to *all* the printers, then we have used a total of $20 + 80 \cdot 20 = 1620$ cables. Clearly this is sufficient, because if computers C_1 through C_{20} need printers, then they can use the printers with the same subscripts, and if any computers with higher subscripts need a printer instead of one or more of these, then they can use the printers that are not being used, since they are connected to all the printers. Now we must show that 1619 cables are not enough. Since there are 1619 cables and 20 printers, the average number of computers per printer is $1619/20$, which is less than 81. Therefore some printer must be connected to fewer than 81 computers (the average of a set of numbers cannot be bigger than each of the numbers in the set). That means it is connected to 80 or fewer computers, so there are at least 20 computers that are not connected to it. If those 20 computers all needed a printer simultaneously, then they would be out of luck, since they are connected to at most the 19 other printers.

41. This problem is similar to Example 10, so we follow the method of solution suggested there. Let a_j be the number of matches held during or before the jth hour. Then $a_1, a_2, \ldots, a_{75}$ is an increasing sequence of distinct positive integers, since there was at least one match held every hour. Furthermore $1 \le a_j \le 125$, since there were only 125 matches altogether. Moreover, $a_1 + 24, a_2 + 24, \ldots, a_{75} + 24$ is also an increasing sequence of distinct positive integers, with $25 \le a_i + 24 \le 149$.

Now the 150 positive integers $a_1, a_2, \ldots, a_{75}, a_1 + 24, a_2 + 24, \ldots, a_{75} + 24$ are all less than or equal to 149. Hence by the pigeonhole principle two of these integers are equal. Since the integers $a_1, a_2, \ldots, a_{75}$ are all distinct, and the integers $a_1 + 24, a_2 + 24, \ldots, a_{75} + 24$ are all distinct, there must be distinct indices i and j such that $a_j = a_i + 24$. This means that exactly 24 matches were held from the beginning of hour $i + 1$ to the end of hour j, precisely the occurrence we wanted to find.

43. This is exactly a restatement of the generalized pigeonhole principle. The pigeonholes are the elements in the codomain (the elements of the set T), and the pigeons are the elements of the domain (the elements of the set S). To say that a pigeon s is in pigeonhole t is just to say that $f(s) = t$. The elements $s_1, s_2, \ldots, s_m$ are just the m pigeons guaranteed by the generalized pigeonhole principle to occupy a common pigeonhole.

45. Let d_j be $jx - N(jx)$, where $N(jx)$ is the integer closest to jx, for $1 \le j \le n$. We want to show that $|d_j| < 1/n$ for some j. Note that each d_j is an irrational number strictly between $-1/2$ and $1/2$ (since jx is irrational and every irrational number is closer than $1/2$ to the nearest integer). The proof is slightly messier if n is odd, so let us assume that n is even. Consider the n intervals

$$\left(0, \frac{1}{n}\right), \left(\frac{1}{n}, \frac{2}{n}\right), \ldots, \left(\frac{(n/2)-1}{n}, \frac{1}{2}\right), \left(-\frac{1}{n}, 0\right), \left(-\frac{2}{n}, -\frac{1}{n}\right), \ldots, \left(-\frac{1}{2}, -\frac{(n/2)-1}{n}\right).$$

The intervals are the pigeonholes and the d_j's are the pigeons. If the interval $(0, 1/n)$ or $(-1/n, 0)$ is occupied, then we are done, since the d_j in that interval tells us which j makes $|d_j| < 1/n$. If not, then there are n pigeons for at most $n-2$ pigeonholes, so by the pigeonhole principle there is some interval, say $\big((k-1)/n, k/n\big)$, with two pigeons in it, say d_r and d_s, with $r < s$. Now we will consider $sx - rx$ and show that it is within $1/n$ of its nearest integer; that will complete our proof, since $sx - rx = (s-r)x$, and $s-r$ is a positive integer less than n.

We can write $rx = N(rx) + d_r$ and $sx = N(sx) + d_s$, where $(k-1)/n \le d_r < k/n$ and $(k-1)/n \le d_s < k/n$. Subtracting, we have that $sx - rx = [N(sx) - N(rx)] + [d_s - d_r]$. Now the quantity in the first pair of square brackets is an integer. Furthermore the quantity in the second pair of square brackets is the difference of two numbers in the interval $\big((k-1)/n, k/n\big)$ and hence has absolute value less than $1/n$ (the extreme case would be when one of them is very close to $(k-1)/n$ and the other is very close to k/n). Therefore, by definition of "closest integer" $sx - rx$ is at most a distance $1/n$ from its closest integer, i.e., $|(sx - rx) - N(sx - rx)| < 1/n$, as desired. (The case in which n is odd is similar, but we need to extend our intervals slightly past $\pm 1/2$, using $n+1$ intervals rather than n. This is okay, since when we subtract 2 from $n+1$ we still have more pigeons than pigeonholes.)

47. **a)** Assuming that each $i_k \le n$, there are only n pigeonholes (namely $1, 2, \ldots, n$) for the n^2+1 numbers $i_1, i_2, \ldots, i_{n^2+1}$. Hence, by the generalized pigeonhole principle at least $\lceil (n^2+1)/n \rceil = n+1$ of the numbers are in the same pigeonhole, i.e., equal.

b) If $a_{k_j} < a_{k_{j+1}}$, then the subsequence consisting of a_{k_j} followed by a maximal increasing subsequence of length $i_{k_{j+1}}$ starting at $a_{k_{j+1}}$ contradicts the fact that $i_{k_j} = i_{k_{j+1}}$. Hence $a_{k_j} > a_{k_{j+1}}$.

c) If there is no increasing subsequence of length greater than n, then parts **(a)** and **(b)** apply. Therefore we have $a_{k_{n+1}} > a_{k_n} > \cdots > a_{k_2} > a_{k_1}$, a decreasing subsequence of length $n+1$.

SECTION 6.3 Permutations and Combinations

In this section we look at counting problems more systematically than in Section 6.1. We have some formulae that apply in many instances, and the trick is to recognize the instances. If an ordered arrangement without repetitions is asked for, then the formula for permutations usually applies; if an unordered selection without repetition is asked for, then the formula for combinations usually applies. Of course the product rule and the sum rule (and common sense and cleverness) are still needed to solve some of these problems—having formulae for permutations and combinations definitely does not reduce the solving of counting problems to a mechanical algorithm.

Again the general comments of Section 6.1 apply. Try to solve problems more than one way and come up with the same answer—you will learn from the process of looking at the same problem from two or more different angles, and you will be (almost) sure that your answer is correct.

1. Permutations are ordered arrangements. Thus we need to list all the ordered arrangements of all 3 of these letters. There are 6 such: a, b, c; a, c, b; b, a, c; b, c, a; c, a, b; and c, b, a. Note that we have listed them in alphabetical order. Algorithms for generating permutations and combinations are discussed in Section 6.6.

3. If we want the permutation to end with a, then we may as well forget about the a, and just count the number of permutations of $\{b, c, d, e, f, g\}$. Each permutation of these 6 letters, followed by a, will be a permutation of the desired type, and conversely. Therefore the answer is $P(6, 6) = 6! = 720$.

5. We simply plug into the formula $P(n,r) = n(n-1)(n-2)\cdots(n-r+1)$, given in Theorem 1. Note that there are r terms in this product, starting with n. This is the same as $P(n,r) = n!/(n-r)!$, but the latter formula is not as nice for computation, since it ignores the fact that each of the factors in the denominator cancels one factor in the numerator. Thus to compute $n!$ and $(n-r)!$ and then to divide is to do a lot of extra arithmetic. Of course if the denominator is 1, then there is no extra work, so we note that $P(n,n) = P(n,n-1) = n!$.
a) $P(6,3) = 6 \cdot 5 \cdot 4 = 120$ **b)** $P(6,5) = 6! = 720$ **c)** $P(8,1) = 8$
d) $P(8,5) = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 6720$ **e)** $P(8,8) = 8! = 40{,}320$ **f)** $P(10,9) = 10! = 3{,}628{,}800$

7. This is $P(9,5) = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15{,}120$ by Theorem 1.

9. We need to pick 3 horses from the 12 horses in the race, and we need to arrange them in order (first, second, and third), in order to specify the win, place, and show. Thus there are $P(12,3) = 12 \cdot 11 \cdot 10 = 1320$ possibilities.

11. **a)** To specify a bit string of length 10 that contains exactly four 1's, we simply need to choose the four positions that contain the 1's. There are $C(10,4) = 210$ ways to do that.
b) To contain at most four 1's means to contain four 1's, three 1's, two 1's, one 1, or no 1's. Reasoning as in part **(a)**, we see that there are $C(10,4)+C(10,3)+C(10,2)+C(10,1)+C(10,0) = 210+120+45+10+1 = 386$ such strings.
c) To contain at least four 1's means to contain four 1's, five 1's, six 1's, seven 1's, eight 1's, nine 1's, or ten 1's. Reasoning as in part **(b)**, we see that there are $C(10,4)+C(10,5)+C(10,6)+C(10,7)+C(10,8)+C(10,9)+C(10,10) = 210+252+210+120+45+10+1 = 848$ such strings. A simpler approach would be to figure out the number of ways not to have at least four 1's (i.e., to have three 1's, two 1's, one 1, or no 1's) and then subtract that from 2^{10}, the total number of bit strings of length 10. This way we get $1024 - (120+45+10+1) = 848$, fortunately the same answer as before. Solving a combinatorial problem in more than one way is a useful check on the correctness of the answer.
d) To have an equal number of 0's and 1's in this case means to have five 1's. Therefore the answer is $C(10,5) = 252$. Incidentally, this gives us another way to do part **(b)**. If we don't have an equal number of 0's and 1's, then we have either at most four 1's or at least six 1's. By symmetry, having at most four 1's occurs in half of these cases. Therefore the answer to part **(b)** is $(2^{10} - C(10,5))/2 = 386$, as above.

13. We assume that the row has a distinguished head. Consider the order in which the men appear relative to each other. There are n men, and all of the $P(n,n) = n!$ arrangements is allowed. Similarly, there are $n!$ arrangements in which the women can appear. Now the men and women must alternate, and there are the same number of men and women; therefore there are exactly two possibilities: either the row starts with a man and ends with a woman ($MWMW\ldots MW$) or else it starts with a woman and ends with a man ($WMWM\ldots WM$). We have three tasks to perform, then: arrange the men among themselves, arrange the women among themselves, and decide which sex starts the row. By the product rule there are $n! \cdot n! \cdot 2 = 2(n!)^2$ ways in which this can be done.

15. We assume that a combination is called for, not a permutation, since we are told to *select a set*, not *form an arrangement*. We need to choose 5 things from 26, so there are $C(26,5) = 26 \cdot 25 \cdot 24 \cdot 23 \cdot 22/5! = 65{,}780$ ways to do so.

17. We know that there are 2^{100} subsets of a set with 100 elements. All of them have more than two elements except the empty set, the 100 subsets consisting of one element each, and the $C(100,2) = 4950$ subsets with two elements. Therefore the answer is $2^{100} - 5051 \approx 1.3 \times 10^{30}$.

19. **a)** Each flip can be either heads or tails, so there are $2^{10} = 1024$ possible outcomes.

b) To specify an outcome that has exactly two heads, we simply need to choose the two flips that came up heads. There are $C(10, 2) = 45$ such outcomes.

c) To contain at most three tails means to contain three tails, two tails, one tail, or no tails. Reasoning as in part **(b)**, we see that there are $C(10, 3) + C(10, 2) + C(10, 1) + C(10, 0) = 120 + 45 + 10 + 1 = 176$ such outcomes.

d) To have an equal number of heads and tails in this case means to have five heads. Therefore the answer is $C(10, 5) = 252$.

21. **a)** If BCD is to be a substring, then we can think of that block of letters as one superletter, and the problem is to count permutations of five items—the letters A, E, F, and G, and the superletter BCD. Therefore the answer is $P(5, 5) = 5! = 120$.

b) Reasoning as in part **(a)**, we see that the answer is $P(4, 4) = 4! = 24$.

c) As in part **(a)**, we glue BA into one item and glue GF into one item. Therefore we need to permute five items, and there are $P(5, 5) = 5! = 120$ ways to do it.

d) This is similar to part **(c)**. Glue ABC into one item and glue DE into one item, producing four items, so the answer is $P(4, 4) = 4! = 24$.

e) If both ABC and CDE are substrings, then $ABCDE$ has to be a substring. So we are really just permuting three items: $ABCDE$, F, and G. Therefore the answer is $P(3, 3) = 3! = 6$.

f) There are no permutations with both of these substrings, since B cannot be followed by both A and E at the same time.

23. First position the men relative to each other. Since there are eight men, there are $P(8, 8)$ ways to do this. This creates nine slots where a woman (but not more than one woman) may stand: in front of the first man, between the first and second men, ..., between the seventh and eight men, and behind the eighth man. We need to choose five of these positions, in order, for the first through fifth woman to occupy (order matters, because the women are distinct people). This can be done is $P(9, 5)$ ways. Therefore the answer is $P(8, 8) \cdot P(9, 5) = 8! \cdot 9!/4! = 609{,}638{,}400$.

25. **a)** Since the prizes are different, we want an ordered arrangement of four numbers from the set of the first 100 positive integers. Thus there are $P(100, 4) = 94{,}109{,}400$ ways to award the prizes.

b) If the grand prize winner is specified, then we need to choose an ordered set of three tickets to win the other three prizes. This can be done is $P(99, 3) = 941{,}094$ ways.

c) We can first determine which prize the person holding ticket 47 will win (this can be done in 4 ways), and then we can determine the winners of the other three prizes, exactly as in part **(b)**. Therefore the answer is $4P(99, 3) = 3{,}764{,}376$.

d) This is the same calculation as in part **(a)**, except that there are only 99 viable tickets. Therefore the answer is $P(99, 4) = 90{,}345{,}024$. Note that this answer plus the answer to part **(c)** equals the answer to part **(a)**, since the person holding ticket 47 either wins a prize or does not win a prize.

e) This is similar to part **(c)**. There are $4 \cdot 3 = 12$ ways to determine which prizes these two lucky people will win, after which there are $P(98, 2) = 9506$ ways to award the other two prizes. Therefore the answer is $12 \cdot 9506 = 114{,}072$.

f) This is like part **(e)**. There are $P(4, 3) = 24$ ways to choose the prizes for the three people mentioned, and then 97 ways to choose the other winner. This gives $24 \cdot 97 = 2328$ ways in all.

g) Here it is just a matter of ordering the prizes for these four people, so the answer is $P(4, 4) = 24$.

h) This is similar to part **(d)**, except that this time the pool of viable numbers has only 96 numbers in it. Therefore the answer is $P(96, 4) = 79{,}727{,}040$.

i) There are four ways to determine the grand prize winner under these conditions. Then there are $P(99,3)$ ways to award the remaining prizes. This gives an answer of $4P(99,3) = 3{,}764{,}376$.

j) First we need to choose the prizes for the holder of 19 and 47. Since there are four prizes, there are $P(4,2) = 12$ ways to do this. Then there are 96 people who might win the remaining prizes, and there are $P(96,2) = 9120$ ways to award these prizes. Therefore the answer is $12 \cdot 9120 = 109{,}440$.

27. **a)** Since the order of choosing the members is not relevant (the offices are not differentiated), we need to use a combination. The answer is clearly $C(25,4) = 12{,}650$.

b) In contrast, here we need a permutation, since the order matters (we choose first a president, then a vice president, then a secretary, then a treasurer). The answer is clearly $P(25,4) = 303{,}600$.

29. **a)** In this part the permutation $5,6,32,7$, for example, is to be counted, since it contains the consecutive numbers 5, 6, and 7 in their correct order (even though separated by the 32). In order to specify such a 4-permutation, we first need to choose the 3 consecutive integers. They can be anything from $\{1,2,3\}$ to $\{98,99,100\}$; thus there are 98 possibilities. Next we need to decide which slot is to contain a number not in this set; there are 4 possibilities. Finally, we need to decide which of the 97 other positive integers not exceeding 100 is to fill this slot, and there are of course 97 choices. Thus our first attempt at an answer gives us, by the product rule, $98 \cdot 4 \cdot 97$.

Unfortunately, this answer is not correct, because we have counted some 4-permutations more than once. Consider the 4-permutation $4,5,6,7$, for example. We cannot tell whether it arose from choosing 4, 5, and 6 as the consecutive numbers, or from choosing 5, 6, and 7. (These are the only two ways it could have arisen.) In fact, every 4-permutation consisting of 4 consecutive numbers, in order, has been double counted. Therefore to correct our count, we need to subtract the number of such 4-permutations. Clearly there are 97 of them (they can begin with any number from 1 to 97). Further thought shows that every other 4-permutation in our collection arises in a unique way (in other words, there is a unique subsequence of three consecutive integers). Thus our final answer is $98 \cdot 4 \cdot 97 - 97 = 37{,}927$.

b) In this part we are insisting that the consecutive numbers be consecutive in the 4-permutation as well. The analysis in part **(a)** works here, except that there are only 2 places to put the fourth number—in slot 1 or in slot 4. Therefore the answer is $98 \cdot 2 \cdot 97 - 97 = 18{,}915$.

31. We need to be careful here, because strings can have repeated letters.

a) We need to choose the position for the vowel, and this can be done in 6 ways. Next we need to choose the vowel to use, and this can be done in 5 ways. Each of the other five positions in the string can contain any of the 21 consonants, so there are 21^5 ways to fill the rest of the string. Therefore the answer is $6 \cdot 5 \cdot 21^5 = 122{,}523{,}030$.

b) We need to choose the position for the vowels, and this can be done in $C(6,2) = 15$ ways (we need to choose two positions out of six). We need to choose the two vowels (5^2 ways). Each of the other four positions in the string can contain any of the 21 consonants, so there are 21^4 ways to fill the rest of the string. Therefore the answer is $15 \cdot 5^2 \cdot 21^4 = 72{,}930{,}375$.

c) The best way to do this is to count the number of strings with no vowels and subtract this from the total number of strings. We obtain $26^6 - 21^6 = 223{,}149{,}655$.

d) As in part **(c)**, we will do this by subtracting from the total number of strings, the number of strings with no vowels and the number of strings with one vowel (this latter quantity having been computed in part **(a)**). We obtain $26^6 - 21^6 - 6 \cdot 5 \cdot 21^5 = 223149655 - 122523030 = 100{,}626{,}625$.

33. We are told that we must select three of the 10 men and three of the 15 women. This can be done in $C(10,3)C(15,3) = 54{,}600$ ways.

35. To implement the condition that every 0 be immediately followed by a 1, let us think of "gluing" a 1 to the right of each 0. Then the objects we have to work with are eight blocks consisting of the string 01 and two 1's. The question is, then, how many strings are there consisting of these ten objects? This is easy to calculate, for we simply have to choose two of the "positions" in the string to contain the 1's and fill the remaining "positions" with the 01 blocks. Therefore the answer is $C(10,2) = 45$.

37. Perhaps the most straightforward way to do this is to look at the several cases. The string might contain three 1's and seven 0's, four 1's and six 0's, five of each, six 1's and four 0's, or seven 1's and three 0's. In each case we can determine the number of strings by calculating a binomial coefficient, since we simply need to choose the positions for the 1's. Therefore the answer is $C(10,3)+C(10,4)+C(10,5)+C(10,6)+C(10,7) = 120+210+252+210+120 = 912$.

39. To specify such a license plate we need to write down a 3-permutation of the set of 26 letters and follow it by a 3-permutation of the set of 10 digits. By the product rule the answer is therefore $P(26,3) \cdot P(10,3) = 26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 = 11{,}232{,}000$.

41. Designate the head of the table and seat the people clockwise. Clearly there are $P(n,r)$ ways to do this. Because rotations of the table do not make for a "different" seating, this overcounts by a factor of r. Therefore the answer is $P(n,r)/r = n!/(r(n-r)!)$.

43. If there are no ties, then there are $3! = 6$ possible finishes. If two of the horses tie and the third has a different time, then there are 3 ways to decide which horse is not tied and then 2 ways to decide whether that horse finishes first or last. That gives $3 \cdot 2 = 6$ possibilities. Finally, all three horses can tie. So the answer is $6+6+1 = 13$.

45. We can solve this problem by breaking it down into cases depending on the ties. There are four basic cases. (1) If there are unique gold and silver winners, then we can choose these winners in $6 \cdot 5 = 30$ ways. Any nonempty subset of the remaining four runners can win the bronze medal. There are $2^4 - 1 = 15$ ways to choose these people, giving us $30 \cdot 15 = 450$ ways in all for this case. (2) If there is a 2-way tie for first place, then there are $C(6,2) = 15$ ways to choose the gold medalists. Any nonempty subset of the remaining four runners can win the bronze medal, so there are $2^4 - 1 = 15$ ways to choose these people, giving us $15 \cdot 15 = 225$ ways in all for this case. (3) If there is a k-way tie for first with $k \geq 3$, then there are $C(6,k)$ ways to choose the gold medalists (there are no other medals in this case). This gives us $C(6,3)+C(6,4)+C(6,5)+C(6,6) = 20+15+6+1 = 42$ more possibilities. (4) The only other case is that there is a single gold medal winner and a k-way tie for second with $k \geq 2$. We can choose the winner in 6 ways and the silver medalists in $2^5 - C(5,1) - C(5,0) = 32-5-1 = 26$ ways. This gives us $6 \cdot 26 = 156$ possibilities. Putting this all together, the answer is $450+225+42+156 = 873$.

SECTION 6.4 Binomial Coefficients and Identities

In this section we usually write the binomial coefficients using the $\binom{n}{r}$ notation, rather than the $C(n,r)$ notation. These numbers tend to come up in many parts of discrete mathematics.

1. **a)** When $(x+y)^4 = (x+y)(x+y)(x+y)(x+y)$ is expanded, all products of a term in the first sum, a term in the second sum, a term in the third sum, and a term in the fourth sum are added. Terms of the form x^4, x^3y, x^2y^2, xy^3 and y^4 arise. To obtain a term of the form x^4, an x must be chosen in each of the sums, and this can be done in only one way. Thus, the x^4 term in the product has a coefficient of 1. (We can think of this coefficient as $\binom{4}{4}$.) To obtain a term of the form x^3y, an x must be chosen in three of the four sums (and consequently a y in the other sum). Hence, the number of such terms is the number of 3-combinations of four objects, namely $\binom{4}{3} = 4$. Similarly, the number of terms of the form x^2y^2 is the number of ways to pick two of the four sums to obtain x's (and consequently take a y from each of the other two factors). This can be done in $\binom{4}{2} = 6$ ways. By the same reasoning there are $\binom{4}{1} = 4$ ways to obtain the xy^3 terms, and only one way (which we can think of as $\binom{4}{0}$) to obtain a y^4 term. Consequently, the product is $x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$.

b) This is explained in Example 2. The expansion is $\binom{4}{0}x^4 + \binom{4}{1}x^3y + \binom{4}{2}x^2y^2 + \binom{4}{3}xy^3 + \binom{4}{4}y^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$. Note that it does not matter whether we think of the bottom of the binomial coefficient expression as corresponding to the exponent on x, as we did in part **(a)**, or the exponent on y, as we do here.

3. The coefficients are the binomial coefficients $\binom{6}{i}$, as i runs from 0 to 6, namely 1, 6, 15, 20, 15, 6, 1. Therefore $(x+y)^6 = \sum_{i=0}^{6} \binom{6}{i}x^{6-i}y^i = x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$.

5. There is one term for each i from 0 to 100, so there are 101 terms.

7. By the binomial theorem the term involving x^9 in the expansion of $(2 + (-x))^{19}$ is $\binom{19}{9}2^{10}(-x)^9$. Therefore the coefficient is $\binom{19}{9}2^{10}(-1)^9 = -2^{10}\binom{19}{9} = -94{,}595{,}072$.

9. Using the binomial theorem, we see that the term involving x^{101} in the expansion of $((2x) + (-3y))^{200}$ is $\binom{200}{99}(2x)^{101}(-3y)^{99}$. Therefore the coefficient is $\binom{200}{99}2^{101}(-3)^{99} = -2^{101}3^{99}C(200, 99)$.

11. Let us apply the binomial theorem to the given binomial:

$$(x^2 - x^{-1})^{100} = \sum_{j=0}^{100} \binom{100}{j}(x^2)^{100-j}(-x^{-1})^j$$

$$= \sum_{j=0}^{100} \binom{100}{j}(-1)^j x^{200-2j-j} = \sum_{j=0}^{100} \binom{100}{j}(-1)^j x^{200-3j}$$

Thus the only nonzero coefficients are those of the form $200 - 3j$ where j is an integer between 0 and 100, inclusive, namely 200, 197, 194, ..., 2, −1, −4, ..., −100. If we denote $200 - 3j$ by k, then we have $j = (200-k)/3$. This gives us our answer. The coefficient of x^k is zero for k not in the list just given (namely those vales of k between −100 and 200, inclusive, that are congruent to 2 modulo 3), and for those values of k in the list, the coefficient is $(-1)^{(200-k)/3}\binom{100}{(200-k)/3}$.

13. We are asked simply to display these binomial coefficients. Each can be computed from the formula in Theorem 2 in Section 6.3. Alternatively, we can apply Pascal's identity to the last row of Figure 1(b), adding successive numbers in that row to produce the desired row. We thus obtain

1 9 36 84 126 126 84 36 9 1.

15. There are many ways to see why this is true. By Corollary 1 the sum of *all* the positive numbers $\binom{n}{k}$, as k runs from 0 to n, is 2^n, so certainly each one of them is no bigger than this sum. Another way to see this is to note that $\binom{n}{k}$ counts the number of subsets of an n-set having k elements, and 2^n counts even more—the number of subsets of an n-set with no restriction as to size; so certainly the former is smaller than the latter.

17. We know that

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k(k-1)(k-2)\cdots 2}.$$

Now if we make the numerator of the right-hand side larger by raising each factor up to n, and make the denominator smaller by lowering each factor to 2, then we have certainly not decreased the value, so the left-hand side is less than or equal to this altered expression. But the result is precisely $n^k/2^{k-1}$, as desired.

19. Using the formula (Theorem 2 in Section 6.3) we have

$$\begin{aligned}\binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!} \\ &= \frac{n!k + n!(n-k+1)}{k!(n-k+1)!} \quad \text{(having found a common denominator)} \\ &= \frac{(n+1)n!}{k!((n+1)-k)!} = \frac{(n+1)!}{k!((n+1)-k)!} = \binom{n+1}{k}.\end{aligned}$$

21. **a)** We show that each side counts the number of ways to choose from a set with n elements a subset with k elements and a distinguished element of that set. For the left-hand side, first choose the k-set (this can be done in $\binom{n}{k}$ ways) and then choose one of the k elements in this subset to be the distinguished element (this can be done in k ways). For the right-hand side, first choose the distinguished element out of the entire n-set (this can be done in n ways), and then choose the remaining $k-1$ elements of the subset from the remaining $n-1$ elements of the set (this can be done in $\binom{n-1}{k-1}$ ways).

b) This is straightforward algebra:

$$k\binom{n}{k} = k \cdot \frac{n!}{k!(n-k)!} = \frac{n \cdot (n-1)!}{(k-1)!(n-k)!} = n\binom{n-1}{k-1}.$$

23. This identity can be proved algebraically or combinatorially. Algebraically, we compute as follows, starting with the right-hand side (we use twice the fact that $(x+1)x! = (x+1)!$):

$$\begin{aligned}\frac{(n+1)\binom{n}{k-1}}{k} &= \frac{(n+1)n!}{(k-1)!(n-(k-1))!k} \\ &= \frac{(n+1)!}{k!(n-(k-1))!} \\ &= \frac{(n+1)!}{k!((n+1)-k)!} \\ &= \binom{n+1}{k}\end{aligned}$$

For a combinatorial argument, we need to construct a situation in which both sides count the same thing. Suppose that we have a set of $n+1$ people, and we wish to choose k of them. Clearly there are $\binom{n+1}{k}$ ways to do this. On the other hand, we can choose our set of k people by first choosing one person to be in the set (there are $n+1$ choices), and then choosing $k-1$ additional people to be in the set, from the n people remaining. This can be done in $\binom{n}{k-1}$ ways. Therefore apparently there are $(n+1)\binom{n}{k-1}$ ways to choose the set of k people. However, we have overcounted: there are k ways that every such set can be chosen, since once we have the set, we realize that any of the k people could have been chosen first. Thus we have overcounted by a factor of k, and the real answer is $(n+1)\binom{n}{k-1}/k$ (we correct for the overcounting by dividing by k). Comparing our two approaches, one yielding the answer $\binom{n+1}{k}$, and the other yielding the answer $(n+1)\binom{n}{k-1}/k$, we conclude that $\binom{n+1}{k} = (n+1)\binom{n}{k-1}/k$.

Finally, we are asked to use this identity to give a recursive definition of the $\binom{n}{k}$'s. Note that this identity expresses $\binom{n}{k}$ in terms of $\binom{i}{j}$ for values of i and j less than n and k, respectively (namely $i = n-1$ and $j = k-1$). Thus the identity will be the recursive part of the definition. We need the base cases to handle $n = 0$ or $k = 0$. Our full definition becomes

$$\binom{n}{k} = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k > 0 \text{ and } n = 0 \\ n\binom{n-1}{k-1}/k & \text{if } n > 0 \text{ and } k > 0. \end{cases}$$

Actually, if we assume (as we usually do) that $k \leq n$, then we do not need the second line of the definition. Note that $\binom{n}{k} = 0$ for $n < k$ under the definition given here, which is consistent with the combinatorial definition, since there are no ways to choose k different elements from a set with fewer than k elements.

25. We use Pascal's identity twice (Theorem 2 of this section) and Corollary 1 of the previous section:

$$\begin{aligned}\binom{2n}{n+1} + \binom{2n}{n} = \binom{2n+1}{n+1} &= \frac{1}{2}\left(\binom{2n+1}{n+1} + \binom{2n+1}{n+1}\right) \\ &= \frac{1}{2}\left(\binom{2n+1}{n+1} + \binom{2n+1}{n}\right) = \frac{1}{2}\left(\binom{2n+2}{n+1}\right)\end{aligned}$$

27. The reason this is called the "hockey stick identity" is that the binomial coefficients being summed lie along a line in Pascal's triangle, and the sum is the entry in the triangle that, together with that line, forms a shape vaguely resembling a hockey stick.

a) We need to find something to count so that the left-hand side of the equation counts it in one way and the right-hand side counts it in a different way. After much thought, we might try the following. We will count the number of bit strings of length $n+r+1$ containing exactly r 0's and $n+1$ 1's. There are $\binom{n+r+1}{r}$ such strings, since a string is completely specified by deciding which r of the bits are to be the 0's. To see that the left-hand side of the identity counts the same thing, let $l+1$ be the position of the last 1 in the string. Since there are $n+1$ 1's, we know that l cannot be less than n. Thus there are disjoint cases for each l from n to $n+r$. For each such l, we completely determine the string by deciding which of the l positions in the string before the last 1 are to be 0's. Since there are n 1's in this range, there are $l-n$ 0's. Thus there are $\binom{l}{l-n}$ ways to choose the positions of the 0's. Now by the sum rule the total number of bit strings will be $\sum_{l=n}^{n+r}\binom{l}{l-n}$. By making the change of variable $k = l-n$, this transforms into the left-hand side, and we are finished.

b) We need to prove this by induction on r; Pascal's identity will enter at the crucial step. We let $P(r)$ be the statement to be proved. The basis step is clear, since the equation reduces to $\binom{n}{0} = \binom{n+1}{0}$, which is the true proposition $1 = 1$. Assuming the inductive hypothesis, we derive $P(r+1)$ in the usual way:

$$\begin{aligned}\sum_{k=0}^{r+1}\binom{n+k}{k} &= \left(\sum_{k=0}^{r}\binom{n+k}{k}\right) + \binom{n+r+1}{r+1} \\ &= \binom{n+r+1}{r} + \binom{n+r+1}{r+1} \quad \text{(by the inductive hypothesis)} \\ &= \binom{n+(r+1)+1}{r+1} \quad \text{(by Pascal's identity)}\end{aligned}$$

29. We will follow the hint and count the number of ways to choose a committee with leader from a set of n people. Note that the size of the committee is not specified, although it clearly needs to have at least one person (its leader). On the one hand, we can choose the leader first, in any of n ways. We can then choose the rest of the committee, which can be any subset of the remaining $n-1$ people; this can be done in 2^{n-1} ways since there are this many subsets. Therefore the right-hand side of the proposed identity counts this.

On the other hand, we can organize our count by the size of the committee. Let k be the number of people who will serve on the committee. The number of ways to select a committee with k people is clearly $\binom{n}{k}$, and once we have chosen the committee, there are clearly k ways in which to choose its leader. By the sum rule the left-hand side of the proposed identity therefore also counts the number of such committees. Since the two sides count the same quantity, they must be equal.

31. Corollary 2 says that $\binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots \pm \binom{n}{n} = 0$. If we put all the negative terms on the other side, we obtain $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = \binom{n}{1} + \binom{n}{3} + \cdots$ (one side ends at $\binom{n}{n}$ and the other side ends at $\binom{n}{n-1}$—which is which depends on whether n is even or odd). Now the left-hand side counts the number of subsets with even cardinality of a set with n elements, and the right-hand side counts the number of subsets with odd cardinality of the same set. That these two quantities are equal is precisely what we wanted to prove.

33. **a)** Clearly a path of the desired type must consist of m moves to the right and n moves up. Therefore each such path can be represented by a bit string consisting of m 0's and n 1's, with the 0's representing moves to the right and the 1's representing moves up. Note that the total length of this bit string is $m + n$.

b) We know from this section that the number of bit strings of length $m + n$ containing exactly n 1's is $\binom{m+n}{n}$, since one need only specify the positions of the 1's. Note that this is the same as $\binom{m+n}{m}$.

35. We saw in Exercise 33 that the number of paths of length n was the same as the number of bit strings of length n, which we know to be 2^n, the right-hand side of the identity. On the other hand, a path of length n must end up at some point the sum of whose coordinates is n, say at $(n-k, k)$ for some k from 0 to n. We saw in Exercise 33 that the number of paths ending up at $(n-k, k)$ was equal to $\binom{n-k+k}{k} = \binom{n}{k}$. Therefore the left-hand side of the identity counts the number of such paths, too.

37. The right-hand side of the identity we are asked to prove counts, by Exercise 33, the number of paths from $(0, 0)$ to $(n+1, r)$. Now let us count these paths by breaking them down into $r+1$ cases, depending on how many steps upward they begin with. Let k be the number of steps upward they begin with before taking a step to the right. Then k can take any value from 0 to r. The number of paths from $(0, 0)$ to $(n+1, r)$ that begin with exactly k steps upward before turning to the right is clearly the same as the number of paths from $(1, k)$ to $(n+1, r)$, since after these k upward steps and the move to the right we have reached $(1, k)$. This latter quantity is the same as the number of paths from $(0, 0)$ to $(n+1-1, r-k) = (n, r-k)$, since we can relabel our diagram to make $(1, k)$ the origin. From Exercise 33, this latter quantity is $\binom{n+r-k}{r-k}$. Therefore the total number of paths is the desired sum

$$\sum_{k=0}^{r} \binom{n+r-k}{r-k} = \sum_{k=0}^{r} \binom{n+k}{k},$$

where the equality comes from changing the dummy variable from k to $r-k$. Since both sides count the same thing, they are equal.

39. **a)** This looks like the third negatively sloping diagonal of Pascal's triangle, starting with the leftmost entry in the second row and reading down and to the right. In other words, the nth term of this sequence is $\binom{n+1}{n-1} = \binom{n+1}{2}$.

b) This looks like the fourth negatively sloping diagonal of Pascal's triangle, starting with the leftmost entry in the third row and reading down and to the right. In other words, the nth term of this sequence is $\binom{n+2}{n-1} = \binom{n+2}{3}$.

c) These seem to be the entries reading straight down the middle of the Pascal's triangle. Only every other row has a middle element. The first entry in the sequence is $\binom{0}{0}$, the second is $\binom{2}{1}$, the third is $\binom{4}{2}$, the fourth is $\binom{6}{3}$, and so on. In general, then, the nth term is $\binom{2n-2}{n-1}$.

d) These seem to be the entries reading down the middle of the Pascal's triangle. Only every other row has an exact middle element, but in the other rows, there are two elements sharing the middle. The first entry in the sequence is $\binom{0}{0}$, the second is $\binom{1}{0}$, the third is $\binom{2}{1}$, the fourth is $\binom{3}{1}$, the fifth is $\binom{4}{2}$, the fourth is $\binom{5}{2}$, and so on. In general, then, the nth term is $\binom{n-1}{\lfloor (n-1)/2 \rfloor}$.

e) One pattern here is a wandering through Pascal's triangle according to the following rule. The nth term is in the nth row of the triangle. We increase the position in that row as we go down, but as soon as we are about to reach the middle of the row, we jump back to the start of the next row. For example, the fifth term is the first entry in row 5; the sixth term is the second entry in row 6; the seventh term is the third entry in row 7; the eighth term is the fourth entry in row 8. The next entry following that pattern would take us to the middle of the ninth row, so instead we jump back to the beginning, and the ninth term is the first entry of row 9. To come up with a formula here, we see that the nth entry is $\binom{n-1}{k-1}$ for a particular k; let us determine k as a function of n. A little playing around with the pattern reveals that k is n minus the largest power of 2 less than n (where for this purpose we consider 0 to be the largest power of 2 less than 1). For example, the 14th term has $k = 14 - 8 = 6$, so it is $\binom{13}{5} = 1287$.

f) The terms seem to come from every third row, so the nth term is $\binom{3n-3}{k}$ for some k. A little observation indicates that in fact these terms are $\binom{0}{0}$, $\binom{3}{1}$, $\binom{6}{2}$, $\binom{9}{3}$, $\binom{12}{4}$, and so on. Thus the nth term is $\binom{3n-3}{n-1}$.

SECTION 6.5 Generalized Permutations and Combinations

As in Section 6.3, we have formulae that give us the answers to some combinatorial problems, if we can figure out which formula applies to which problem, and in what way it applies. Here, even more than in previous sections, the ability to see a problem from the right perspective is the key to solving it. Expect to spend several minutes staring at each problem before any insight comes. Reread the examples in the section and try to imagine yourself going through the thought processes explained there. Gradually your mind will begin to think in the same terms. In particular, ask yourself what is being selected from what, whether ordered or unordered selections are to be made, and whether repetition is allowed. In most cases, after you have answered these questions, you can find the appropriate formula from Table 1.

1. Since order is important here, and since repetition is allowed, this is a simple application of the product rule. There are 3 ways in which the first element can be selected, 3 ways in which the second element can be selected, and so on, with finally 3 ways in which the fifth element can be selected, so there are $3^5 = 243$ ways in which the 5 elements can be selected. The general formula is that there are n^k ways to select k elements from a set of n elements, in order, with unlimited repetition allowed.

3. Since we are considering strings, clearly order matters. The choice for each position in the string is from the set of 26 letters. Therefore, using the same reasoning as in Exercise 1, we see that there are $26^6 = 308{,}915{,}776$ strings.

5. We assume that the jobs and the employees are distinguishable. For each job, we have to decide which employee gets that job. Thus there are 5 ways in which the first job can be assigned, 5 ways in which the second job can be assigned, and 5 ways in which the third job can be assigned. Therefore, by the multiplication principle (just as in Exercise 1) there are $5^3 = 125$ ways in which the assignments can be made. (Note that we do not require that every employee get at least one job.)

7. Since the selection is to be an unordered one, Theorem 2 applies. We want to choose $r = 3$ items from a set with $n = 5$ elements. Theorem 2 tells us that there are $C(5 + 3 - 1, 3) = C(7, 3) = 7 \cdot 6 \cdot 5/(3 \cdot 2) = 35$ ways to do so. (Equivalently, this problem is asking us to count the number of nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 + x_5 = 3$, where x_i represents the number of times that the ith element of the 5-element set gets selected.)

9. Let $b_1, b_2, \ldots, b_8$ be the number of bagels of the 8 types listed (in the order listed) that are selected. Order does not matter: we are presumably putting the bagels into a bag to take home, and the order in which we put them there is irrelevant.

a) If we want to choose 6 bagels, then we are asking for the number of nonnegative solutions to the equation $b_1 + b_2 + \cdots + b_8 = 6$. Theorem 2 applies, with $n = 8$ and $r = 6$, giving us the answer $C(8 + 6 - 1, 6) = C(13, 6) = 1716$.

b) This is the same as part **(a)**, except that $r = 12$ rather than 6. Thus there are $C(8 + 12 - 1, 12) = C(19, 12) = C(19, 7) = 50{,}388$ ways to make the selection. (Note that $C(19, 7)$ was easier to compute than $C(19, 12)$, and since they are equal, we chose the latter form.)

c) This is the same as part **(a)**, except that $r = 24$ rather than 6. Thus there are $C(8 + 24 - 1, 24) = C(31, 24) = C(31, 7) = 2{,}629{,}575$ ways to make the selection.

d) This one is more complicated. Here we want to solve the equation $b_1 + b_2 + \cdots + b_8 = 12$, subject to the constraint that each $b_i \geq 1$. We reduce this problem to the form in which Theorem 2 is applicable with the following trick. Let $b'_i = b_i - 1$; then b'_i represents the number of bagels of type i, in excess of the required 1, that are selected. If we substitute $b_i = b'_i + 1$ into the original equation, we obtain $(b'_1 + 1) + (b'_2 + 1) + \cdots + (b'_8 + 1) = 12$, which reduces to $b'_1 + b'_2 + \cdots + b'_8 = 4$. In other words, we are asking how many ways are there to choose the 4 extra bagels (in excess of the required 1 of each type) from among the 8 types, repetitions allowed. By Theorem 2 the number of solutions is $C(8 + 4 - 1, 4) = C(11, 4) = 330$.

e) This final part is even trickier. First let us ignore the restriction that there can be no more than 2 salty bagels (i.e., that $b_4 \leq 2$). We will take into account, however, the restriction that there must be at least 3 egg bagels (i.e., that $b_3 \geq 3$). Thus we want to count the number of solutions to the equation $b_1 + b_2 + \cdots + b_8 = 12$, subject to the condition that $b_i \geq 0$ for all i and $b_3 \geq 3$. As in part **(d)**, we use the trick of choosing the 3 egg bagels at the outset, leaving only 9 bagels free to be chosen; equivalently, we set $b'_3 = b_3 - 3$, to represent the extra egg bagels, above the required 3, that are chosen. Now Theorem 2 applies to the number of solutions of $b_1 + b_2 + b'_3 + b_4 + \cdots + b_8 = 9$, so there are $C(8 + 9 - 1, 9) = C(16, 9) = C(16, 7) = 11{,}440$ ways to make this selection.

Next we need to worry about the restriction that $b_4 \leq 2$. We will impose this restriction by subtracting from our answer so far the number of ways to violate this restriction (while still obeying the restriction that $b_3 \geq 3$). The difference will be the desired answer. To violate the restriction means to have $b_4 \geq 3$. Thus we want to count the number of solutions to $b_1 + b_2 + \cdots + b_8 = 12$, with $b_3 \geq 3$ and $b_4 \geq 3$. Using the same technique as we have just used, this is equal to the number of nonnegative solutions to the equation $b_1 + b_2 + b'_3 + b'_4 + b_5 + \cdots + b_8 = 6$ (the 6 on the right being $12 - 3 - 3$). By Theorem 2 there are $C(8 + 6 - 1, 6) = C(13, 6) = 1716$ ways to make this selection. Therefore our final answer is $11440 - 1716 = 9724$.

11. This can be solved by common sense. Since the pennies are all identical and the nickels are all identical, all that matters is the number of each type of coin selected. We can select anywhere from 0 to 8 pennies (and the rest nickels); since there are nine numbers in this range, the answer is 9. (The number of pennies and nickels is irrelevant, as long as each is at least eight.) If we wanted to use a high-powered theorem for this problem, we could observe that Theorem 2 applies, with $n = 2$ (there are two types of coins) and $r = 8$. The formula gives $C(2 + 8 - 1, 8) = C(9, 8) = 9$.

13. Assuming that the warehouses are distinguishable, let w_i be the number of books stored in warehouse i. Then we are asked for the number of solutions to the equation $w_1 + w_2 + w_3 = 3000$. By Theorem 2 there are $C(3 + 3000 - 1, 3000) = C(3002, 3000) = C(3002, 2) = 4{,}504{,}501$ of them.

15. a) Let $x_1 = x'_1 + 1$; thus x'_1 is the value that x_1 has in excess of its required 1. Then the problem

asks for the number of nonnegative solutions to $x_1' + x_2 + x_3 + x_4 + x_5 = 20$. By Theorem 2 there are $C(5 + 20 - 1, 20) = C(24, 20) = C(24, 4) = 10{,}626$ of them.

b) Substitute $x_i = x_i' + 2$ into the equation for each i; thus x_i' is the value that x_i has in excess of its required 2. Then the problem asks for the number of nonnegative solutions to $x_1' + x_2' + x_3' + x_4' + x_5' = 11$. By Theorem 2 there are $C(5 + 11 - 1, 11) = C(15, 11) = C(15, 4) = 1365$ of them.

c) There are $C(5 + 21 - 1, 21) = C(25, 21) = C(25, 4) = 12650$ solutions with no restriction on x_1. The restriction on x_1 will be violated if $x_1 \geq 11$. Following the procedure in part **(a)**, we find that there are $C(5 + 10 - 1, 10) = C(14, 10) = C(14, 4) = 1001$ solutions in which the restriction is violated. Therefore there are $12650 - 1001 = 11{,}649$ solutions of the equation with its restriction.

d) First let us impose the restrictions that $x_3 \geq 15$ and $x_2 \geq 1$. Then the problem is equivalent to counting the number of solutions to $x_1 + x_2' + x_3' + x_4 + x_5 = 5$, subject to the constraints that $x_1 \leq 3$ and $x_2' \leq 2$ (the latter coming from the original restriction that $x_2 < 4$). Note that these two restrictions cannot be violated simultaneously. Thus if we count the number of solutions to $x_1 + x_2' + x_3' + x_4 + x_5 = 5$, subtract the number of its solutions in which $x_1 \geq 4$, and subtract the numbers of its solutions in which $x_2' \geq 3$, then we will have the answer. By Theorem 2 there are $C(5 + 5 - 1, 5) = C(9, 5) = 126$ solutions of the unrestricted equation. Applying the first restriction reduces the equation to $x_1' + x_2' + x_3' + x_4 + x_5 = 1$, which has $C(5 + 1 - 1, 1) = C(5, 1) = 5$ solutions. Applying the second restriction reduces the equation to $x_1 + x_2'' + x_3' + x_4 + x_5 = 2$, which has $C(5 + 2 - 1, 2) = C(6, 2) = 15$ solutions. Therefore the answer is $126 - 5 - 15 = 106$.

17. Theorem 3 applies here, with $n = 10$ and $k = 3$. The answer is therefore

$$\frac{10!}{2!3!5!} = 2520.$$

19. Theorem 3 applies here, with $n = 14$, $n_1 = n_2 = 3$ (the triplets), $n_3 = n_4 = n_5 = 2$ (the twins), and $n_6 = n_7 = 1$. The answer is therefore

$$\frac{14!}{3!3!2!2!2!1!1!} = 302{,}702{,}400.$$

21. If we think of the balls as doing the choosing, then this is asking for the number of ways to choose six bins from the nine given bins, with repetition allowed. (The number of times each bin is chosen is the number of balls in that bin.) By Theorem 2 with $n = 9$ and $r = 6$, this choice can be made in $C(9 + 6 - 1, 6) = C(14, 6) = 3003$ ways.

23. There are several ways to count this. We can first choose the two objects to go into box #1 ($C(12, 2)$ ways), then choose the two objects to go into box #2 ($C(10, 2)$ ways, since only 10 objects remain), then choose the two objects to go into box #3 ($C(8, 2)$ ways), and so on. So the answer is $C(12, 2) \cdot C(10, 2) \cdot C(8, 2) \cdot C(6, 2) \cdot C(4, 2) \cdot C(2, 2) = (12 \cdot 11/2)(10 \cdot 9/2)(8 \cdot 7/2)(6 \cdot 5/2)(4 \cdot 3/2)(2 \cdot 1/2) = 12!/2^6 = 7{,}484{,}400$. Alternatively, just line up the 12 objects in a row (12! ways to do that), and put the first two into box #1, the next two into box #2, and so on. This overcounts by a factor of 2^6, since there are that many ways to swap objects in the permutation without affecting the result (swap the first and second objects or not, and swap the third and fourth objects or not, and so on). So this results in the same answer. Here is a third way to get this answer. First think of pairing the objects. Think of the objects as ordered (a first, a second, and so on). There are 11 ways to choose a mate for the first object, then 9 ways to choose a mate for the first unused object, then 7 ways to choose a mate for the first still unused object, and so on. This gives $11 \cdot 9 \cdot 7 \cdot 5 \cdot 3$ ways to do the pairing. Then there are 6! ways to choose the boxes for the pairs. So the answer is the product of these two quantities, which is again 7,484,400.

25. Let $d_1, d_2, \ldots, d_6$ be the digits of a natural number less than 1,000,000; they can each be anything from 0 to 9 (in particular, we may as well assume that there are leading 0's if necessary to make the number exactly 6 digits long). If we want the sum of the digits to equal 19, then we are asking for the number of solutions to the equation $d_1 + d_2 + \cdots + d_6 = 19$ with $0 \le d_i \le 9$ for each i. Ignoring the upper bound restriction, there are, by Theorem 2, $C(6 + 19 - 1, 19) = C(24, 19) = C(24, 5) = 42504$ of them. We must subtract the number of solutions in which the restriction is violated. If the digits are to add up to 19 and one or more of them is to exceed 9, then exactly one of them will have to exceed 9, since $10 + 10 > 19$. There are 6 ways to choose the digit that will exceed 9. Once we have made that choice (without loss of generality assume it is d_1 that is to be made greater than or equal to 10), then we count the number of solutions to the equation by counting the number of solutions to $d_1' + d_2 + \cdots + d_6 = 19 - 10 = 9$; by Theorem 2 there are $C(6 + 9 - 1, 9) = C(14, 9) = C(14, 5) = 2002$ of them. Thus there are $6 \cdot 2002 = 12012$ solutions that violate the restriction. Subtracting this from the 42504 solutions altogether, we find that $42504 - 12012 = 30{,}492$ is the answer to the problem.

27. We assume that each problem is worth a whole number of points. Then we want to find the number of integer solutions to $x_1 + x_2 + \cdots + x_{10} = 100$, subject to the constraint that each $x_i \ge 5$. Letting x_i' be the number of points assigned to problem i in excess of its required 5, and substituting $x_i = x_i' + 5$ into the equation, we obtain the equivalent equation $x_1' + x_2' + \cdots + x_{10}' = 50$. By Theorem 2 the number of solutions is given by $C(10 + 50 - 1, 50) = C(59, 50) = C(59, 9) = 12{,}565{,}671{,}261$.

29. There are at least two good ways to do this problem. First we present a solution in the spirit of this section. Let us place the 1's and some gaps in a row. A 1 will come first, followed by a gap, followed by another 1, another gap, a third 1, a third gap, a fourth 1, and a fourth gap. Into the gaps we must place the 12 0's that are in this string. Let g_1, g_2, g_3, and g_4 be the numbers of 0's placed in gaps 1 through 4, respectively. The only restriction is that each $g_i \ge 2$. Thus we want to count the number of solutions to the equation $g_1 + g_2 + g_3 + g_4 = 12$, with $g_i \ge 2$ for each i. Letting $g_i = g_i' + 2$, we want to count, equivalently, the number of nonnegative solutions to $g_1' + g_2' + g_3' + g_4' = 4$. By Theorem 2 there are $C(4 + 4 - 1, 4) = C(7, 4) = C(7, 3) = 35$ solutions. Thus our answer is 35.

Here is another way to solve the problem. Since each 1 must be followed by two 0's, suppose we glue 00 to the right end of each 1. This uses up 8 of the 0's, leaving 4 unused 0's. Now we have 8 objects, namely 4 0's and 4 100's. We want to find the number of strings we can form with these 8 objects, starting with a 100. After placing the 100 first, there are 7 places left for objects, 3 of which have to be 100's. Clearly there are $C(7, 3) = 35$ ways to choose the positions for the 100's, so our answer is 35.

31. This is a direct application of Theorem 3, with $n = 11$, $n_1 = 5$, $n_2 = 2$, $n_3 = n_4 = 1$, and $n_5 = 2$ (where n_i represents the number of A's, etc.). Thus the answer is $11!/(5!2!1!1!2!) = 83{,}160$.

33. We need to use the sum rule at the outermost level here, adding the number of strings using each subset of letters. There are quite a few cases. First, there are 3 strings of length 1, namely O, R, and N. There are several strings of length 2. If the string uses no O's, then there are 2; if it uses 1 O, then there are 2 ways to choose the other letter, and 2 ways to permute the letters in the string, so there are 4; and of course there is just 1 string of length 2 using 2 O's. Strings of length 3 can use 1, 2, or 3 O's. A little thought shows that the number of such strings is $3! = 6$, $2 \cdot 3 = 6$, and 1, respectively. There are 3 possibilities of the choice of letters for strings of length 4. If we omit an O, then there are $4!/2! = 12$ strings; if we omit either of the other letters (2 ways to choose the letter), then there are 4 strings. Finally, there are $5!/3! = 20$ strings of length 5. This gives a total of $3 + 2 + 4 + 1 + 6 + 6 + 1 + 12 + 2 \cdot 4 + 20 = 63$ strings using some or all of the letters.

35. We need to consider the three cases determined by the number of characters used in the string: 7, 8, or 9. If all nine letters are to be used, then Theorem 3 applies and we get

$$\frac{9!}{4!2!1!1!1!} = 7560$$

strings. If only eight letters are used, then we need to consider which letter is left out. In each of the cases in which the V, G, or N is omitted, Theorem 3 tells us that there are

$$\frac{8!}{4!2!1!1!} = 840$$

strings, for a total of 2520 for these cases. If an R is left out, then Theorem 3 tells us that there are

$$\frac{8!}{4!1!1!1!} = 1680$$

strings, and if an E is left out, then Theorem 3 tells us that there are

$$\frac{8!}{3!2!1!1!} = 3360$$

strings. This gives a total of $2520 + 1680 + 3360 = 7560$ strings of length 8. (It was not an accident that there are as many strings of length 8 as there are of length 9, since there is a one-to-one correspondence between these two sets, given by associating with a string of length 9 its first 8 characters.) For strings of length 7 there are even more cases. We tabulate them here:

omitting VG	$7!/(4!2!1!) = 105$ strings
omitting VN	$7!/(4!2!1!) = 105$ strings
omitting GN	$7!/(4!2!1!) = 105$ strings
omitting VR	$7!/(4!1!1!1!) = 210$ strings
omitting GR	$7!/(4!1!1!1!) = 210$ strings
omitting NR	$7!/(4!1!1!1!) = 210$ strings
omitting RR	$7!/(4!1!1!1!) = 210$ strings
omitting EV	$7!/(3!2!1!1!) = 420$ strings
omitting EG	$7!/(3!2!1!1!) = 420$ strings
omitting EN	$7!/(3!2!1!1!) = 420$ strings
omitting ER	$7!/(3!1!1!1!1!) = 840$ strings
omitting EE	$7!/(2!2!1!1!1!) = 1260$ strings

Adding up these numbers we see that there are 4515 strings of length 7. Thus the answer is $7560 + 7560 + 4515 = 19{,}635$.

37. We assume that all the fruit is to be eaten; in other words, this process ends after 7 days. This is a permutation problem since the order in which the fruit is consumed matters (indeed, there is nothing else that matters here). Theorem 3 applies, with $n = 7$, $n_1 = 3$, $n_2 = 2$, and $n_3 = 2$. The answer is therefore $7!/(3!2!2!) = 210$.

39. We can describe any such travel in a unique way by a sequence of 4 x's, 3 y's, and 5 z's. By Theorem 3 there are

$$\frac{12!}{4!3!5!} = 27720$$

such sequences.

41. This is like Example 8. If we approach it as is done there, we see that the answer is

$$C(52,7)C(45,7)C(38,7)C(31,7)C(24,7) = \frac{52!}{7!45!} \cdot \frac{45!}{7!38!} \cdot \frac{38!}{7!31!} \cdot \frac{31!}{7!24!} \cdot \frac{24!}{7!17!} = \frac{52!}{7!7!7!7!7!17!} \approx 7.0 \times 10^{34}.$$

Applying Theorem 4 will yield the same answer; in this approach we think of the five players and the undealt cards as the six distinguishable boxes.

43. We assume that we are to care about which player gets which cards. For example, a deal in which Laurel gets a royal flush in spades and Blaine gets a royal flush in hearts will be counted as different from a deal in which Laurel gets a royal flush in hearts and Blaine gets a royal flush in spades (and the other four players get the same cards each time). The order in which a player receives his or her cards is not relevant, however, so we are dealing with combinations. We can look at one player at a time. There are $C(48,5)$ ways to choose the cards for the first player, then $C(43,5)$ ways to choose the cards for the second player (because five of the cards are gone), and so on. So the answer, by the multiplication principle, is $C(48,5) \cdot C(43,5) \cdot C(38,5) \cdot C(33,5) \cdot C(28,5) \cdot C(23,5) = 649{,}352{,}163{,}073{,}816{,}339{,}512{,}038{,}979{,}194{,}880 \approx 6.5 \times 10^{32}$.

45. **a)** All that matters is how many copies of the book get placed on each shelf. Letting x_i be the number of copies of the book placed on shelf i, we are asking for the number of solutions to the equation $x_1 + x_2 + \cdots + x_k = n$, with each x_i a nonnegative integer. By Theorem 2 this is $C(k+n-1, n)$.

b) No generality is lost if we number the books b_1, b_2, ..., b_n and think of placing book b_1, then placing b_2, and so on. There are clearly k ways to place b_1, since we can put it as the first book (for now) on any of the shelves. After b_1 is placed, there are $k+1$ ways to place b_2, since it can go to the right of b_1 or it can be the first book on any of the shelves. We continue in this way: there are $k+2$ ways to place b_3 (to the right of b_1, to the right of b_2, or as the first book on some shelf), $k+3$ ways to place b_4, ..., $k+n-1$ ways to place b_n. Therefore the answer is the product of these numbers, which can more easily be expressed as $(k+n-1)!/(k-1)!$.

Another, perhaps easier, way to obtain this answer is to think of first choosing the locations for the books, which is what we counted in part **(a)**, and then choose a permutation of the n books to put into those locations (shelf by shelf, from the top down, and from left to right on each shelf). Thus the answer is $C(k+n-1, n) \cdot n!$, which evaluates to the same thing we obtained with our other analysis.

47. The first box holds n_1 objects, and there are $C(n, n_1)$ ways to choose those objects from among the n objects in the collection. Once these objects are chosen, we can choose the objects to be placed in the second box in $C(n-n_1, n_2)$ ways, since there are $n-n_1$ objects not yet placed, and we need to put n_2 of them into the second box. Similarly, there are then $C(n-n_1-n_2, n_3)$ ways to choose objects for the third box. We continue in this way, until finally there are $C(n-n_1-n_2-\cdots-n_{k-1}, n_k)$ ways to choose the objects to put in the last (k^{th}) box. Note that this last expression equals $C(n_k, n_k) = 1$, since $n_1 + n_2 + \cdots + n_k = n$. Now by the product rule the number of ways to make the entire assignment is

$$C(n, n_1) \cdot C(n-n_1, n_2) \cdot C(n-n_1-n_2, n_3) \cdots C(n-n_1-n_2-\cdots-n_{k-1}, n_k).$$

We use the formula for combinations to write this as

$$\frac{n!}{n_1!(n-n_1)!} \cdot \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdot \frac{(n-n_1-n_2)!}{n_3!(n-n_1-n_2-n_3)!} \cdots \frac{(n-n_1-n_2-\cdots-n_{k-1})!}{n_k!(n-n_1-n_2-\cdots-n_{k-1}-n_k)!},$$

which simplifies after the telescoping cancellation to

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

(we use the fact that $n-n_1-n_2-\cdots-n_{k-1} = n_k$, since $n_1 + n_2 + \cdots + n_k = n$), as desired.

49. **a)** The sequence was nondecreasing to begin with. By adding $k-1$ to the kth term, we are adding more to each term than was added to the previous term. Hence even if two successive terms in the sequence were originally equal, the second term must strictly exceed the first after this addition is completed. Therefore the sequence is made up of distinct numbers. The smallest can be no smaller than $1 + (1-1) = 1$, and the largest can be no larger than $n + (r-1) = n + r - 1$; therefore the terms all come from T.

b) If we are given an increasing sequence of r terms from T, then by subtracting $k-1$ from the kth term we have a nondecreasing sequence of r terms from S, repetitions allowed. (The kth term in the original sequence

must be between k and $n+r-1-(r-k)=n+(k-1)$, so subtracting $k-1$ leaves a number between 1 and n, inclusive. Furthermore, only 1 more is subtracted from a term than is subtracted from the previous term; thus no term can become strictly smaller than its predecessor, since it exceeded it by at least 1 to start with.) This operation exactly inverts the operation described in part **(a)**, so the correspondence is one-to-one.

c) The first two parts show that there are exactly as many r-combinations with repetition allowed from S as there are r-combinations (without repetition) from T. Since T has $n+r-1$ elements, this latter quantity is clearly $C(n+r-1,r)$.

51. We use the formula for the Stirling numbers of the second kind stated near the end of this section, which gives the number of ways to distribute n distinguishable objects into j indistinguishable boxes with no box empty:

$$S(n,j)=\frac{1}{j!}\sum_{i=0}^{j-1}(-1)^i\binom{j}{i}(j-i)^n$$

In this case, $n=6$ and $j=4$, so we have

$$S(6,4)=\frac{1}{4!}\left(\binom{4}{0}4^6-\binom{4}{1}3^6+\binom{4}{2}2^6-\binom{4}{3}1^6\right)=\frac{1}{4!}(4096-2916+384-4)=65.$$

If we want to work this out from scratch, we can argue as follows. There are two patterns possible. We can put three of the objects into one box and each of the remaining objects into a separate box; there are $C(6,3)=20$ ways to choose the objects for the crowded box. Alternatively, we can choose a pair of objects for one box ($C(6,2)=15$ ways) and a pair of remaining objects for the second box ($C(4,2)=6$ ways) and put the other two objects into separate boxes, but divide by 2 because of the overcounting caused by the indistinguishability of the first two boxes, for a total of 45 ways. Therefore the answer is $20+45=65$.

53. We assume that people are distinguishable, so this problem is identical to Exercise 51. There are 65 ways to place the employees.

55. Since each box has to contain at least one object, we might as well put one object into each box to begin with. This leaves us with just two more objects, and there are only two choices: we can put them both into the same box (so that the partition we end up with is $6=3+1+1+1$), or we can put them into different boxes (so that the partition we end up with is $6=2+2+1+1$). So the answer is 2.

57. Since each box has to contain at least two DVDs, we might as well put two DVDs into each box to begin with. This leaves us with just three more DVDs, and there are only three choices: we can put them all into the same box (so that the partition we end up with is $9=5+2+2$), or we can put two into one box and one into another (so that the partition we end up with is $9=4+3+2$), or we can put them all into different boxes (so that the partition we end up with is $9=3+3+3$). So the answer is 3.

59. To begin, notice that because each box must have at least one ball, there are only two basic arrangements: to put three balls into one box and one ball into each of the other two boxes (denoted 3-1-1), or to put one ball into one box and two balls into each of the other two boxes (denoted 1-2-2).

a) For the 3-1-1 arrangement, there are 3 ways to choose the crowded box, $C(5,3)=10$ ways to choose the balls to be put there, and 2 ways to decide where the other balls go, for a total of $3\cdot 10\cdot 2=60$ possibilities. For the 1-2-2 arrangement, there are 3 ways to choose the box that will have just one ball, 5 ways to choose which ball goes there, and $C(4,2)=6$ ways to decide which two balls go into the lower-numbered remaining box, for a total of $3\cdot 5\cdot 6=90$ possibilities. Thus the answer is $60+90=150$.

b) There are $C(5,3)=10$ ways to choose the balls for the crowded box in the 3-1-1 arrangement. For the 1-2-2 arrangement there are 5 ways to choose the lonely ball and 3 ways to choose the partner of the lowest-numbered remaining ball. Therefore the answer is $10+5\cdot 3=25$.

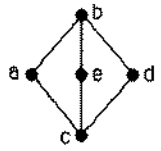
that $P(i)$ is maximized and following the C links.

SECTION 10.7 Planar Graphs

As with Euler and Hamilton circuits and paths, the topic of planar graphs is a classical one in graph theory. The theory (Euler's formula, Kuratowski's theorem, and their corollaries) is quite beautiful. It is easy to ask extremely difficult questions in this area, however—see Exercise 27, for example. In practice, there are very efficient algorithms for determining planarity that have nothing to do with Kuratowski's theorem, but they are quite complicated and beyond the scope of this book. For the exercises here, the best way to show that a graph is planar is to draw a planar embedding; the best way to show that a graph is nonplanar is to find a subgraph homeomorphic to K_5 or $K_{3,3}$. (Usually it will be $K_{3,3}$.)

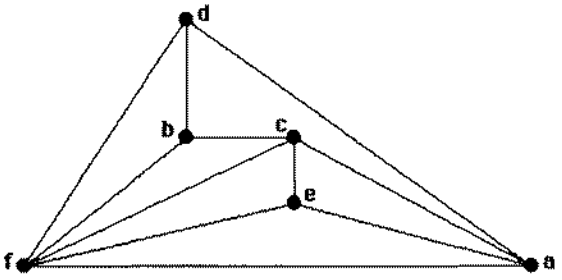
1. The question is whether $K_{5,2}$ is planar. It clearly is so, since we can draw it in the xy-plane by placing the five vertices in one part along the x-axis and the other two vertices on the positive and negative y-axis.

3. For convenience we label the vertices a, b, c, d, e, starting with the vertex in the lower left corner and proceeding clockwise around the outside of the figure as drawn in the exercise. This graph is just $K_{2,3}$; the picture below shows it redrawn by moving vertex c down.



5. This is $K_{3,3}$, with parts $\{a, d, f\}$ and $\{b, c, e\}$. Therefore it is not planar.

7. This graph can be untangled if we play with it long enough. The following picture gives a planar representation of it.



9. If one has access to software such as *The Geometer's Sketchpad*, then this problem can be solved by drawing the graph and moving the points around, trying to find a planar drawing. If we are unable to find one, then we look for a reason why—either a subgraph homeomorphic to K_5 or one homeomorphic to $K_{3,3}$ (always try the latter first). In this case we find that there is a homeomorphic copy of $K_{3,3}$, with vertices b, g, and i in one set and a, f, and h in the other; all the edges are there except for the edge bh, and it is represented by the path beh.

11. We give a proof by contradiction. Suppose that there is a planar representation of K_5, and let us call the vertices v_1, v_2, ..., v_5. There must be an edge from every vertex to every other. In particular, v_1, v_2, v_3, v_4, v_5, v_1 must form a pentagon. The pentagon separates the plane into two regions, an inside and an outside. The edge from v_1 to v_3 must be present, and without loss of generality let us assume it is drawn on the inside. Then there is no way for edges $\{v_2, v_4\}$ and $\{v_2, v_5\}$ to be in the inside, so they must be in the outside region. Now this prevents edges $\{v_1, v_4\}$ and $\{v_3, v_5\}$ from being on the outside. But they cannot both be on the inside without crossing. Therefore there is no planar representation of K_5.

13. We apply Euler's formula $r = e - v + 2$. Here we are told that $v = 6$. We are also told that each vertex has degree 4, so that the sum of the degrees is 24. Therefore by the handshaking theorem there are 12 edges, so $e = 12$. Solving, we find $r = 8$.

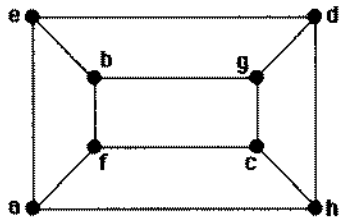
15. The proof is very similar to the proof of Corollary 1. First note that the degree of each region is at least 4. The reason for this is that there are no loops or multiple edges (which would give regions of degree 1 or 2) and no simple circuits of length 3 (which would give regions of degree 3); and the degree of the unbounded region is at least 4 since we are assuming that $v \geq 3$. Therefore we have, arguing as in the proof of Corollary 1, that $2e \geq 4r$, or simply $r \leq e/2$. Plugging this into Euler's formula, we obtain $e - v + 2 \leq e/2$, which gives $e \leq 2v - 4$ after some trivial algebra.

17. The proof is exactly the same as in Exercise 15, except that this time the degree of each region must be at least 5. Thus we get $2e \geq 5r$, which after the same algebra as before, gives the desired inequality.

19. **a)** If we remove a vertex from K_5, then we get K_4, which is clearly planar.
b) If we remove a vertex from K_6, then we get K_5, which is not planar.
c) If we remove a vertex from $K_{3,3}$, then we get $K_{3,2}$, which is clearly planar.
d) We assume the question means "Is it the case that for every v, the removal of v makes the graph planar?" Then the answer is no, since we can remove a vertex in the part of size 4 to leave $K_{3,3}$, which is not planar.

21. This graph is planar and hence cannot be homeomorphic to $K_{3,3}$.

23. The instructions are really not fair. It is hopeless to try to use Kuratowski's theorem to prove that a graph *is* planar, since we would have to check hundreds of cases to argue that there is no subgraph homeomorphic to K_5 or $K_{3,3}$. Thus we will show that this graph is planar simply by giving a planar representation. Note that it is Q_3.



25. This graph is nonplanar, since it contains $K_{3,3}$ as a subgraph: the parts are $\{a, g, d\}$ and $\{b, c, e\}$. (Actually it contains $K_{3,4}$, and it even contains a subgraph homeomorphic to K_5.)

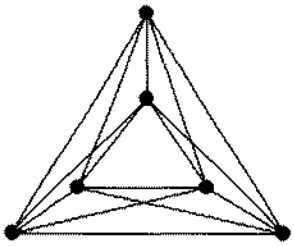
27. This is an extremely hard problem. We will present parts of the solution; the reader should consult a good graph theory book, such as Gary Chartrand, Linda Lesniak and Ping Zhang's *Graphs & Digraphs*, fifth edition (Chapman & Hall/CRC Press, 2011), for references and further details.

First we will state, without proof, what is known about crossing numbers for complete graphs (much is still not known about crossing numbers). If $n \leq 10$, then the crossing number of K_n is given by the following product

$$\frac{1}{4}\left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor \left\lfloor \frac{n-2}{2} \right\rfloor \left\lfloor \frac{n-3}{2} \right\rfloor.$$

Thus the answers for parts **(a)**, **(b)**, and **(c)** are 1, 3, and 9, respectively. The figure below shows K_6 drawn in the plane with three crossings, which at least proves that the crossing number of K_6 is at most 3. The

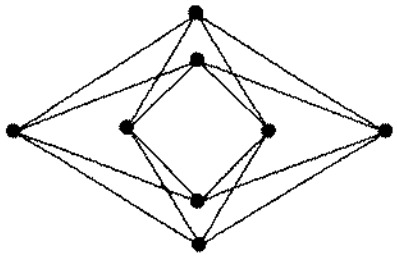
proof that it is not less than 3 is not easy. The embedding of K_5 with one crossing can be seen in this same picture, by ignoring the vertex at the top.



Second, for the complete bipartite graphs, what is known is that if the smaller of m and n is at most 6, then the crossing number of $K_{m,n}$ is given by the following product

$$\left\lfloor \frac{m}{2} \right\rfloor \left\lfloor \frac{m-1}{2} \right\rfloor \left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Thus the answers for parts **(d)**, **(e)**, and **(f)** are 2, 4, and 16, respectively. The figure below shows $K_{4,4}$ drawn in the plane with four crossings, which at least proves that the crossing number of $K_{4,4}$ is at most 4. The proof that it is not less than 4 is, again, difficult. It is also easy to see from this picture that the crossing number of $K_{3,4}$ is at most 2 (by ignoring the top vertex).



29. Let us follow the hint, and draw all the edges with straight line segments. This clearly produces a drawing of $K_{m,n}$. We will show that the number of crossings is $mn(m-2)(n-2)/16$, and that will complete the proof. (Incidentally, it is not known whether this upper bound is actually the crossing number. No one has found an embedding with fewer crossings, but only in the case in which the smaller of m and n is at most 6 has it been proved that it cannot be done. See the comments in the solution to Exercise 27.) In order to count the crossings, it is enough to count the crossings occurring in the first quadrant and multiply by 4. Let us label the points on the positive x-axis with the numbers 1 through $m/2$, and those on the y-axis with the numbers 1 through $n/2$. If we choose any two distinct numbers, say a and b with $a < b$, from 1 to $m/2$, and any two distinct numbers, say r and s with $r < s$, from 1 through $n/2$, then we get exactly one crossing in our graph, namely between the edges as and br. (There is no crossing between ar and bs.) So the number of crossings in the first quadrant is the same as the number of ways to make these choices, which is clearly $C(m/2,2) \cdot C(n/2,2)$. So the total number of crossings is 4 times this quantity, namely

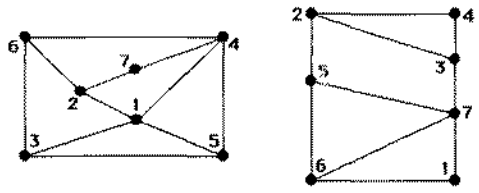
$$4 \cdot C(m/2,2) \cdot C(n/2,2) = 4 \cdot \frac{\frac{m}{2}\left(\frac{m}{2}-1\right)}{2} \cdot \frac{\frac{n}{2}\left(\frac{n}{2}-1\right)}{2},$$

which easily simplifies to

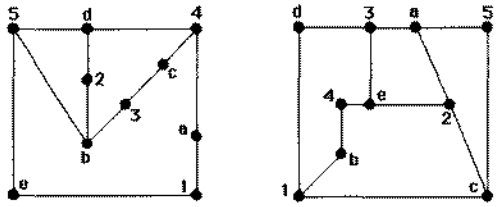
$$\frac{mn(m-2)(n-2)}{16}.$$

31. Each of these graphs is nonplanar; the first three contain K_5, and the last three contain $K_{3,3}$. Thus if we can show how to draw each of the graphs in two planes, then we will have shown that the thickness is 2 in each

case. The following picture shows that K_7 can be drawn in 2 planes, so this takes care of part **(a)**, part **(b)**, and part **(c)**.



The following picture shows that $K_{5,5}$ can be drawn in 2 planes, so this takes care of part **(d)**, part **(e)**, and part **(f)**.



33. The formula is certainly valid for $n \leq 4$, so let us assume that $n > 4$. By Exercise 32, the thickness of K_n is at least

$$\frac{C(n,2)}{3n-6} = \frac{n(n-1)/2}{3n-6} = \frac{n(n-1)}{6(n-2)} = \frac{1}{6}\left(n+1+\frac{2}{n-2}\right)$$

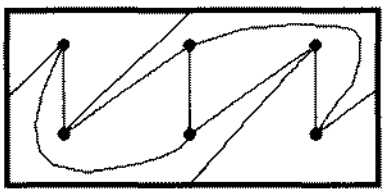
rounded up. Since this quantity is never an integer, it equals one more than itself rounded down, namely

$$\frac{1}{6}\left(n+1+\frac{2}{n-2}\right)+1 = \frac{n+7}{6}+\frac{2}{6(n-2)}$$

rounded down. The last term can be ignored: it is always less than $1/6$ and therefore will not influence the rounding process (since the first term has denominator 6). Thus we have proved that the thickness of K_n is at least $\lfloor (n+7)/6 \rfloor$.

35. This follows immediately from Exercise 34, since $K_{m,n}$ has mn edges and $m+n$ vertices and, being bipartite, has no triangles.

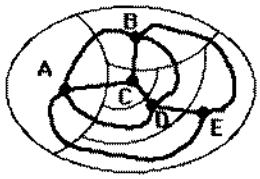
37. We can represent the surface of a torus with a rectangle, thinking of the right-hand edge as being equal to the left-hand edge, and the top edge as being equal to the bottom edge. For example, if we travel out of the rectangle across the right-hand edge about a third of the way from the top, then we immediately reenter the rectangle across the left-hand edge about a third of the way from the top. The picture below shows $K_{3,3}$ drawn on this surface. Note that the edges that seem to leave the rectangle really reenter it from the opposite side.



SECTION 10.8 Graph Coloring

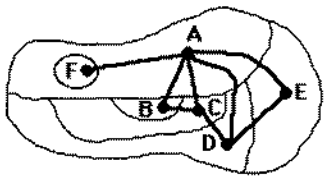
Like the problem of finding Hamilton paths, the problem of finding colorings with the fewest possible colors probably has no good algorithm for its solution. In working these exercises, for the most part you should proceed by trial and error, using whatever insight you can gain by staring at the graph (for instance, finding large complete subgraphs). There are also some interesting exercises here on coloring the edges of graphs—see Exercises 21–26. Exercises 29–31 are worth looking at, as well: they deal with a fast algorithm for coloring a graph that is not guaranteed to produce an optimal coloring.

1. We construct the dual graph by putting a vertex inside each region (but not in the unbounded region), and drawing an edge between two vertices if the regions share a common border. The easiest way to do this is illustrated in our answer. First we draw the map, then we put a vertex inside each region and make the connections. The dual graph, then, is the graph with heavy lines.



The number of colors needed to color this map is the same as the number of colors needed to color the dual graph. Since A, B, C, and D are mutually adjacent, at least four colors are needed. We can color each of the vertices (i.e., regions) A, B, C, and D a different color, and we can give E the same color as we give C.

3. We construct the dual as in Exercise 1.



As in Exercise 1, the number of colors needed to color this map is the same as the number of colors needed to color the dual graph. Three colors are clearly necessary, because of the triangle ABC, for instance. Furthermore three colors suffice, since we can color vertex (region) A red, vertices B, D, and F blue, and vertices C and E green.

5. For Exercises 5–11, in order to prove that the chromatic number is k, we need to find a k-coloring and to show that (at least) k colors are needed. Here, since there is a triangle, at least 3 colors are needed. Clearly 3 colors suffice, since we can color a and d the same color.

7. Since there is a triangle, at least 3 colors are needed. Clearly 3 colors suffice, since we can color a and c the same color.

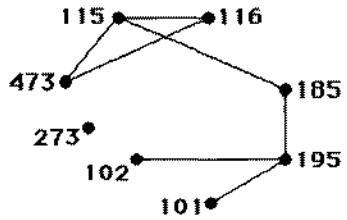
9. Since there is an edge, at least 2 colors are needed. The coloring in which b, d, and e are red and a and c blue shows that 2 colors suffice.

11. Since there is a triangle, at least 3 colors are needed. It is not hard to construct a 3-coloring. We can let a, f, h, j, and n be blue; let b, d, g, k, and m be green; and let c, e, i, l, and o be yellow.

13. If a graph has an edge (not a loop, since we are assuming that the graphs in this section are simple), then its chromatic number is at least 2. Conversely, if there are no edges, then the coloring in which every vertex receives the same color is proper. Therefore a graph has chromatic number 1 if and only if it has no edges.

15. In Example 4 we saw that the chromatic number of C_n is 2 if n is even and 3 if n is odd. Since the wheel W_n is just C_n with one more vertex, adjacent to all the vertices of the C_n along the rim of the wheel, W_n clearly needs exactly one more color than C_n (for that middle vertex). Therefore the chromatic number of W_n is 3 if n is even and 4 if n is odd.

17. Consider the graph representing this problem. The vertices are the 8 courses, and two courses are joined by an edge if there are students taking both of them. Thus there are edges between every pair of vertices except the 7 pairs listed. It is much easier to draw the complement than to draw this graph itself; it is shown below.



We want to find the chromatic number of the graph whose complement we have drawn; the colors will be the time periods for the exams. First note that since Math 185 and the four CS courses form a K_5 (in other words, there are no edges between any two of these in our picture), the chromatic number is at least 5. To show that it equals 5, we just need to color the other three vertices. A little trial and error shows that we can make Math 195 the same color as (i.e., have its final exam at the same time as) CS 101; and we can make Math 115 and 116 the same color as CS 473. Therefore five time slots (colors) are sufficient.

19. We model the problem with the intersection graph of these sets. Note that every pair of these intersect except for C_4 and C_5. Thus the graph is K_6 with that one edge deleted. Clearly its chromatic number is 5, since we need to color all the vertices different colors, except that C_4 and C_5 may have the same color. In other words, 5 meeting times are needed, since only committees C_4 and C_5 can meet simultaneously.

21. Note that the number of colors needed to color the edges is at least as large as the largest degree of a vertex, since the edges at each vertex must all be colored differently. Hence if we can find an edge coloring with that many colors, then we know we have found the answer. In Exercise 5 there is a vertex of degree 3, so the edge chromatic number is at least 3. On the other hand, we can color $\{a, c\}$ and $\{b, d\}$ the same color, so 3 colors suffice. In Exercise 6 the 6 edges incident to g must all get different colors. On the other hand, it is not hard to complete a proper edge coloring with only these colors (for example, color edge $\{a, f\}$ with the same color as used on $\{d, g\}$), so the answer is 6. In Exercise 7 the answer must be at least 3; it is 3 since edges that appear as parallel line segments in the picture can have the same color. In Exercise 8 clearly 4 colors are required, since the vertices have degree 4. In fact 4 colors are sufficient. Here is one proper 4-coloring (we denote edges in the obvious shorthand notation): color 1 for ac, be, and df; color 2 for ae, bd, and cf; color 3 for ab, cd, and ef; and color 4 for ad, bf, and ce. In Exercise 9 the answer must be at least 3; it is easy to construct a 3-coloring of the edges by inspection: $\{a, b\}$ and $\{c, e\}$ have the same color, $\{a, d\}$ and $\{b, c\}$ have the same color, and $\{a, e\}$ and $\{c, d\}$ have the same color. In Exercise 10 the largest degree is 6 (vertex i has degree 6); therefore at least 6 colors are required. By trial and error we come up with this coloring using 6 colors (we use the obvious shorthand notation for edges); there are many others, of course. Assign color 1 to ag, cd, and hi; color 2 to ab, cf, dg, and ei; color 3 to bh, cg, di, and ef; color 4 to ah, ci, and de; color 5 to bi, ch, and fg; and color 6 to ai, bc, and gh. Finally, in Exercise 11 it is easy to construct an edge-coloring with 4 colors; again the edge chromatic number is the maximum degree of a vertex.

Despite the appearances of these examples, it is not the case that the edge chromatic number of a graph is always equal to the maximum degree of the vertices in the graph. The simplest example in which this is not

true is K_3. Clearly its edge chromatic number is 3 (since all three edges are adjacent to each other), but its maximum degree is 2. There is a theorem, however, stating that the edge chromatic number is always equal to either the maximum degree or one more than the maximum degree.

23. **a)** The n-cycle's edges are just like the n-cycle's vertices (each adjacent to the next as we go around the cycle), so the edge chromatic number is the same, namely 2 if n is even and 3 if n is odd, as in Example 4. **b)** The edge chromatic number is at least n, because the radial edges are all pairwise adjacent and therefore must all have distinct colors. Suppose we call these colors 1 through n proceeding clockwise. We need no additional colors for the edges of the cycle, because we can color the edge adjacent to the spokes colored 1 and 2 with color 3 and proceed clockwise with colors 4, 5, ..., $n-1$, n, 1, and 2. Therefore $\chi'(W_n) = n$.

25. Two edges that have the same color share no endpoints. Therefore if more than $n/2$ edges were colored the same, the graph would have more than $2(n/2) = n$ vertices.

27. This problem can be modeled with the intersection graph of the sets of steps during which the variables must be stored. This graph has 7 vertices, t through z; there is an edge between two vertices if the two variables they represent must be stored during some common step. The answer to the problem is the chromatic number of this graph. Rather than considering this graph, we look at its complement (it has a lot fewer edges). Here two vertices are adjacent if the sets (of steps) do not intersect. The only edges are $\{u,w\}$, $\{u,x\}$, $\{u,y\}$, $\{u,z\}$, $\{v,x\}$, $\{x,z\}$. Note that there are no edges in the complement joining any two of $\{t,v,w,y,z\}$, so that these vertices form a K_5 in the original graph. Thus the chromatic number of the original graph is at least 5. To see that it is 5, note that vertex u can have the same color as w, and x can have the same color as z (these pairs appear as edges in the complement). Since the chromatic number is 5, we need 5 registers, with variables u and w sharing a register, and vertices x and z sharing one.

29. First we need to list the vertices in decreasing order of degree. This ordering is not unique, of course; we will pick e,a,b,c,f,h,i,d,g,j. Next we assign color 1 to e, and then to f and d, in that order. Now we assign color 2 to a, c, i, and g, in that order. Finally, we assign color 3 to b, h and j, in that order. Thus the algorithm gives a 3-coloring. Since the graph contains triangles, we know that this is the best possible, so the algorithm "worked" here (but it need not always work—see Exercise 27).

31. A simple example in which the algorithm may fail to provide a coloring with the minimum number of colors is C_6, which of course has chromatic number 2. Since all the vertices are of degree 2, we may order them v_1, v_4, v_2, v_3, v_5, v_6, where the edges are $\{v_1,v_2\}$, $\{v_2,v_3\}$, $\{v_3,v_4\}$, $\{v_4,v_5\}$, $\{v_5,v_6\}$, and $\{v_1,v_6\}$. Then v_1 gets color 1, as does v_4. Next v_2 and v_5 get color 2; and then v_3 and v_6 must get color 3.

33. We need to show that the wheel W_n when n is an odd integer greater than 1 can be colored with four colors, but that any graph obtained from it by removing one edge can be colored with three colors. Four colors are needed to color this graph, because three colors are needed for the rim (see Example 4), and the center vertex, being adjacent to all the rim vertices, will require a fourth color. To complete the proof that W_n is chromatically 4-critical, we must show that the graph obtained from W_n by deleting one edge can be colored with three colors. There are two cases. If we remove a rim edge, then we can color the rim with two colors, by starting at an endpoint of the removed edge and using the colors alternately around the portion of the rim that remains. The third color is then assigned to the center vertex. On the other hand, if we remove a spoke edge, then we can color the rim by assigning color #1 to the rim endpoint of the removed edge and colors #2 and #3 alternately to the remaining vertices on the rim, and then assign color #1 to the center.

35. We give a proof by contradiction. Suppose that G is chromatically k-critical but has a vertex v of degree $k-2$ or less. Remove from G one of the edges incident to v. By definition of "k-critical," the resulting graph can be colored with $k-1$ colors. Now restore the missing edge and use this coloring for all vertices except v. Because we had a proper coloring of the smaller graph, no two adjacent vertices have the same color. Furthermore, v has at most $k-2$ neighbors, so we can color v with an unused color to obtain a proper $(k-1)$-coloring of G. This contradicts the fact that G has chromatic number k. Therefore our assumption was wrong, and every vertex of G must have degree at least $k-1$.

37. **a)** Note that vertices d, e, and f are mutually adjacent. Therefore six different colors are needed in a 2-tuple coloring, since each of these three vertices needs a disjoint set of two colors. In fact it is easy to give a coloring with just six colors: Color a, d, and g with $\{1,2\}$; color c and e with $\{3,4\}$; and color b and f with $\{5,6\}$. Thus $\chi_2(G) = 6$.

b) This one is trickier than part **(a)**. There is no coloring with just six colors, since if there were, we would be forced (without loss of generality) to color d with $\{1,2\}$; e with $\{3,4\}$; f with $\{5,6\}$; then g with $\{1,2\}$, b with $\{5,6\}$, and c with $\{3,4\}$. This gives no free colors for vertex a. Now this may make it appear that eight colors are required, but a little trial and error shows us that seven suffice: Color a with $\{2,4\}$; color b and f with $\{5,6\}$; color d with $\{1,2\}$; color c with $\{3,7\}$; color e with $\{3,4\}$; and color g with $\{1,7\}$. Thus $\chi_2(H) = 7$.

c) This is similar to part **(a)**. Here nine colors are necessary and sufficient, since a, d, and g can get one set of three colors; b and f can get a second set; and c and e can get a third set. Clearly nine colors are necessary to color the triangles.

d) First we construct a coloring with 11 colors: Color a with $\{3,6,11\}$; color b and f with $\{7,8,9\}$; color d with $\{1,2,3\}$; color c with $\{4,5,10\}$; color e with $\{4,6,11\}$; and color g with $\{1,2,5\}$. To prove that $\chi_3(H) = 11$, we must show that it is impossible to give a 3-tuple coloring with only ten colors. If such a coloring were possible, without loss of generality we could color d with $\{1,2,3\}$, e with $\{4,5,6\}$, f with $\{7,8,9\}$, and g with $\{1,2,10\}$. Now nine colors are needed for the three vertices a, b, and c, since they form a triangle; but colors 1 and 2 are already used in vertices adjacent to all three of them. Therefore at least $9+2 = 11$ colors are necessary.

39. The frequencies are the colors, the zones are the vertices, and two zones that are so close that interference would be a problem are joined by an edge in the graph. Then it is clear that a k-tuple coloring is exactly an assignment of frequencies that avoids possible interference.

41. We use induction on the number of vertices of the graph. Every graph with five or fewer vertices can be colored with five or fewer colors, since each vertex can get a different color. That takes care of the basis case(s). So we assume that all graphs with k vertices can be 5-colored and consider a graph G with $k+1$ vertices. By Corollary 2 in Section 10.7, G has a vertex v with degree at most 5. Remove v to form the graph G'. Since G' has only k vertices, we 5-color it by the inductive hypothesis. If the neighbors of v do not use all five colors, then we can 5-color G by assigning to v a color not used by any of its neighbors. The difficulty arises if v has five neighbors, and each has a different color in the 5-coloring of G'. Suppose that the neighbors of v, when considered in clockwise order around v, are a, b, c, m, and p. (This order is determined by the clockwise order of the curves representing the edges incident to v.) Suppose that the colors of the neighbors are azure, blue, chartreuse, magenta, and purple, respectively. Consider the azure-chartreuse subgraph (i.e., the vertices in G colored azure or chartreuse and all the edges between them). If a and c are not in the same component of this graph, then in the component containing a we can interchange these two colors (make the azure vertices chartreuse and vice versa), and G' will still be properly colored. That makes a chartreuse, so we can now color v azure, and G has been properly colored. If a and c are in the same component, then there is a path of vertices alternately colored azure and chartreuse joining a and c. This path together with

edges av and vc divides the plane into two regions, with b in one of them and m in the other. If we now interchange blue and magenta on all the vertices in the same region as b, we will still have a proper coloring of G', but now blue is available for v. In this case, too, we have found a proper coloring of G. This completes the inductive step, and the theorem is proved.

43. We follow the hint. Because the measures of the interior angles of a pentagon total 540°, there cannot be as many as three interior angles of measure more than 180° (reflex angles). If there are no reflex angles, then the pentagon is convex, and a guard placed at any vertex can see all points. If there is one reflex angle, then the pentagon must look essentially like figure (a) below, and a guard at vertex v can see all points. If there are two reflex angles, then they can be adjacent or nonadjacent (figures (b) and (c)); in either case, a guard at vertex v can see all points. (In figure (c), choose the reflex vertex closer to the bottom side.) Thus for all pentagons, one guard suffices, so $g(5) = 1$.



45. The figure suggested in the hint (generalized to have k prongs for any $k \geq 1$) has $3k$ vertices. Consider the set of points from which a guard can see the tip of the first prong, the set of points from which a guard can see the tip of the second prong, and so on. These are disjoint triangles (together with their interiors). Therefore a separate guard is needed for each of the k prongs, so at least k guards are needed. This shows that $g(3k) \geq k = \lfloor 3k/3 \rfloor$. To handle values of n that are not multiples of 3, let $n = 3k + i$, where $i = 1$ or 2. Then obviously $g(n) \geq g(3k) \geq k = \lfloor n/3 \rfloor$.

GUIDE TO REVIEW QUESTIONS FOR CHAPTER 10

1. **a)** See pp. 641–642 and Table 1 in Section 10. **b)** See Exercise 1 in Section 10.1.

2. See all the examples Section 10.1.

3. See Theorem 1 in Section 10.2.

4. See Theorem 2 in Section 10.2.

5. See Theorem 3 in Section 10.2.

6. **a)** See Example 5 in Section 10.2. **b)** See Example 13 in Section 10.2.
c) See Example 6 in Section 10.2. **d)** See Example 7 in Section 10.2.
e) See Example 8 in Section 10.2.

7. **a)** n, $C(n,2)$ **b)** $m+n$, mn **c)** n, n **d)** $n+1$, $2n$ **e)** 2^n, $n2^{n-1}$

8. **a)** See p. 656. **b)** K_2 and C_{2m}
c) (See also Example 12 and Exercise 66 in Section 10.2.) The following algorithm is an efficient way to determine whether a connected graph can be 2-colored (which is the same thing as saying that it is bipartite); apply it to each component of the given graph. First color any vertex red. Then color all vertices adjacent to this vertex blue. Then look at all vertices adjacent to these just-colored blue vertices. If any of them are already colored blue, then stop and declare the graph not to be bipartite; otherwise color all the uncolored ones red. Next look at all vertices adjacent to all the vertices just colored red. If any of them are already

colored red, then stop and declare the graph not to be bipartite; otherwise color all the uncolored ones blue. Continue in this way until no more vertices can be colored. If we get this far, then (this component of) the graph is bipartite. (If uncolored vertices remain, they are in a different component, so we can repeat the entire process starting with any uncolored vertex.)

9. **a)** adjacency lists, adjacency matrices, incidence matrices

b) Look at Figure 1 in Section 10.3, and let G be the graph consisting of the vertices and edges shown there, together with edges $\{b, c\}$ and $\{b, e\}$. Its adjacency lists are shown on p. 668 (Table 1), once we add c and e to the list of adjacent vertices of b, and add b to the list for c and for e. Its adjacency matrix and incidence matrix are as follows (using alphabetical order):

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

10. **a)** See p. 672.

b) See p. 672; number of vertices, number of edges, degrees of vertices, existence of triangle (C_3 as subgraph), and existence of Hamilton circuit are all invariants.

c) See Figure 10 in Section 10.3. **d)** no

11. **a)** See p. 681. **b)** See p. 682.

12. **a)** See p. 669. **b)** See p. 673. **c)** See Theorem 2 in Section 10.4.

13. **a)** See p. 694. **b)** See pp. 693–694.

c) All edges are in the same component and there are at most two vertices of odd degree (see Theorems 1 and 2 in Section 10.5).

d) All edges are in the same component and there are no vertices of odd degree (see Theorem 1 in Section 10.5).

14. **a)** See p. 698. **b)** the existence of a cut edge or a cut vertex

15. finding the shortest highway route between two cities; finding the cheapest way to lease telephone lines to join two communications centers, using intermediate switching centers

16. **a)** See pp. 709–712. **b)** See Exercise 5, third part, in Section 10.6.

17. **a)** See p. 719. **b)** K_6

18. **a)** If the planar graph has v vertices, e edges, and c components, and is embedded in the plane so as to form r regions, then $v - e + r = 1 + c$. (The theorem as stated in the text—Theorem 1 in Section 10.7—assumes that $c = 1$, but this is the more general statement.)

b) By Corollary 1 in Section 10.7, for every planar graph with at least three vertices we know that $e \leq 3v - 6$ (connectivity need not be assumed). Thus we can show that a graph is nonplanar by showing that it has too many edges, namely more than $3v - 6$ edges. For example, K_6 is nonplanar, since it has 15 edges, and $15 \not\leq 3 \cdot 6 - 6$.

19. See Theorem 2 in Section 10.7. A graph is planar if and only if it does not contain a subgraph homeomorphic to K_5 or $K_{3,3}$.

20. **a)** See p. 728. **b)** n **c)** 2 if n is even, 3 if n is odd **d)** 2

21. Every planar graph can be 4-colored (i.e., has chromatic number at most 4), but K_6, for instance, requires six colors.

22. See Examples 5–7 in Section 10.8.

SUPPLEMENTARY EXERCISES FOR CHAPTER 10

1. Every vertex has degree 50. Thus the sum of the degrees must be $50 \cdot 100 = 5000$. By the handshaking theorem, the graph therefore has $5000/2 = 2500$ edges.

3. Both graphs have a lot of symmetry to them, and the degrees of the vertices are the same, so we might hope that they are isomorphic. Let us try to form the correspondence f. First note that there is a 4-cycle u_1, u_5, u_2, u_6, u_1 in the first graph. Suppose that we try letting it correspond to the 4-cycle v_1, v_2, v_3, v_4, v_1 in the second graph. Thus we let $f(u_1) = v_1$, $f(u_5) = v_2$, $f(u_2) = v_3$, and $f(u_6) = v_4$. The rest of the assignments are forced; since u_7 is the other vertex adjacent to u_1, we must let $f(u_7) = v_6$, since v_6 is the other vertex adjacent to v_1 (which is $f(u_1)$). Similarly, $f(u_3) = v_7$, $f(u_8) = v_8$, and $f(u_4) = v_5$. Now we just have to check that the vertices corresponding to the vertices in the 4-cycle v_5, v_6, v_7, v_8, v_5 in the second graph form a 4-cycle in that order. Since these vertices form the 4-cycle u_4, u_7, u_3, u_8, u_4, our correspondence works.

5. These graphs are isomorphic, although the isomorphism is hard to find. One approach that can lead to the isomorphism is to draw the complement of each graph. The complements are a little simpler than the original graphs, since they have fewer edges. When we do this, it is easy to give a planar representation of each. Then by looking at the sizes of the regions we can find an isomorphism. One such correspondence is $u_1 \leftrightarrow v_5$, $u_2 \leftrightarrow v_4$, $u_3 \leftrightarrow v_7$, $u_4 \leftrightarrow v_3$, $u_5 \leftrightarrow v_8$, $u_6 \leftrightarrow v_2$, $u_7 \leftrightarrow v_6$, and $u_8 \leftrightarrow v_1$. We just need to check that all the edges are preserved by this correspondence.

7. It follows immediately from the definition that the complete m-partite graph with parts $n_1, n_2, \ldots, n_m$ has $n_1 + n_2 + \cdots + n_m = \sum_{i=1}^{m} n_i$ vertices. We will organize the count of the edges by looking at which parts the edges join. Fix $1 \leq i < j \leq m$, and consider the edges between the ith part and the jth part. It is easy to see from the product rule that there are $n_i n_j$ edges. Therefore to get all the edges, we have to add all these products, for all possible pairs (i, j). Thus the number of edges is $\sum_{1 \leq i < j \leq m} n_i n_j$.

9. **a)** We prove that each set is a subset of the other. If $x \in N(A \cup B)$, then x is adjacent to some vertex $v \in A \cup B$. If $v \in A$, then $x \in N(A)$ and therefore also in $N(A) \cup N(B)$; the other case is similar. Conversely, if $x \in N(A) \cup N(B)$, then without loss of generality assume $x \in N(A)$. That means that x is adjacent to some vertex $v \in A$. Since $v \in A \cup B$, we have $x \in N(A \cup B)$.

b) If $x \in N(A \cap B)$, then x is adjacent to some vertex $v \in A \cap B$. Since both $v \in A$ and $v \in B$, it follows that $x \in N(A)$ and $x \in N(B)$, whence $x \in N(A) \cap N(B)$. For an example to show that equality may fail, consider $K_{1,2}$, let A be the set consisting of one of the vertices of degree 1, and let B be the set consisting of the other vertex of degree 1. Then $A \cap B = \emptyset$, so $N(A \cap B) = \emptyset$, but $N(A) \cap N(B)$ is the vertex of degree 2.

11. There are many possible SDR's here, which can be found by trial and error. One is (m, e, p, x, n, a).

13. **a)** There are actually 15 subsets of $\{1, 2, 3, 4\}$ to check here (letting $I = \emptyset$ always gives the trivial true fact that $0 \leq 0$), but since each set S_i has at least three elements, the inequality is automatically true for all I of size less than 4. The only other case is $I = \{1, 2, 3, 4\}$ in which case both sides of the inequality are $|\{a, b, c, d\}|$.

b) Many choices are possible, such as (a, b, d, c).

15. Consider each of the ordered triples of vertices (u, v, w) of G. It may happen that u and v are neighbors and that v and w are neighbors; for all triples for which this is so, we want to know how many also have u adjacent to w. The corresponding fraction is by definition the clustering coefficient of G. Each triangle in G gives rise to 6 ordered triples that contribute to the numerator of this fraction. Each path of length 2 in G gives rise to one ordered triple that contribute to the denominator of this fraction. Thus $C(G)$ equals 6 times the number of triangles divided by the number of paths of length 2. For example, in K_3, there is one triangle and there are 6 paths of length 2, so $C(K_3) = 6 \cdot 1/6 = 1$. Similarly, $C(W_4) = 6 \cdot 4/36 = 2/3$.

17. Each of these can be phrased in terms of a probability—how likely something is to happen. It is a little subtle, though. In each case, we have to start with the common neighbor. See part **(a)** for the more precise wording; the others are similar.

a) How likely is it that two actors will have appeared in a film together if we know that there is a third actor who has appeared in a film with each of them? To be more precise, if we randomly pick an actor and two actors who have appeared in a film with that actor, how likely is it that those two actors have appeared in a film together. This is not the same as starting with two random actors who have a common co-star, because if there is more than one common co-star, the probability goes up.

b) How likely is it that two people who have a Facebook friend in common are themselves Facebook friends? A little more precisely: How likely is it that two of a randomly chosen person's friends are themselves friends?

c) How likely is it that two coauthors of a randomly chosen author are themselves coauthors?

d) How likely is it that two proteins that each interact with a randomly chosen protein interact with each other?

e) How likely is it that two routers each of which has a communications link to a randomly chosen router are themselves linked?

19. In general it is no easy task to find cliques. We need to be careful not to overlook things. We will denote a clique simply by listing the vertices in it, without punctuation. There is one K_4, namely $bcef$, and it is the largest clique. There are several K_3's not contained in this K_4, and they are all cliques: abg, adg, beg, and deg. Since every edge is contained in one of these five cliques (and there are no isolated vertices), there are no smaller cliques, so this list is complete.

21. See the comments for Exercise 19. We find the cliques by brute force and careful looking, hoping that we do not miss any. Some staring at the graph convinces us that there are no K_6's. There is one K_5, namely the clique $bcdjk$. There are two K_4's not contained in this K_5, which therefore are cliques: $abjk$, and $efgi$. All the K_3's not contained in any of the cliques listed so far are also cliques. We find abi, aij, bde, bei, bij, ghi, and hij. All the edges are in at least one of the cliques listed so far (and there are no isolated vertices), so we are done.

23. Clearly no single vertex is dominating by itself, but $\{c, d\}$ dominates, so it is a minimum dominating set (there are lots of others).

25. These graphs are quite a mess to draw (but see the answer section of the textbook for pictures), since they contain so many edges. Instead of drawing them, we describe them in set-theoretic terms. Note that we do not consider a queen on a square to control that square itself, since to do so would give us loops in the graph.

a) The vertex set consists of all pairs (i, j) with $1 \le i \le 3$ and $1 \le j \le 3$. Since a queen in any square controls all the squares in the same row and column, there are edges $\{(i, j), (i, j')\}$ and $\{(i, j), (i', j)\}$ for all i, j, i' and j' between 1 and 3 inclusive, with $i \ne i'$ and $j \ne j'$. These are not all, though, since we have to put in the diagonal controls. There are 10 such edges: $(1, 1)$, $(2, 2)$, and $(3, 3)$ are all joined to each other;